

The (Mis)Allocation of Corporate News*

Xing Guo
Bank of Canada

Alistair Macaulay
University of Surrey

Wenting Song
UC Davis

September 1, 2026

Abstract

This paper studies how the distribution of media's reporting of firm news affects the macroeconomy. We document three connected facts on media's reporting of firm news: corporate news coverage is highly concentrated among the largest firms; equity financing and investment rise after media coverage; and yet these increases are largest among small, rarely covered firms. We develop a heterogeneous-firm model with a media sector that matches these facts, and use it to quantify the aggregate effects of the distribution of media coverage. In the model, asymmetric information between firms and investors leads to financial frictions that constrain firms' investments. Media may alleviate these information frictions, but its effects are limited by its focus on large and financially unconstrained firms. Reallocating coverage towards financially constrained firms can increase the aggregate effect of media by a factor of more than twenty.

JEL: E22, G32, D82, L82

Keywords: news media, asymmetric information, firm heterogeneity, macro-finance

*Guo (guoxing.econ@gmail.com): Bank of Canada; Macaulay (a.macaulay@surrey.ac.uk): Department of Economics, University of Surrey; and Song (wtgsong@ucdavis.edu): Department of Economics, University of California, Davis. We thank Yu-Ting Chiang, Oleksiy Kryvtsov, John Leahy, Jordi Mondria, Kristoffer Nimark, Pablo Ottonello, Francisco Queirós, Víctor Ríos-Rull, Kjetil Storesletten, and participants at several conferences and seminars for helpful discussions. Maude Ouellet provided excellent research assistance. The views expressed herein are those of the authors and not necessarily those of the Bank of Canada.

1. Introduction

Information asymmetries between firms and potential investors distort resource allocation and restrict firm growth (Myers and Majluf, 1984). Growing evidence suggests that news media is a key information source for investors (e.g., Dougal et al., 2012; Peress, 2014). However, media coverage on firms is not randomly distributed. Rather, editors and journalists selectively report on the firms they consider most newsworthy. This raises several questions. At the micro level, which firms are considered newsworthy, and how does media coverage relate to a firm’s financing and investment? At the macro level, how does the distribution of media coverage affect aggregate outcomes?

We answer using new firm-level data on media coverage, and a macro-finance model with heterogeneous firms and a media sector. Empirically, we document that news coverage is highly concentrated among the largest firms. Although firms’ equity financing and investment increase after they receive coverage, the firms receiving the most coverage respond the least. This negative relationship between media coverage and firms’ responsiveness points to a potential misallocation of coverage.

The model quantifies the aggregate importance of the cross-sectional *distribution* of corporate news. In the model, media reports can alleviate information asymmetry between firms and investors, and thus reduce financial frictions and increase investment. However, the aggregate effect of media is limited because the firms receiving the majority of coverage are large, and thus do not rely on external financing. In a constrained-efficient counterfactual that reallocates reports toward firms with the largest *gains* from coverage, the aggregate effects of media on firm values, investment, and output increase by a factor of more than twenty, and almost all aggregate losses from information frictions are eliminated. This result highlights the allocation of reports across firms as a dominant determinant of media’s aggregate relevance.

Empirical evidence We begin by constructing a new dataset on firm-level media coverage in the U.S., consisting of firm news in three major U.S. newspapers (*The Wall Street Journal*, *The New York Times*, and *USA Today*) and social media coverage on Twitter (now X). We combine this with balance-sheet data from Compustat to form a dataset of the timing and

frequency of firm news covering the universe of publicly traded firms in the U.S. over a 30-year period. We use this to document three connected facts on the distribution of news coverage.

The first fact is that corporate news coverage is highly concentrated. The variation in news coverage can be mostly accounted for by firm-specific factors. Among the set of observable firm characteristics, news coverage displays a particularly strong nonlinear relationship with firm size. The largest 10% of firms account for more than 85% of all news coverage. This concentration is unique to firm size; media coverage is substantially less concentrated by other firm characteristics.

The second fact concerns the relationship between media coverage and firms' financing and investment. The causation here is likely to go two ways: highly active firms tend to attract media coverage, and media coverage might affect firms' outcomes. We first measure the total association between the two, combining these effects. In the quarters following the media coverage, firms have a greater likelihood of raising equity financing and invest at a higher rate. At its peak, a one-standard deviation increase in media coverage is associated with a 0.25 percentage point greater probability that the firm issues new equity, and a 0.5 percentage point increase in the probability of "investment surges" (with an investment rate greater than 20%). These associations are not present for coverage on social media, and do not depend on news sentiment, indicating that the results are not due to pure salience or investor co-ordination mechanisms.

We provide further evidence that this positive correlation might be causal, using strikes in the media sector to introduce variation in news coverage unrelated to individual firms' outcomes. For this analysis, we complement the U.S. data with evidence from France, where strikes are more common. We collect news coverage data from four major newspapers in France and link them with firms' financial data from Compustat Global. Among the firms that have issued equity, those that did so in quarters with a media strike invest 16% less in the subsequent year compared to firms that issued equity outside of strikes, suggesting that the lack of information during media strikes has made the equity issuance more costly. Furthermore, firms that received high coverage before the strike were more severely affected by the strikes than firms without previous coverage, consistent with these firms being reliant on the media to communicate information to investors.

Finally, the last empirical fact concerns the joint distribution of media coverage and firm responsiveness. Recent studies suggest that the macroeconomic impact of micro-level heterogeneity depends heavily on the distribution (e.g., [Auclert, 2019](#); [Alves et al., 2020](#)). Ranking firms by size, we document that the association between news coverage and firm responses is strongest for small firms and almost negligible for large firms. Therefore, firms that receive the most coverage are those that respond the least to it.

Taken together, our empirical results suggest that media reporting helps alleviate information asymmetries between firms and investors.¹ Under this interpretation, media’s reporting reveals firm information that investors are previously unaware of. This channel is most relevant for retail investors, who make up more than a quarter of the trading volume in the U.S. and have large effects on asset prices.² Compared to retail investors, newspapers plausibly have better information about firms’ conditions. Indeed, we document a set of additional findings consistent with this information channel. In particular, the association of media coverage with firm financing is specific to *equity* financing, but is not present for cash or debt, which are much less sensitive to information ([Gorton and Ordóñez, 2014](#)).

Quantitative analysis Our findings on the negative correlation between media coverage and firm responsiveness suggest that the firms that receive the most coverage are the least responsive to it. This highlights the importance of considering the distribution of media coverage when evaluating its aggregate consequences. We use the model both to isolate the causal effects of reports, which are difficult to recover empirically, and to quantify the aggregate consequences of their allocation across firms.

To quantify these effects, we introduce a media sector into a macro-finance model with heterogeneous firms. Firm managers maximize the value accruing to existing shareholders and can raise external equity from retail investors to finance investment. Retail investors, however, face asymmetric information about the quality of firms’ heterogeneous assets. Without media reporting, adverse-selection concerns limit equity issuance, as in the literature pioneered by [Myers and Majluf \(1984\)](#). Media outlets observe firm information but, because

¹Similarly, [Tetlock \(2010\)](#) documents several empirical features of equity prices around news-coverage events that support the view that a media report removes information asymmetries.

²JP Morgan research suggests retail investors accounted for 20-30% of U.S. market volume in the first months of 2025 ([Elder, 2025](#)). For the effects of retail investors on asset prices, see e.g. [Kumar and Lee \(2006\)](#); [Greenwood et al. \(2023\)](#). Although we focus on retail investors, there are examples of the media drawing attention to information institutional investors had been ignoring (e.g. [Huberman and Regev, 2001](#)).

reporting capacity is limited, can cover only a subset of firms. A report reveals information to investors and relaxes the resulting financing friction, but only for the firms selected for coverage.

The model therefore combines three elements: heterogeneous firms, asymmetric information, and endogenous editorial choices. Firm heterogeneity makes the distribution of coverage consequential; information asymmetry creates a role for reporting; and limited reporting capacity makes its allocation across firms an economic choice.

We discipline the model using empirical moments on firms’ financing and investment, together with the observed distribution of media reports across firms. Our quantitative analysis finds that, under the baseline allocation, media reporting plays only a limited role in reducing the aggregate cost of information frictions: it eliminates 4.5% of the aggregate firm-value loss from asymmetric information. This limited role reflects the distribution of reports across firms. Media outlets allocate disproportionate coverage to large, mature firms. These firms tend to be less financially constrained and therefore benefit less from reporting. By contrast, smaller firms are more likely to be financially constrained and to benefit more from reporting, yet receive little coverage.

To isolate the role of allocation, we construct a constrained-efficient counterfactual that holds reporting capacity fixed at its baseline level and allocates reports to firms with the largest gains in firm value from coverage. This reallocation makes media’s aggregate role more than 20 times as large as under the baseline allocation and eliminates almost all aggregate losses from asymmetric information. This result highlights the paper’s central quantitative message: conditional on the amount of media coverage, its allocation across firms is the dominant determinant of its aggregate relevance.

Related literature Our paper is related to four strands of literature. First, we contribute to the literature on the macroeconomic consequences of the news media.³ Most of this literature focuses on the role of the media in reporting macroeconomic news, demonstrating that both the selection of the macroeconomic news and the way it is reported can affect aggregate dynamics (e.g., [Nimark, 2014](#); [Bybee et al., 2020](#); [Larsen et al., 2021](#); [Macaulay and Song,](#)

³A related but distinct strand of literature studies news shocks, where news typically refers to signals obtained by agents about future productivity, with the signals arriving from an unspecified source (see [Beaudry and Portier, 2014](#), for a review).

2022). Beyond macroeconomic news, [Chahrour et al. \(2021\)](#) study the reporting of sectoral news and find that it plays a substantial role in driving the business cycle. We contribute to this literature by studying the aggregate consequences of firm-level news, which we show varies substantially even within sectors. Closer to us, [Hu \(2024\)](#) provides empirical evidence that the media’s production of firm news responds to macroeconomic conditions and that this variation in news production amplifies aggregate fluctuations. We instead focus on cross-sectional heterogeneity in media coverage among firms, and our quantitative analysis shows that the distribution of the media coverage influences long-term macroeconomic outcomes.

Second, we document that both empirically and theoretically, there is a selectivity in media’s reporting of firms, which shapes the distribution of information supplied by the media. Therefore, we also relate to the broader literature studying the editorial decisions of news media. [Gentzkow and Shapiro \(2008\)](#), [Nimark and Pitschner \(2019\)](#), among others, document and model the selectivity in media reporting across political and other forms of news. [Chiang \(2020\)](#), [Martineau and Mondria \(2022\)](#), [Perego and Yuksel \(2022\)](#), [Denti and Nimark \(2022\)](#), among others, provide theoretical foundations for incentives in the news industry. We extend this literature by documenting that the selectivity is also present in the reporting of firm-level news, and characterizing the makings of newsworthy firms.⁴

Third, our findings highlight the macroeconomic importance of micro-level heterogeneity. In our model, firms have heterogeneous investment responsiveness to media coverage. As a result, redistributing media reporting to more responsive firms can significantly raise aggregate investment. This mechanism is analogous to the crucial mechanism in the literature studying household heterogeneity (e.g., [Auclert, 2019](#); [Alves et al., 2020](#); [Auclert et al., 2024](#)), which highlights the aggregate relevance of redistributing financial resources across households with heterogeneous marginal propensities to consume. This mechanism also relates to the literature studying the importance of information frictions for firms’ choices and resource allocations ([Gorton and Ordóñez, 2014](#); [Asriyan, 2021](#); [Coibion et al., 2020, 2023](#)).

Finally, our paper relates to the broader literature on the effects of financial frictions on firm dynamics and investment (e.g., [Cooley and Quadrini, 2001](#), [Serk et al., 2021](#), and see [Brunnermeier et al., 2012](#) for a survey). [Eisfeldt \(2004\)](#), [Kurlat \(2013\)](#), [Bigio \(2015\)](#), and [Caramp \(2024\)](#) study the macroeconomic and financial implications of financial frictions that

⁴Related, a large body of empirical work in finance documents that media reporting affects equity markets (e.g., [Cutler et al., 1988](#); [Chan, 2003](#); [Engelberg and Parsons, 2011](#); [Dougal et al., 2012](#)).

arise from asymmetric information. We particularly build the firm and investor blocks of our model on [Guo et al. \(2025\)](#), extending their model to incorporate a media sector. Media generates endogenous variation in the degree of asymmetric information, and consequently financial frictions across firms. We find that the allocation of media-reporting resources plays a substantial role in shaping the firm distribution and dynamics.

2. Empirical Evidence

This section documents three interrelated facts on corporate news coverage: news coverage is concentrated among the largest firms, is associated with real effects on firm outcomes, and is allocated to the least responsive firms.

2.1. Illustrative framework: a decomposition

To begin with, we present a simple model, in which media coverage interacts with firms' financing cost. This highlights that there are three moments needed to measure the aggregate consequences of media coverage: the average level of the coverage, the average firm response to the coverage, and the covariance between the news coverage and the firms' responses.

The model is static and there is a unit mass of firms. Firm i has production technology $f(I_i) = \frac{1}{\theta} I_i^\theta$. To finance its investment, the firm raises external equity from a frictional market. The news coverage of the firm, $m_i \in \{0, 1\}$, is considered exogenous to the firm and interacts with financial frictions. The marginal cost of investment is given by $\log c_i = a + a_i m_i$, where $a \in \mathbb{R}$ denotes the component of the financing costs that does not interact with the media coverage (assumed constant across firms), and $a_i \in \mathbb{R}$ denotes the component that does interact with the financing costs. This set-up allows us to study the potential role of news reporting on aggregate investment, our object of interest.

Firms choose I_i to maximize $f(I_i)$ net of the investment costs. The first-order condition of firm i 's optimal investment leads to $\log I_i^* = \psi(a + a_i m_i)$, where $\psi = -\frac{1}{1-\theta}$. Aggregating individual firms' investments implies that aggregate investment is given by $I = \int_{i \in [0,1]} I_i^* di = \mathbb{E}(I_i^*) = \exp(\psi a) \mathbb{E}(\exp(\psi a_i m_i))$. Denoting the log of aggregate investment without a media sector as $\log I^0 \equiv \psi a$, the effects of the media sector on aggregate

investment can be characterized as

$$\begin{aligned}
\log I - \log I^0 &= \log \mathbb{E}(\exp(\psi a_i m_i)) \\
&\approx \mathbb{E}(\psi a_i m_i) \\
&= \mathbb{E}(m_i) \mathbb{E}(\psi a_i) + \text{Cov}(m_i, \psi a_i) \\
&= \mathbb{E}(m_i) \mathbb{E}\left(\frac{\partial \log I_i}{\partial m_i}\right) + \text{Cov}\left(m_i, \frac{\partial \log I_i}{\partial m_i}\right), \tag{1}
\end{aligned}$$

where the second line uses a first-order Taylor approximation, the third line uses properties of an expectations operator, and the last line substitutes for ψa_i with $\frac{\partial \log I_i}{\partial m_i}$, which follows directly from differentiating firm i 's optimal investment with respect to m_i .

The decomposition in (1) shows that the aggregate effects of the media depend not only on the average level of the coverage, $\mathbb{E} m_i$, and the average investment responses to the coverage, $\mathbb{E} \frac{\partial \log I_i}{\partial m_i}$, but also on the distribution of the media coverage, consistent with the broader literature on the macroeconomic implications of micro-level heterogeneity (e.g., Auclert, 2019; Alves et al., 2020). Specifically, the covariance term indicates that when the media coverage is negatively correlated with firms' responsiveness, this dampens the aggregate investment response, whereas when the media coverage is positively correlated with firms' responsiveness, it amplifies it. Motivated by this decomposition, we now measure each component in (1) in turn.

2.2. Data

We collect data from Dow Jones Factiva on the frequency of firm news coverage by three of the largest U.S. newspapers by circulation: *The Wall Street Journal*, *The New York Times*, and *USA Today*.⁵ The news-coverage frequency is matched to firms' quarterly balance sheet data from Compustat, using a fuzzy matching algorithm (Levenshtein, 1966) based on firm names.⁶ The resulting dataset contains firm-level media coverage for the universe of publicly

⁵Factiva is a widely used database for measuring the frequency of news coverage (see, e.g., Chahrour et al., 2021; Bui et al., 2022). Our search parameters follow those in Chahrour et al. (2021), which provides media-coverage frequencies, for the top 100 firms, by news coverage in each newspaper and each quarter.

⁶Factiva provides named entity tags identifying the entities mentioned in each news article. These entities include not only firms but also organizations such as the United Nations and Harvard University. Using a fuzzy matching algorithm based on the Levenshtein distance, we match the firm names in Factiva with those of publicly traded U.S. firms in Compustat. Factiva-named entities often include slight variants of the same firm (e.g., "AT&T Inc" and "AT&T Inc."). Our algorithm recognizes that both names refer to the same

traded firms in the U.S., consisting of 385,698 articles on 18,809 firms from 1990 to 2021. On average, news about publicly traded firms in the U.S. constitutes 9% of total news articles in these newspapers over the sample period.

We complement the main data on news frequency with three additional datasets. The first contains the full texts of the news articles obtained from Dow Jones Data, News, and Analytics. This contains the detailed content of the subset of the news articles that the Dow Jones is licensed to redistribute (representing 54% of the full coverage sample).

The second is the firms’ social media coverage on Twitter (now X), which allows us to compare the role of the curated news coverage with the social media coverage. We identify over 3,000 publicly traded firms with official accounts on this social media platform. We then collect the frequency with which each firm is mentioned (e.g., *@Microsoft*) each quarter from 2014, when Twitter became a popular platform, to 2021.

Finally, we use news-coverage data from France, where periods of media strikes introduce variation in the media coverage. We use Factiva to collect the frequency of firms’ news coverage from 2005 to 2021 in four major French newspapers: *Les Echos*, *Le Monde*, *La Tribune*, and *Le Figaro*, and link the coverage data with firm variables from Compustat Global as in the U.S. analysis.

2.3. Distribution of media coverage

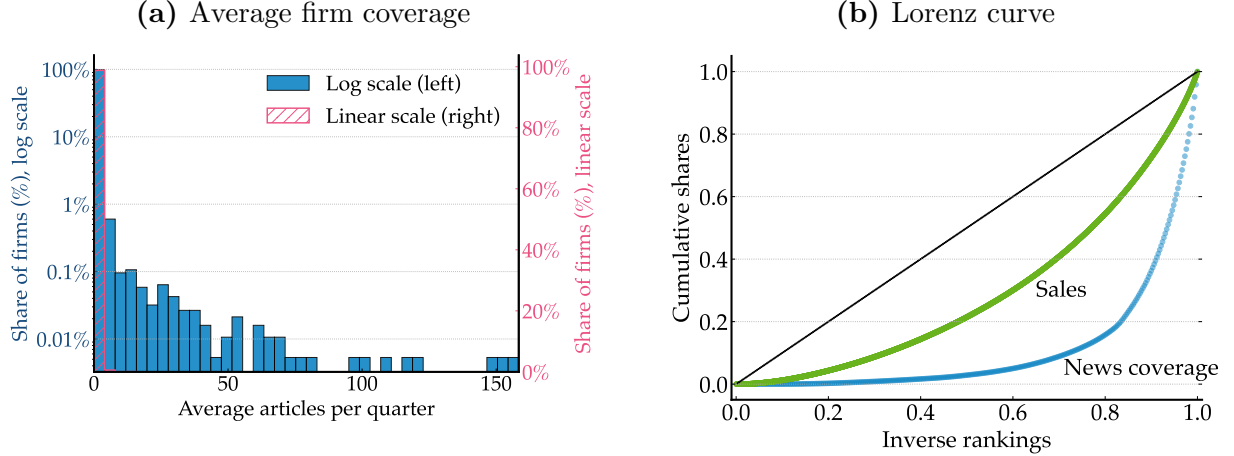
Panel (a) in Figure 1 shows the distribution of firms’ average per-quarter article counts over the sample period. The right (dashed red) axis plots the share of firms with each level of average coverage at the untransformed linear scale. To display the right tail of the distribution, the left (solid blue) axis plots the share at log scale.

The distribution is highly skewed, with news coverage concentrated among a small number of firms. While most firms receive no media coverage, those in the top 1% appear in an average of 21 articles per quarter in major newspapers.⁷ Appendix Figure A.1a restricts the sample to firms with non-zero coverage, and Appendix Figure A.1b reports the distribution of firm news coverage within industries, which suggests that the skewed pattern of media

firm. To ensure match quality, we perform manual checks on each of the matches.

⁷Table A.1 in the Appendix lists the top 10 firms by total media coverage. The top firms are household names such as BlackRock, AT&T, and General Motors, whose brand recognition may attract attention from readers who do not necessarily have a specific interest in business news.

Figure 1: Distribution of corporate news coverage



Notes: Panel (a) reports the histogram of the average number of articles that firms receive per quarter over the sample period. The left (solid blue) axis plots the share of firms with each level of average coverage at log scale, and the right (dashed red) axis plots the share at the untransformed linear scale. Panel (b) reports the Lorenz curve of media coverage compared with that of real sales. Firms are ranked by the share of their media coverage (or real sales), measured as a firm’s log article counts (or real sales) divided by the sum across all firms over the sample period. The blue line plots the cumulative share of log article counts against firms inversely ranked by media coverage share, and the green line plots the same for log real sales. We invert the ranking in this panel to follow the standard Lorenz curve convention, so that firms with the least media coverage or sales appear on the left. The black line represents the 45-degree line of perfect equality.

coverage is not driven by firms without any coverage or industry-specific media attention.

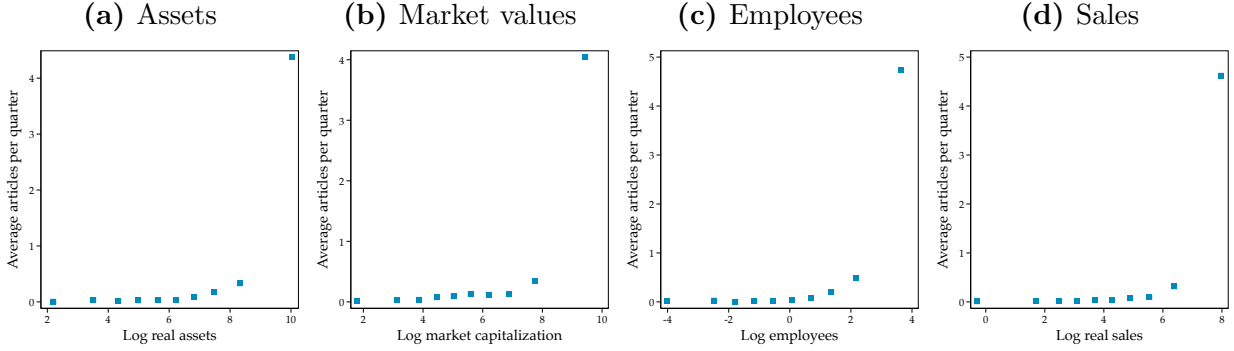
To assess the degree of concentration in corporate news coverage, Panel (b) in Figure 1 presents the Lorenz curve of firm-level media coverage shares. The blue line shows the cumulative share of log article counts across firms, ranked in ascending order by their media coverage share. We invert the ranking in this panel to follow the standard Lorenz curves convention, so that firms with the least media coverage appear on the left. The figure illustrates the high concentration in firm news coverage. For comparison, the green line plots the Lorenz curve for firms’ log real sales, which is itself highly concentrated (e.g., [Autor et al., 2020](#)), yet notably less so than media coverage.

Appendix Table A.2 decomposes the variation in media coverage. Firm fixed effects account for 60% of the variation, while the firm’s industry explains an additional 4%. The time dimension plays a minor role. The remaining 34% reflects variation over time at the firm level.

We next explore the association between media coverage and firm characteristics.⁸ In particular, Figure 2 shows the relationship between media coverage and firm size, measured

⁸Firm-level variables are constructed as standard in the literature (details in Appendix A.1).

Figure 2: Firm size and media coverage



Notes: This figure reports binned scatterplots of average news articles per quarter. Each bin represents a decile of firms by size, sorted from the smallest to largest. Firm size is measured by log real assets in panel (a), market capitalization (prices per share times shares outstanding) in panel (b), log employee numbers in panel (c), and log real sales in panel (d).

four different ways: log real assets (panel a), market value (b), log number of employees (c), and log real sales (d). Across all measures, media coverage is highly concentrated among the largest 10% of firms. Each panel is a binned scatter plot, in which each bin represents a decile of firm-quarter observations. Appendix Figure A.3 confirms that this concentration is also present within 4-digit NAICS industries.

This sharp concentration turns out to be unique to firm size. Appendix Figure A.2 plots the equivalent associations between media coverage and firm investment, R&D, age, and leverage. While some correlations are apparent, they are substantially weaker than that with firm size. This suggests that media’s attention to the largest firms is a specific function of their size, not a byproduct of their activity levels, age, or financial conditions.⁹

2.4. Firm responses to media coverage

Next, we study the relationship between news coverage and firm outcomes. Both media coverage and firm outcomes are endogenously determined: firms with high investment and financing activities tend to attract media coverage, and media coverage may affect firms’ investment and financing. In this subsection, we first estimate the correlation between media coverage and firm outcomes. Then, we focus on a purged coverage measure, removing news

⁹We choose age and leverage in particular as, alongside firm size, they are often considered important for business cycle fluctuations and the transmission of macroeconomic policy (e.g. Gertler and Gilchrist, 1993; Cloyne et al., 2023; Ottonello and Winberry, 2020). The relationship of media coverage with age, like that with size, is monotonically increasing, but the concentration is still much weaker, with mid-aged firms appearing regularly in the news.

on investment and financing. Lastly, we provide evidence on the causal mechanism using media strikes.

The main measure of coverage for this analysis is the *within-firm* variation in media coverage, $(\nu_{it} - \mathbb{E}_i \nu_{it})$, where ν_{it} denotes the number of mentions of firm i in major U.S. newspapers in quarter t , and $\mathbb{E}_i \nu_{it}$ denotes the average number of mentions of the firm over the sample. We focus on within-firm variation because it accounts for the time-invariant differences in media coverage across firms documented in Section 2.3, and because it is consistent with the model mechanism in Section 3, which studies ex-post heterogeneity. We complement it with a second measure, $\mathbb{1}(\nu_{it} > 0)$, which is an indicator variable that takes the value of 1 if firm i has any news coverage in quarter t and 0 otherwise. We report robustness with this measure because of its simplicity, which provides a particularly transparent interpretation of the estimate.

2.4.1. Media coverage and firm investment

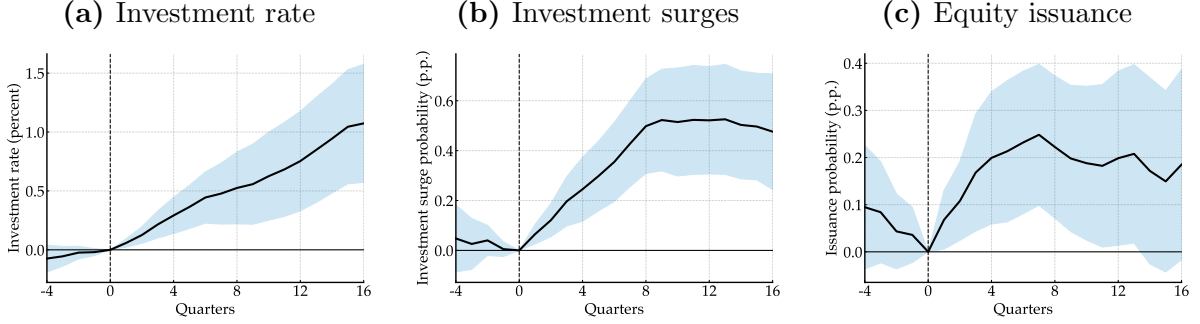
We first estimate the correlation between media coverage and firm outcomes using local projections (Jordà, 2005). For firm i in quarter t , we estimate for each horizon $-4 \leq h \leq 16$

$$\Delta_h y_{it} = \alpha_{st} + \alpha_i + \beta_h(\nu_{it} - \mathbb{E}_i \nu_{it}) + \Gamma' Z_{it} + u_{ith}, \quad (2)$$

where $\Delta_h y_{it}$ is the firm variable of interest; ν_{it} denotes the number of mentions of firm i in major U.S. newspapers in quarter t , and $\mathbb{E}_i \nu_{it}$ denotes the average number of mentions of the firm over the sample; $\{\alpha_{st}, \alpha_i\}$ are sector-by-quarter and firm fixed effects; Z_{it} is a vector of firm controls including size, age, and real sales growth; and u_{ith} is a random error.

The firm variables of interest include (i) the investment rate, $\Delta_h \log k_{it}$, defined as the log change in the book value of the firm's tangible capital stock; (ii) the cumulative probability of investment surges, $\max_{t \leq \tau \leq t+h} \mathbb{1}(\Delta \log k_{i\tau} > 20\%)$, defined as an indicator equal to 1 if a firm has a quarterly investment rate higher than 20% between quarters t and $t+h$, and 0 otherwise; and (iii) the cumulative probability of equity issuance, E_{it} , an indicator equal to 1 if a firm issues new equity between quarters t and $t+h$ and 0 otherwise. We study investment surges in addition to investment rates because investment tends to be infrequent and lumpy (e.g., Doms and Dunne, 1998). We double cluster the standard errors by firm and quarter.

Figure 3: News coverage, firm investment, and financing



Notes: This figure reports the results from estimating the local projections in equation (2) for quarters $-4 \leq h \leq 16$: $\Delta_h y_{it} = \alpha_{st} + \alpha_i + \beta_h(\nu_{it} - \mathbb{E}_i \nu_{it}) + \Gamma' Z_{it} + u_{ith}$, where $\{\alpha_{st}, \alpha_i\}$ denote the sector-by-quarter and firm fixed effects; ν_{it} denotes the number of mentions of firm i in major U.S. newspapers in quarter t ; $\mathbb{E}_i \nu_{it}$ denotes the average number of mentions of the firm over the sample; and Z_{it} is a vector of firm controls including their size, age, and real sales growth. The dependent variable y_{it} includes (a) the investment rate ($\Delta_h \log k_{it}$), defined as the log change in the book value of the firm's tangible capital stock; (b) the cumulative probability of investment surges ($\max_{t \leq \tau \leq t+h} \mathbb{1}(\Delta \log k_{i\tau} > 20\%)$), defined as an indicator variable that takes the value of 1 if a firm has a quarterly investment rate higher than 20% between quarters t and $t+h$, and 0 otherwise; and (c) the cumulative probability of equity issuance (E_{it}), defined as an indicator variable that takes the value of 1 if a firm issues new equity between quarters t and $t+h$ and 0 otherwise. We have standardized $(\nu_{it} - \mathbb{E}_i \nu_{it})$ over the entire sample. Standard errors are double clustered by firm and quarter. 90% confidence intervals are reported.

To help interpret the estimates, we standardize firms' demeaned coverage, $\nu_{it} - \mathbb{E}_i \nu_{it}$, over the entire sample, so that the unit can be interpreted as one standard-deviation within-firm change in media coverage.

Figure 3 reports our baseline findings. Panel (a) shows that more coverage is associated with a subsequent increase in investment. A one standard deviation higher media coverage is associated with 0.05% higher investment in the quarter after the coverage. The positive association rises gradually over the estimation horizon, reaching a peak of approximately 1%. Panel (b) takes into account the lumpy nature of investment. It shows that media coverage is also associated with a higher subsequent likelihood of investment surges (defined as an investment rate greater than 20%). The peak association occurs around 8 quarters after the coverage, with a one standard deviation higher coverage associated with a 0.5 percentage point higher probability of investment surges. Panel (c) shows that media coverage is associated with a higher probability of raising financing from the equity market. In the quarter after the news coverage, a one standard deviation higher media coverage is associated with a 0.07 percentage point greater probability of a firm issuing equity. The association rises gradually to a peak of around 0.25 percentage points after 6 quarters.

Table 1: Media coverage and firm investment: Summary

		Impact	Peak	Average	Obs.	Dynamics
(a) Alternative measures of coverage or investment						
<i>Dependent var.:</i>	<i>Independent var.:</i>					
Investment rate	Media coverage	0.058*** (0.022)	1.073*** (0.305)	0.564	385,715	Fig. 3a (baseline)
Investment surges	Media coverage	0.065*** (0.024)	0.526*** (0.134)	0.393	384,653	Fig. 3b
Investment rate	Coverage indicator	0.669*** (0.225)	6.124** (2.407)	3.637	385,715	Fig. A.4a
Investment surges	Coverage indicator	0.796*** (0.294)	1.734 (1.048)	1.033	384,653	Fig. A.4b
(b) Robustness and additional analyses						
Excluding common ownership		0.060*** (0.022)	1.089*** (0.305)	0.573	383,815	Fig. A.5
Excluding media firms		0.059*** (0.022)	1.074*** (0.305)	0.564	383,979	Fig. A.6
Trimming outlier firms		0.076*** (0.022)	0.896*** (0.332)	0.527	381,470	Fig. A.14
Post-2005 sample		0.077** (0.033)	1.007*** (0.329)	0.612	155,965	Fig. A.7
(c) Information content in media coverage						
By financing types	Equity financing	0.068* (0.038)	0.248*** (0.091)	0.184	377,708	Fig. A.8
	Debt financing	-0.003 (0.003)	-0.003 (0.003)	0.002	287,263	
	Cash holdings	0.001 (0.003)	0.010** (0.005)	0.005	388,255	
Twitter coverage		-0.024 (0.043)	-0.454 (0.347)	-0.269	42,942	Fig. A.9
(d) Purged measure of media coverage						
Ex. articles on investment and financing		0.071*** (0.026)	1.544*** (0.451)	0.809	385,715	Fig. A.11
(e) By sentiment						
Effects by sentiment	Coverage	0.060** (0.023)	1.109*** (0.322)	0.588	385,715	Fig. A.12
	Coverage $\times$ Sentiment	-0.000 (0.004)	0.008 (0.039)	0.007		

Notes: This table estimates variants of (2): $\Delta_h y_{it} = \alpha_{st} + \alpha_i + \beta_h(\nu_{it} - \mathbb{E}_i \nu_{it}) + \Gamma' Z_{it} + u_{ith}$. “Impact” denotes estimates at $h = 1$; “peak” denotes the largest estimates in absolute value for $1 \leq h \leq 16$; and “average” denotes the average of coefficients over $1 \leq h \leq 16$. The first two rows of panel (a) repeat the baseline estimates in Figure 3. The last two rows of panel (a) use $\mathbb{1}(\nu_{it} > 0)$ as the independent variable, which is an indicator variable that equals 1 if firm i has any coverage in quarter t . Panel (b) excludes firms that share common ownership with the three newspapers in the U.S. sample, those in the media sector (NAICS 5418), and firms with sizes above the 99th percentile; and restricts to the post-2005 sample. The first three rows of Panel (c) use as the dependent variable the cumulative probabilities of equity issuance, debt issuance, and increasing cash holdings, respectively. The last row of Panel (c) uses as the independent variable Twitter coverage, $(\tau_{it} - \mathbb{E}_i \tau_{it})$. Panel (d) estimates (2) where the independent variable is $\nu_{it}^{\text{purged}} - \mathbb{E}_i \nu_{it}^{\text{purged}}$, a purged measure of media coverage defined in Section 2.4.2 that excludes articles on investment or financing. Panel (e) reports $\beta_{\nu h}$ and $\beta_{\nu sh}$ estimated from $\Delta_h y_{it} = \alpha_{st} + \alpha_i + \beta_{\nu h}(\nu_{it} - \mathbb{E}_i \nu_{it}) + \beta_{\nu sh}(\nu_{it} - \mathbb{E}_i \nu_{it}) \times \text{sent}_{it} + \Gamma' Z_{it} + u_{ith}$, where sent_{it} denotes the news sentiment of firm i in quarter t defined in Section 2.4.2. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table 1 presents a summary of the robustness tests and additional analysis. We report the estimated coefficients and standard errors for the on-impact and peak effects from estimating variants of (2), along with hyperlinks to the full estimated dynamics in the appendix.

Panel (a) of Table 1 presents results using a more transparent measure of media coverage. Rather than using within-firm variation, we use a binary indicator, $\mathbf{1}(\nu_{it} > 0)$, which takes the value of 1 if a firm has any coverage in a quarter and 0 otherwise. Firms' investment rate and investment surges remain positively correlated with media coverage under this alternative measure.

Panel (b) of Table 1 contains four sets of robustness and additional analyses. The first two rows account for the possibility that publicly traded firms may be related to the newspapers in our sample, by sharing the same owners or operating in the same industry. To address this, we exclude firms that share common owners with the three newspapers, or firms in the media sector (NAICS 5418), and re-estimate the local projection in (2). The estimates are little changed from the baseline, which suggests that the positive relationship between media coverage and firms' investment and equity issuance is not driven by the potential connections between non-media and media firms. The third row ensures that the results are not driven by the largest firms by excluding the largest 1% of firms from the estimation. The last row of panel (b) examines whether the importance of media coverage has changed in recent years. Figure A.7 shows that over the 2005–2021 sample, which coincides with the rise of social media, media coverage continues to be positively associated with firm investment. However, its correlation with equity issuance declines to nearly zero, suggesting that the influence of media coverage may have weakened in more recent years.

Panel (c) of Table 1 presents two sets of evidence that media coverage reveals information on firms. First, Appendix Figure A.8 shows that the association with news coverage is specific to equity financing: the correlations of media coverage with debt financing and cash financing are much smaller in magnitude and statistically insignificant. This is consistent with equity being an informationally sensitive form of financing, whereas debt and cash financing are much less sensitive to information (Gorton and Pennacchi, 1990; Gorton and Ordonez, 2014; Hoberg and Maksimovic, 2015).

Second, we compare the curated news from newspapers with the social media platform Twitter (now X), which has become a major alternative to traditional news media over the

last decade. While newspaper articles are produced by trained journalists and curated by editors, tweets are produced by individual users and are largely unmoderated. Panel (c) of Table 1 (and the full dynamics in Appendix Figure A.9) shows that unlike newspaper coverage, there is no significant association of Twitter coverage with investment and equity issuance probability. Indeed, if anything, the associations are negative. This suggests that the positive association with firms’ outcomes is specific to curated news.

These findings motivate our modeling choices in Section 3, where we assume that media affects firms by providing information to investors. The findings that Twitter coverage does not have similar effects to newspaper reporting provide evidence against alternative theories, such as media acting as a coordination device for investors who already hold information, or media affecting investors through behavioral channels such as increasing salience.

2.4.2. Purged measure of media coverage

Having documented the positive association between media coverage and firm outcomes, we now focus on isolating the effects that media coverage has on firms. In this subsection, we address a selection bias where firms with imminent investment or financing tend to attract media attention. Leveraging detailed text of newspaper articles, we identify and remove news articles covering these events from our measure.

We proceed in two steps. First, we use the latent Dirichlet allocation (LDA) model to extract 20 distinct “topics” that represent the articles.¹⁰ The resulting topics are reported in Appendix Figure A.10. The news coverage about firms falls into three broad categories: the news related to overall financial conditions (e.g., “stock markets”), the news related to firms’ industries (e.g., “technology” and “automobiles”), and firm-specific news (e.g., “investment,” “financing,” “litigation,” and “employees”). Then, we exclude news articles that have any positive loadings on topics related to firm investment and financing: topic 5 (“investment”), topic 16 (“financing”), and topic 8 (“financing from international markets”) in Appendix Figure A.10. We then re-estimate the baseline local projection

$$\Delta_h y_{it} = \alpha_{st} + \alpha_i + \beta_h(\nu_{it}^{\text{purged}} - \mathbb{E}_i \nu_{it}^{\text{purged}}) + \Gamma' Z_{it} + u_{ith} \quad (3)$$

¹⁰Appendix A.3 details the text pre-processing and LDA estimation.

using this purged measure of media coverage, $\nu_{it}^{\text{purged}} - \mathbb{E}_i \nu_{it}^{\text{purged}}$, which excludes the coverage of firm investment and financing.

Panel (d) of Table 1 reports the estimates (with the full dynamics reported in Appendix Figure A.11). We find that firms' coverage unrelated to investment or financing remains associated with a higher probability of equity issuance and a higher rate of investment, with somewhat stronger effects than those estimated using the total news frequency. This suggests that direct reverse causality is unlikely to be driving our results.

Using detailed texts of news articles, we also analyze the effect of *news sentiment*. Sentiment is measured using three approaches: a keyword-based measure (Loughran and McDonald, 2011), a FinBERT-based measure (Araci, 2019), and a GPT-based measure (Ye et al., 2023), detailed in Appendix A.4. Since each methodology has strengths and weaknesses, we measure news sentiment, sent_{it} , with the first principal component of the three measures. We then interact the frequency and the sentiment of media coverage and estimate

$$\Delta_h y_{it} = \alpha_{st} + \alpha_i + \beta_{\nu h}(\nu_{it} - \mathbb{E}_i \nu_{it}) + \beta_{sh} \text{sent}_{it} + \beta_{\nu sh}(\nu_{it} - \mathbb{E}_i \nu_{it}) \times \text{sent}_{it} + \Gamma' Z_{it} + u_{ith}, \quad (4)$$

where sent_{it} denotes the average sentiment of firm i 's media coverage in quarter t .

Panel (e) of Table 1 reports the estimates. The coefficient on news coverage frequency remains positive and statistically significant. In contrast, the coefficient on news sentiment is close to zero and statistically insignificant. These findings suggest that, in the context of firm-specific news, all coverage is good coverage.

2.4.3. Evidence from media strikes

To complete the analysis on the effects of media coverage, we provide evidence on the causal mechanism behind the relationship between coverage and investment. In this subsection, we use episodes of media strikes to introduce variation in the news coverage that is unrelated to firm choices. During strikes, journalists stop reporting, reducing the amount and quality of information provided by the media sector, for reasons that are unrelated to individual non-media firms (Peress, 2014).

Media strikes in the U.S. have been rare in recent history. However, in France we identify 6 episodes of large-scale media strikes, using the criteria developed by Peress (2014),

detailed in Appendix Table A.3.¹¹ We focus on sector-wide strikes and exclude strikes against individual newspapers, to ensure that these strikes occur not because of individual newspaper or non-media firm factors but rather as a response to the government’s policy changes (such as Nicolas Sarkozy’s broadcasting-advertising reform and Emmanuel Macron’s pension reform).

To facilitate comparison with the U.S. evidence, we first report the distribution and the effects of media coverage in France. Panel (a) of Appendix Figure A.13 shows that the distribution of corporate news coverage in France displays a similar concentration to the U.S. coverage. Panels (b) and (c) report estimates of the relationship between media coverage and firms’ investment and equity issuance, respectively, using the local projection specified in (2).¹² The results are consistent with the U.S. evidence: greater media coverage in France is associated with a higher probability of equity issuance and investment.¹³

Our main analysis tests whether the reduction in media coverage during media strikes affects firms’ outcomes. We restrict our focus to firms that issued equity during the sample period. To the extent that equity issuances are pre-scheduled and not strategically timed to coincide with or avoid media strikes, differences in post-issuance investment depending on firms’ exposure to media strikes are informative about the effect of media coverage. Indeed, Appendix Table A.4 reports descriptive statistics on firm news, showing that both the number of firm news articles and their share of total news coverage are lower during media strikes than in adjacent quarters without strikes. Further, Appendix Table A.5 shows that the probability of equity issuance does not vary with media strikes, which verifies that there is no evidence of strategic timing in firms’ equity issuance related to media strikes.

¹¹We search Factiva for keywords containing (i) “strike” and “journalist,” or (ii) “strike” and “broadcaster,” as well as their French translations. Using Factiva’s tagging, we restrict the region to France, the industry to Media/Entertainment, and the subject to Labor Dispute. We focus on national strikes and exclude strikes against individual newspapers. The 6 strike episodes are reported in Appendix Table A.3. They are concentrated in 5 quarters: 2005Q4, 2008Q1, 2008Q4, 2013Q1, and 2018Q2.

¹²For horizons $-4 \leq h \leq 12$, we estimate $\Delta_h y_{it+h} = \alpha_{st} + \alpha_i + \beta_h(\nu_{it} - \mathbb{E}_i \nu_{it}) + \Gamma' Z_{it} + u_{ith}$. As with the U.S. analysis, the dependent variables consist of cumulative changes in investment and equity issuance probabilities, and the explanatory variable, ν_{it} , measures firm coverage in the four major French newspapers and is demeaned at the firm level and standardized. α_i and α_{st} refer to the firm and sector-by-quarter fixed effects. We classify sectors using the 2-digit rather than the 4-digit NAICS levels because the French equity market is far smaller than the U.S. market (there are 959 unique publicly traded firms in our French sample compared to 13,207 firms in our U.S. sample). The vector Z_{it} controls for firms’ size (log real assets), age, and real sales growth.

¹³Similar to the U.S., retail investors in France also play an increasingly important role in the equity market. According to a report by the Autorité des Marchés Financiers (AMF, 2026), which regulates the French financial market place, over 1.9 million French people bought or sold shares at least once, up 21% on the previous year, and 56 million equity transactions were carried out by retail investors in 2025. Over the entire 2021-2025 period, the equity markets attracted more than 3.8 million retail investors.

Having verified our key assumptions, we estimate

$$\log k_{it+4} - \log k_{it} = \alpha_s + \beta S_t + \delta \theta_{it} + \gamma \theta_{it} S_t + \Gamma' Z_{it} + u_{it}, \quad (5)$$

where $\log k_{it+4} - \log k_{it}$ denotes firm i 's cumulative investment a year after its equity issuance, α_s denotes the sector fixed effect; S_t is an indicator for media strikes in quarter t ; θ_{it} denotes firm i 's average news coverage in the year before the strike; and Z_{it} is a vector of firm and macroeconomic controls, including firm size, age, real sales growth, fiscal year end, real GDP growth, inflation, and aggregate investment.¹⁴

The parameter β estimates the average effect of media strikes by comparing the investment of firms that issue equity during media strikes with that of firms issuing equity outside strike periods. However, the timing of media strikes may be influenced by broader economic conditions. In particular, strikes tend to happen in economic downturns, which have confounding effects on firm investment. To address this concern, we further exploit cross-sectional variation in firms' exposure to the strikes, measured by each firm's past media coverage. Firms with greater past media coverage rely more heavily on the media to disseminate information to investors and are therefore more exposed to the disruptive effects of media strikes.

The parameter of interest is γ . Among firms that issued equity during media strikes, γ measures the differential impact of a strike on a firm's investment, depending on the firm's past coverage. If the news media disseminates firm news to investors, firms that tend to receive more coverage are expected to suffer bigger impacts during strikes compared to their peers with little coverage to begin with. The specification in (5) allows for the possibility that strikes tend to happen in economic downturns, since the source of the variation in this regression is the cross-sectional variation in firms' exposure to the same strike.

Table 2 reports the results. Column 1 estimates the average effect of media strikes and finds that firms that issue equity during media strikes invest less in the subsequent year, compared to firms that issue equity during quarters without media strikes. Columns 2 through 5 further exploit the cross-sectional variation in firms' past news coverage to study the effects of the exposure to strikes. Column 2 reports the baseline estimates of (5) without

¹⁴We retrieve GDP (CLVMNACSCAB1GQFR), inflation (CPHPTT01FRM659N), and aggregate investment (PRMNVG01FRQ661S) from FRED.

Table 2: Equity issuance during media strikes and exposure to media coverage

	(1)	(2)	(3)	(4)	(5)
	Investment after issuance (1yr)				
Strike	-0.140*	-0.173	-0.162*	-0.162*	-0.025
	(0.078)	(0.106)	(0.096)	(0.096)	(0.049)
Past coverage		0.004	0.006	0.005	0.005
		(0.004)	(0.005)	(0.005)	(0.005)
Strike $\times$ Past coverage		-0.042*	-0.039*	-0.039*	-0.061***
		(0.021)	(0.020)	(0.020)	(0.012)
Observations	1072	1024	882	881	842
R^2	0.029	0.039	0.056	0.056	0.052
Industry FE	yes	yes	yes	yes	yes
Double-clustered SE	yes	yes	yes	yes	yes
Firm + macro controls	no	no	yes	yes	yes
Ex. common ownership	no	no	no	yes	yes
Ex. 2008	no	no	no	no	yes

Notes: This table reports the coefficients β, δ, γ from estimating equation (5): $\log k_{it+4} - \log k_{it} = \alpha_j + \beta S_t + \delta \theta_{it} + \gamma \theta_{it} S_t + \Gamma' Z_{it} + u_{it}$, where t is the quarter in which a firm issues equity, $\log k_{it+4} - \log k_{it}$ is the cumulative investment 4 quarters after the equity issuance, α_j is a sector fixed effect, S_t is an indicator for media strikes, θ_{it} is the average media coverage of firm i 4 quarters before the strike at time t , and Z_{it} is a vector of controls containing size, age, real sales growth, real GDP growth, inflation, and aggregate investment. Column 5 excludes 2008. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

any controls. Column 3 controls for firm performance and macroeconomic fluctuations. Column 4 excludes firms that share a common owner with a major newspaper, to account for a possible direct effect of the labor disputes behind the media strikes on the investments of the firms in our sample.¹⁵ Strikes in newspapers can arise from disputes with their owners, which potentially affect the investment decisions of their non-media subsidiaries for reasons other than media coverage, and we account for this possibility by removing these subsidiaries. Finally, Column 5 excludes episodes from 2008, to ensure that our results are not driven by outliers from the Global Financial Crisis.

We focus our discussion on Column 4 in Table 2, which provides the most conservative estimates using the full sample. Firms that issue equity during media strikes invest 16%

¹⁵Specifically, *Les Echos* and *Le Figaro* are owned by LVMH and the Dassault Group, respectively. These groups are also the parent companies of some of the non-media firms in our sample. Subsidiaries of the Dassault Group (parent of *Le Figaro*) include Dassault Aviation and Dassault Systems; and the subsidiaries of LVMH (parent of *Les Echos*) include Bulgari and Moët. *La Tribune* was owned by LVMH from 1993 to 2007 and is currently owned by individual investors. *Le Monde* belongs to Groupe Le Monde, which does not have other subsidiaries in our sample.

less in the year following the issuance compared to those issuing during non-strike quarters. This effect is more pronounced for firms with historically higher media coverage. Among firms that issue equity during strikes, those with a historical coverage that is one standard deviation higher invest 4% less post-issuance. The economic magnitude is one quarter of the average effects from the strikes. Remaining columns in Table 2 show that the qualitative effects we document are not sensitive to controls or driven by outliers from the financial crisis.

Appendix A.5 documents similar patterns for firm equity issuance following media strikes. Table A.6 shows that firms are less likely to issue equity in the quarters following a media strike, with the largest decline occurring in the second and third quarters after the strike. Among firms that do issue equity, issuance amounts also appear to be smaller, although the sample of issuers is too small to draw definite conclusions.

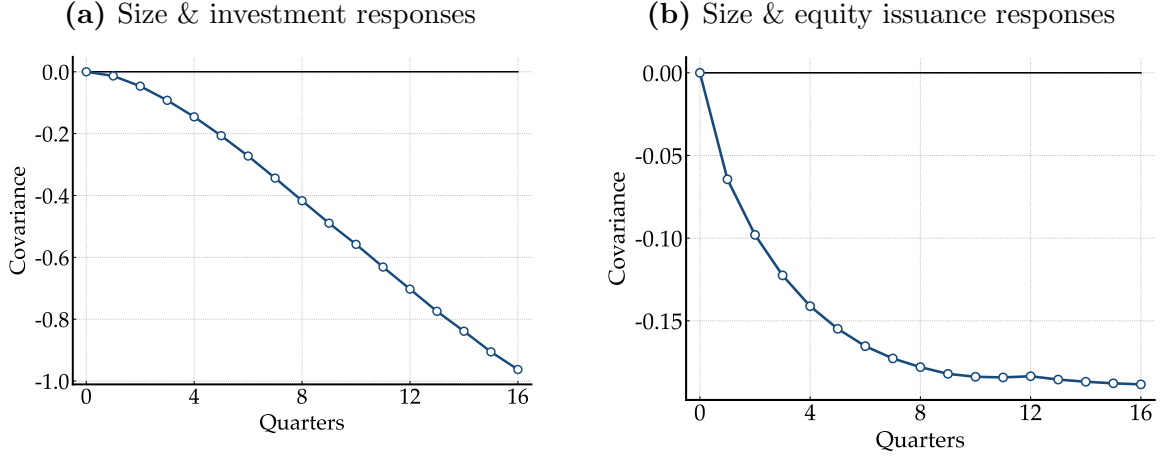
The results suggest that firms that rely more on media coverage to disseminate firm news reduce investment more sharply because of the strikes, which is consistent with the interpretation that media reports can alleviate the information friction firms face and facilitate their financing and investment.

2.5. Firm responsiveness and media coverage

The decomposition in Section 2.1 highlights that the aggregate effects of corporate news depend on how this news coverage is distributed across firms. Specifically, aggregate investment depends on whether the coverage is correlated with firm-level responsiveness to this news. Motivated by this, we now study the joint distribution between news coverage and the firm responsiveness. To do so, we return to the U.S. data, which provide the richest set of firm-level information. We focus on the size dimension, the strongest observed driver of these firms' news coverage.

We first report the covariance between a firm's size and its responses to media coverage. Firm size is residualized with respect to sector and year fixed effects. Firm-level response to media coverage is measured by the fitted value implied by the baseline local-projection specification in equation (2). Figure 4 reports the covariance between firm size and the fitted firm-level investment and equity responses to media coverage, $\text{Cov}(s_{it}, \widehat{\Delta_h y_{it}})$ for each horizon h . The covariance is negative across all estimation horizons. This pattern indicates

Figure 4: Covariance: firm size and responsiveness to coverage



Notes: This figure reports the covariance, $\text{Cov}(s_{it}, \widehat{\Delta_h y_{it}})$, between firm size, s_{it} , for firm i in quarter t and its cumulative responses to media coverage, $\widehat{\Delta_h y_{it}}$, measured by the fitted value implied by the baseline local-projection specification in equation (2) for horizon h : $\Delta_h y_{it} = \alpha_{st} + \alpha_i + \beta_h(\nu_{it} - \mathbb{E}_i \nu_{it}) + \Gamma' Z_{it} + u_{it}$. In panel (a), $\Delta_h y_{it}$ denotes the investment rate ($\Delta_h \log k_{it}$), defined as the log change in the book value of the firm's tangible capital stock. In panel (b), $\Delta_h y_{it}$ denotes the cumulative probability of equity issuance (E_{it}), defined as an indicator variable that takes the value of 1 if a firm issues new equity between quarters t and $t + h$ and 0 otherwise.

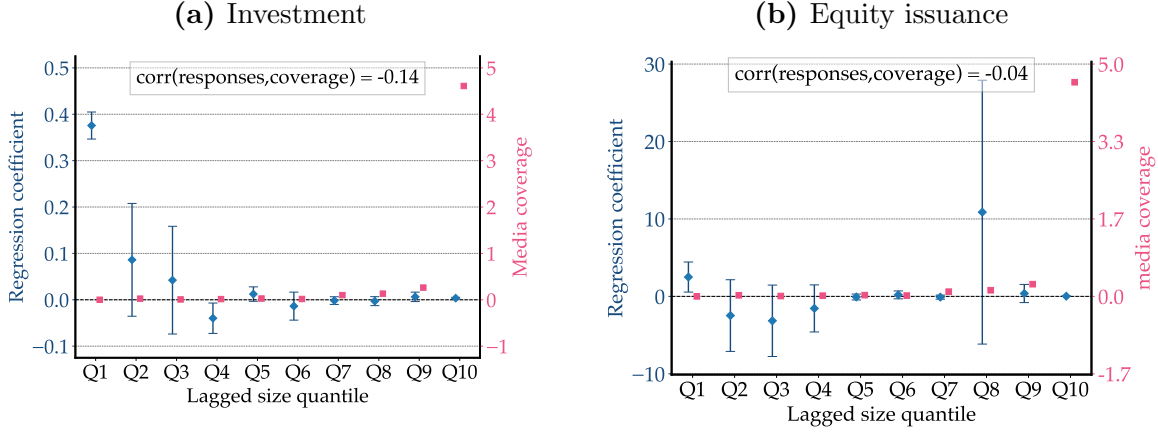
that larger firms' investment and equity issuances tend to vary less with media coverage than those of the smaller firms.

We next estimate the responsiveness by size decile. Sorting firms into size deciles, ranked from smallest to largest as $q = 1, \dots, 10$, we estimate

$$\Delta y_{it} = \alpha_{st} + \alpha_i + \beta_q \cdot \mathbb{1}(Q_{it-1} = q)(\nu_{it} - \mathbb{E}_i \nu_{it}) + \Gamma' Z_{it} + u_{it}, \quad (6)$$

where y_{it} is the firm variable of interest; $\mathbb{1}(Q_{it-1} = q)$ is an indicator variable that takes the value of 1 if a firm's size decile in the previous quarter Q_{it-1} belongs to decile q ; $\nu_{it} - \mathbb{E}_i \nu_{it}$ is the news coverage of firm i that major U.S. newspapers mention in quarter t , demeaned at the firm level and standardized; $\{\alpha_{st}, \alpha_i\}$ are sector-by-quarter and firm fixed effects; Z_{it} represents a vector of standard firm controls that include size, age, real sales growth; and u_{it} is a random error. The firm variables of interest include (i) the cumulative investment rate, defined as the log change in the book value of the firms' tangible capital stock one year from coverage, and (ii) the firms' equity issuance one year from coverage, defined as the log equity issuance scaled by the firms' tangible capital stock. We double cluster the standard errors by firm and quarter.

Figure 5: Media coverage and firm responsiveness by size decile



Notes: This figure reports the results from estimating equation (6): $\Delta y_{it} = \alpha_{st} + \alpha_i + \beta_q \cdot \mathbf{1}(Q_{it-1} = q) \times (\nu_{it} - \mathbb{E}_i \nu_{it}) + \Gamma' Z_{it} + u_{it}$, where $\{\alpha_{st}, \alpha_i\}$ denote the sector-by-quarter and firm fixed effects; $\mathbf{1}_{Q_{it-1}=q}$ is an indicator variable that takes the value of 1 if a firm's size decile within quarter Q_{it-1} belongs to decile q ; ν_{it} is firm i 's news coverage mentioned in major U.S. newspapers in quarter t , demeaned at the firm level and standardized; and Z_{it} is a vector of firm controls including size, age, and real sales growth. The dependent variable Δy_{it} includes the cumulative investment rate in panel (a), defined as the log change in the book value of the firm's tangible capital stock one year from coverage, and the equity issuance one year from coverage in panel (b), defined as the log equity issuance scaled by the firm's tangible capital stock. Both panels are estimated on the same sample of firms. Standard errors are double clustered by firm and quarter. "Q1" in the figure denotes the smallest 10% of firms, and "Q10" denotes the largest 10% of firms. 90% confidence intervals are reported.

The estimates for β_q are reported in Figure 5 in blue, along with 90% confidence intervals. Firms are ordered from smallest to largest, with "Q1" denoting the smallest 10% of firms and "Q10" denoting the largest 10% of firms. We overlay the estimated coefficients with the average level of media coverage from Figure 2a, reported in red on the right-hand axis.

Panel (a) shows that the smallest 10% of firms are the most responsive to media coverage. A one standard deviation higher media coverage is associated with a 36% higher investment in the year after the coverage. However, these small firms receive close to zero coverage from news outlets. In contrast, the largest 10% of firms receive substantial news coverage, but their investment does not seem to vary with it.

Panel (b) finds a similar pattern for equity issuance. Among firms with equity issuance one year from the news coverage, the smallest firms issue the most equity after having received higher news coverage. For larger firms, equity issuance does not vary significantly with coverage.

The joint distributions in Figure 5 reveal that the correlation between a firm's news coverage and its responses to this coverage is negative, measured at -0.14 for investment

and -0.04 for equity issuance. The firms that receive the most coverage are those that respond the least to it, which, as equation (1) shows, will reduce the aggregate effect of news coverage on firm investment. In fact, these correlations likely understate this effect, since they measure the linear relationship between variables, and we quantify the full effect of the nonlinear relationship between coverage and firm responsiveness in Section 4.

A potential explanation for our findings is that because large firms are *granular* and have disproportionately large effects on the macroeconomy (Gabaix, 2011), media outlets devote disproportional attention to them. Appendix A.6 examines this interpretation by excluding the largest firms in our sample, whose news may be viewed as particularly informative by the media. Our empirical results remain both qualitatively and quantitatively unchanged after excluding the largest 1% of firms, suggesting that the media’s preference for large firms is not driven by firm granularity.

The evidence in this section suggests that news coverage has positive effects on firms’ financing and investment. However, the large firms that news outlets focus on are the least responsive to this coverage. In the next section, we incorporate these features in a macro-finance model with a media sector to quantify how the distribution of curated media reporting shapes corporate finance and firms’ life cycles.

3. A Model of Corporate News Reporting

The model has two connected blocks. The first is a heterogeneous-firm block in which firms finance investment using internal funds and external equity, which is priced by investors under asymmetric information. The second is a media-reporting block in which media outlets choose which firms to report. Together, these allow us to study the aggregate consequences of media report allocation. Appendix C develops a simplified two-type model that works through the key intuitions underlying the model’s mechanisms.

3.1. Setup

Time is discrete and there is no aggregate uncertainty.

3.1.1. Equity issuance under asymmetric information

Firms There is a continuum of firms, indexed by $j \in [0, 1]$, that are heterogeneous in their capital quantity k , productivity z , and “capital quality” a . Capital quantity and productivity are public information, while capital quality is privately observed by the firm.

At the beginning of each period, firm j inherits capital k_{jt} and observes its idiosyncratic productivity z_{jt} , which evolves according to

$$\log z_{jt} = \rho_z \log z_{jt-1} + \epsilon_{jt}^z, \quad \epsilon_{jt}^z \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_z^2). \quad (7)$$

The firm then receives an i.i.d. exit shock $\epsilon_{jt}^{\text{exit}} \sim \text{Bernoulli}(\xi)$. Exiting firms liquidate their assets and are replaced by an equal mass of entrants drawn from $\mathcal{F}^{\text{entrant}}(k, z)$. Firms that remain produce using

$$y_{jt} = Z z_{jt} k_{jt}, \quad (8)$$

where Z denotes aggregate productivity.

After production, the firm receives an i.i.d. capital-quality shock $a_{jt} \stackrel{i.i.d.}{\sim} \mathcal{G}$ to its assets in place, where $\mathcal{G}$ is a discrete distribution with support $\mathcal{A} = \{a_1, \dots, a_n\}$, $a_1 < \dots < a_n$.¹⁶ Because the shock arrives after production and is independent over time, investors cannot infer it from the firm’s previously observed information.

After observing a_{jt} , the firm chooses how much external equity to raise to maximize the value retained by existing shareholders. Taking the firm’s media-reporting status m as given, they solve

$$V_t(k, z, a, m) = \max_{e \geq 0} \frac{P_t(k, z, a, m, e)}{P_t(k, z, a, m, e) + e} W_t(k, z, a, e), \quad (9)$$

where $m = 1$ if the firm is reported and $m = 0$ otherwise. The stock price $P_t(k, z, a, m, e)$ determines the dilution required to raise e . Normalizing existing shares to one, the firm issues e/P_t new shares, so existing shareholders retain the fraction $P_t/(P_t + e)$ of the post-issuance

¹⁶We parameterize the discrete distribution $\mathcal{G}$ as a discretized mean-one lognormal distribution, with $\log a \sim \mathcal{N}(-\sigma_a^2/2, \sigma_a^2)$, where σ_a governs the dispersion of capital quality. We use $n = 20$ support points, evenly spaced in capital-quality levels over the interval $[a_{\min}, a_{\max}]$, where $a_{\min} = \exp(-\sigma_a^2/2 - 3\sigma_a)$ and $a_{\max} = \exp(-\sigma_a^2/2 + 3\sigma_a)$. Each support point receives the probability mass of its surrounding interval. See Appendix D.3 for details.

value $W_t(k, z, a, e)$. The post-issuance value is determined by the firms' choice of dividends and investment, as described below:

$$W_t(k, z, a, e) = \max_{div \geq 0, x \geq 0} div + \beta \mathbb{E}_t [\xi k' + (1 - \xi) V_{t+1}(k', z', a', \mathbf{m}_{t+1}(k', z', a', \kappa')) \mid z] \quad (10)$$

$$\text{s.t.} \quad x = Zzk + e - \mathbb{1}_{e>0}\phi^e - div, \quad (11)$$

$$k' = (1 - \delta)ak + x^\theta. \quad (12)$$

Here, $\mathbf{m}_{t+1}$ is the aggregate reporting policy determined in the selective media-reporting block. Capital quality determines how much of the firm's assets in place carries into future capital (as in e.g., [Bigio, 2015](#); [Gertler et al., 2019](#)).¹⁷ We denote the resulting equity-issuance, dividend, and investment policies by $\mathbf{e}_t(k, z, a, m)$, $\mathbf{div}_t(k, z, a, m)$, and $\mathbf{x}_t(k, z, a, m)$.

Investors There is a continuum of competitive, risk-neutral retail investors. Investors observe each firm's capital k , productivity z , reporting status m , and equity issuance e , but they do not observe capital quality a unless it is reported. For a reported firm, investors price equity using the firm's true capital quality. For an unreported firm, they form a posterior belief $\mathcal{B}_t(a \mid k, z, e)$ from its issuance choice. Appendix [B.3](#) specifies this belief system, including off-path beliefs.

Because investors are competitive, the price of any positive issuance is such that they break even in expectation. Equivalently, the stock price equals the expected post-issuance firm value net of the funds raised:

$$P_t(k, z, a, m, e) = \begin{cases} W_t(k, z, a, e) - e, & m = 1, \\ \int W_t(k, z, \tilde{a}, e) \mathcal{B}_t(\tilde{a} \mid k, z, e) d\tilde{a} - e, & m = 0. \end{cases} \quad (13)$$

¹⁷Because private information concerns the firm's existing capital, equity issuance is a signal of low quality. If private information instead concerned new investment opportunities, firms with better opportunities would have the strongest incentive to issue equity. As noted by [Myers and Majluf \(1984\)](#), equity issuance would then be associated on average with stock-price increases, contrary to empirical evidence (e.g., [Asquith and Mullins, 1986](#); [Korajczyk et al., 1991](#); [Choe et al., 1993](#)).

The resulting stock-market value is

$$\text{MV}_t(k, z, a, m) = \begin{cases} P_t(k, z, a, m, \mathbf{e}_t(k, z, a, m)), & \mathbf{e}_t(k, z, a, m) > 0, \\ \frac{\int V_t(k, z, \tilde{a}, m) \mathbb{1}_{\mathbf{e}_t(k, z, \tilde{a}, m)=0} \mathcal{G}(\tilde{a}) d\tilde{a}}{\int \mathbb{1}_{\mathbf{e}_t(k, z, \tilde{a}, m)=0} \mathcal{G}(\tilde{a}) d\tilde{a}}, & \mathbf{e}_t(k, z, a, m) = 0. \end{cases} \quad (14)$$

How asymmetric information distorts firms' investment To see the intuition, suppose for a moment that $n = 2$, so there are only two capital-quality types, $a_L < a_H$. If investors had full information about capital quality, the two types would choose their full-information issuance levels, e_L^* and e_H^* , respectively, where $e_s^* \equiv \arg \max_{e \geq 0} \{W(k, z, a_s, e) - e\}$ for $s \in \{L, H\}$.¹⁸

However, if investors do not observe capital quality, the low-quality type a_L can mimic the high-quality type a_H 's issuance to obtain more favorable pricing. To deter such mimicry, the high-quality type cannot issue as much as it would like. Its issuance is capped at $\bar{e}_H$, the largest issuance that prevents the low-quality type from mimicking it:

$$W(k, z, a_L, e_L) - e_L = W(k, z, a_L, \bar{e}_H) - \bar{e}_H \frac{W(k, z, a_L, \bar{e}_H)}{W(k, z, a_H, \bar{e}_H)}. \quad (15)$$

The equality makes the low-quality type indifferent between the payoff from its own issuance and the payoff from mimicking the high-quality type at issuance $\bar{e}_H$. Appendix B.3 generalizes to multiple types and provides the full characterization of the equity-market equilibrium.¹⁹

The high-quality type's equilibrium issuance is therefore $e_H = \min\{e_H^*, \bar{e}_H\}$. When the non-mimicking constraint binds—which is always the case for positive issuance in our calibrated model—asymmetric information depresses the high-quality firm's issuance and

¹⁸To be precise, this is a one-shot full-information benchmark: current equity is priced using the firm's true capital quality, while its continuation value under asymmetric information is held fixed.

¹⁹The relationship between the low- and high-quality types' issuance reflects the general recursive structure of the equilibrium allocation with multiple types. For ordered types $a_1 < \dots < a_N$, the issuance of type a_i is bounded by the non-mimicking constraint of the adjacent lower-quality type a_{i-1} , so the separating allocation is solved recursively from the lowest- to the highest-quality type. It can also be shown that the equity-market equilibrium must be separating for types with positive issuance; pooling at a positive issuance level cannot be sustained. See Guo et al. (2025) and Appendix B.3 for the full characterization.

investment relative to the full-information allocation.²⁰

How media reporting affects firms' investment When the H-type firm is reported by the media, its capital quality becomes observable to investors, who then price its shares according to the firm's true type. The firm can therefore issue e_H^* rather than e_H , allowing it to increase investment toward the full-information level. Because media reporting alleviates the financing distortion only for the firms it covers, its aggregate effect depends on how limited coverage is allocated across firms. We therefore turn to the model's second block: selective media reporting.

3.1.2. Selective media reporting

Media Outlets A continuum of media outlets, indexed by $i \in [0, 1]$, have full information on firm fundamentals, including capital qualities a_{jt} . Each outlet is owned by a corresponding forecaster, who reads the news in that outlet but not in other outlets. An outlet selects firms to report on to maximize its forecaster's expected utility, specified below.²¹

Outlets face constraints, such as physical newspaper space or limited reader attention, so can report on only a fraction $r \in (0, 1)$ of firms each period:

$$\int_0^1 m_{ijt}^o dj = r, \quad (16)$$

where $m_{ijt}^o \in \{0, 1\}$ indicates whether outlet i reports firm j 's capital quality in period t . Outlet i therefore solves

$$\max_{m_{ijt}^o} \quad \mathbb{E} \int_0^1 U_{it}(\mathcal{I}_{it}^{\text{news}}) dj - \int_0^1 \kappa_{jt} m_{ijt}^o dj \quad (17)$$

$$\text{s.t.} \quad \mathcal{I}_{it}^{\text{news}} = \{a_{jt} : m_{ijt}^o = 1\}, \quad (18)$$

$$r = \int_0^1 m_{ijt}^o dj, \quad (19)$$

²⁰Because no lower-quality type can mimic a_L , its issuance is not constrained by a no-mimicking condition. Thus, in the separating equilibrium, the low-quality type issues at its full-information level, $e_L = e_L^*$. Appendix C develops a simplified two-type model that works through all of the key intuitions underlying this discussion.

²¹See Armona et al. (2026) for another model with this feature and a discussion of how such "direct maximization" incentives may arise.

where U_{it} is forecaster i 's utility and $\kappa_{jt} \sim \mathcal{H}(\kappa)$ is a firm- and period-specific reporting cost independent of firm fundamentals.²²

Investors observe the information published by all outlets.²³ Let $\mathcal{I}_t^{\text{news}} = \{a_{jt} : m_{jt} = 1\}$ denote total media information, where

$$m_{jt} = \begin{cases} 0, & m_{ijt}^o = 0 \text{ for all } i, \\ 1, & \text{otherwise.} \end{cases} \quad (20)$$

Thus, if at least one outlet reports on firm j , investors observe a_{jt} . We summarize aggregate reporting by $\mathbf{m}_t(k_{jt}, z_{jt}, a_{jt}, \kappa_{jt}) = m_{jt}$.

Forecasters Forecasters' utility depends on the forecast errors they make about firm market values when using the information provided by their media outlet. This captures the idea that outlets value a reputation for providing timely and relevant information to readers.²⁴

Formally, forecaster i uses k_{jt} , z_{jt} , and $\mathcal{I}_{it}^{\text{news}}$ to predict firm market values before equity markets open, and thus before equity issuance e_{jt} is realized. Forecasters observe other outlets' reporting decisions m_{ijt}^o , and hence aggregate reporting m_{jt} , but not the contents of those reports. They therefore observe a_{jt} only when their own outlet reports it.

As in the literature on forecaster incentives (reviewed by [Marinovic et al., 2013](#)), forecasters value accuracy relative to their peers:

$$U_{it}(\mathcal{I}_{it}^{\text{news}}) \equiv - \int_0^1 [\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) - \overline{\text{FE}}_{-it}(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{-it}^{\text{news}})] dj, \quad (21)$$

where forecaster i 's squared forecast error, based on their best prediction of market value given their information $\text{Pred}_t(k_{jt}, z_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}})$, is

$$\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) \equiv [\text{Pred}_t(k_{jt}, z_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt})]^2, \quad (22)$$

²²These can be interpreted as cognitive or effort costs, similar to information-processing costs in the rational-inattention literature (surveyed by [Maćkowiak et al., 2023](#)). They are distinct from the reader-attention constraint that motivates (16). The reporting-cost shock delivers a continuous reporting-probability function but is not essential to the mechanism.

²³Because market prices contain no noise in the baseline model (unlike e.g., [Grossman and Stiglitz, 1980](#)), prices aggregate the information used by informed investors.

²⁴The Financial Times, for example, used the slogan “No FT, no comment” from 1982–2006 and later “Make sense of it all.” Both suggest the paper provides information relevant for understanding market news.

and the average squared forecast error of other forecasters is

$$\overline{\text{FE}}_{-it}(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{-it}^{\text{news}}) \equiv \int_{i' \neq i} [\text{Pred}_t(k_{jt}, z_{jt}, m_{jt}, \mathcal{I}_{i',t}^{\text{news}}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt})]^2 di'. \quad (23)$$

Since a forecaster's prediction does not affect realized market values or others' predictions, the optimal forecast $\text{Pred}_t(k_{jt}, z_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}})$ is the conditional expectation of the firm's market value.

3.2. Equilibrium

The equilibrium consists of paths for the firm distribution $\mathcal{F}_t(k, z)$, aggregate media reporting $\mathbf{m}_t(k, z, a, \kappa)$, firm value functions $V_t(k, z, a, m)$, policy functions for equity issuance $\mathbf{e}_t(k, z, a, m)$, dividends $\mathbf{div}_t(k, z, a, m)$, and investment $\mathbf{x}_t(k, z, a, m)$, investor beliefs $\mathcal{B}_t(a | k, z, e)$, equity prices $P_t(k, z, a, m, e)$, and stock-market values $\text{MV}_t(k, z, a, m)$ such that:

1. given the firm distribution $\mathcal{F}_t(k, z)$ and stock-market values $\text{MV}_t(k, z, a, m)$, media outlets choose $\{m_{ijt}^o\}$, which determine aggregate reporting $\mathbf{m}_t(k, z, a, \kappa)$;
2. given the equity-price function $P_t(k, z, a, m, e)$, firms choose equity issuance $\mathbf{e}_t(k, z, a, m)$, dividends $\mathbf{div}_t(k, z, a, m)$, and investment $\mathbf{x}_t(k, z, a, m)$ optimally;
3. given $\mathbf{m}_t(k, z, a, \kappa)$ and $\mathbf{e}_t(k, z, a, m)$, investors form posterior beliefs $\mathcal{B}_t(a | k, z, e)$ about the capital quality of unreported firms;
4. given the beliefs and firms' policies, equity prices satisfy the break-even condition (13);
5. given firm values and equity prices, stock-market values satisfy (14); and
6. the firm distribution evolves according to

$$\mathcal{F}_{t+1}(k', z') = \xi \mathcal{F}^{\text{entrant}}(k', z') + (1 - \xi) \int \Gamma^z(z' | z) \mathbb{1}\{\mathbf{k}'_t = k'\} \mathcal{F}_t(k, z, a) \mathcal{G}(a) dk dz da,$$

where $\Gamma^z(z' | z)$ is the transition probability for idiosyncratic productivity and $\mathbf{k}'_t(k, z, a, m) \equiv (1 - \delta)ak + \mathbf{x}_t(k, z, a, m)^\theta$.

The firm-side financing and investment allocation is characterized in Section 3.1.1; Appendix B.3 provides the supporting beliefs and prices. We now characterize the equilibrium selection of media reports.

Equilibrium selection of media reporting Because outlets and forecasters are ex-ante identical, we focus on symmetric equilibria in pure strategies.²⁵

Proposition 1. *There is a unique news-reporting policy that can be sustained in a symmetric equilibrium:*

$$m_{jt} = \mathbb{1} (N_t(k_{jt}, z_{jt}, a_{jt}) - \kappa_{jt} \geq N_t^*), \quad (24)$$

where newsworthiness $N_t(k_{jt}, z_{jt}, a_{jt})$ is defined as

$$N_t(k_{jt}, z_{jt}, a_{jt}) = \mathbb{V} [MV_t(k_{jt}, z_{jt}, a_{jt}, 1) \mid k_{jt}, z_{jt}], \quad (25)$$

and the threshold N_t^* is determined by the space constraint (16).

Proof. Appendix B.1. □

Newsworthiness is therefore defined by the uncertainty that exists about a firm's reported market value, conditional on its public characteristics. The equilibrium is such that outlets report the firms with the largest newsworthiness net of reporting costs, up to the point that the space constraint binds.

This characterization explains why larger firms receive more coverage. Capital quality operates on assets in place in (12); hence, the same uncertainty about a_{jt} creates more uncertainty about future capital, and therefore market value, when current capital k_{jt} is high. All else equal, larger firms are therefore more newsworthy and more likely to be reported, as in the data.

Integrating the threshold rule over reporting costs gives the probability that firm j is

²⁵Because forecasters are identical, the motives for media specialization studied in Nimark and Pitschner (2019) and Perego and Yuksel (2022) are absent. Mixed-strategy equilibria may also exist. We focus on pure strategies to make the economic mechanism transparent, as does Armona et al. (2026).

reported:

$$R_t(k_{jt}, z_{jt}, a_{jt}) = \Pr(\kappa_{jt} \leq N_t(k_{jt}, z_{jt}, a_{jt}) - N_t^*) \quad (26)$$

$$= \int_0^{N_t(k_{jt}, z_{jt}, a_{jt}) - N_t^*} \mathcal{H}(\kappa) d\kappa. \quad (27)$$

Newsworthiness determines the ranking of firms, reporting capacity determines the threshold, and the distribution of reporting costs smooths this ranking into reporting probabilities. Because (25) depends only on the public states (k_{jt}, z_{jt}) , we write newsworthiness and reporting probability as $N_t(k, z)$ and $R_t(k, z)$. Non-reporting therefore conveys no additional information about realized capital quality.

We choose a distribution for κ_{jt} , detailed in Appendix B.4, such that

$$R_t(k, z) = \lambda_p + (1 - \lambda_p) \left[1 - \exp \left(- \left(\frac{N_t(k, z)}{\lambda_\alpha} \right)^{\lambda_\xi} \right) \right]. \quad (28)$$

The parameters $(\lambda_p, \lambda_\alpha, \lambda_\xi)$ control the level, scale, and shape of the reporting-probability function. This function determines how scarce coverage is allocated across firms and is a key object disciplined by the quantitative analysis.

3.3. Model assumptions and robustness

We close the model section by discussing the assumptions underlying the media-reporting block and summarizing the existing robustness exercises.

First, we assume that the media objective is to provide readers with better information, consistent with the notion of newsworthiness in Armona et al. (2026) and the contest model of Ottaviani and Sørensen (2006). Appendix E considers an alternative microfoundation in which outlets maximize newspaper sales to investors. Noise traders (as in Grossman and Stiglitz, 1980) prevent prices from perfectly aggregating information and generate non-degenerate investor demand for news. This investor-led model produces the same equilibrium reporting function. Because solving it requires abstracting from much of the firm side, we retain the forecaster model in the quantitative analysis.

Second, outlet i takes other outlets' reporting decisions $m_{-i,j,t}^o$ as given. Although its objective depends on other outlets through relative forecast errors and realized market values,

each individual outlet is atomistic because there is a continuum of outlets. This assumption removes strategic reporting incentives, as is common in media models; related approaches use atomistic outlets (Eliaz and Spiegler, 2024), monopoly media supply (Martineau and Mondria, 2022), or restrictions on media demand (Gentzkow and Shapiro, 2010).

Third, the outlet maximizes expected U_{it} before its forecaster observes information and makes a prediction, so the baseline reporting objective is not conditional on realized a_{jt} . The outlet can therefore be interpreted as directly valuing low relative forecast errors. We retain a separate forecaster because alternative timing makes the distinction relevant. Appendix B.2 considers, among other cases, outlets that maximize realized utility after observing all firm states. This changes the exact reporting rule but preserves its key qualitative properties.

Finally, Appendix F relaxes the assumption that a report reveals a_{jt} exactly. We allow noise in outlets' observations of capital quality and in forecasters' processing of media signals. Processing noise scales every firm's newsworthiness by a common factor and is absorbed by the scale parameter of the reporting-probability function, leaving the allocation of coverage unchanged. Noisy outlet information weakens the firm-level effect of a report and reduces the aggregate role of media. Nevertheless, the importance of report allocation survives across the calibrated range of outlet precision: reallocating coverage continues to amplify the aggregate effect of media by a factor of more than twenty.

4. Quantitative Analysis

This section quantifies the aggregate economic relevance of corporate news reporting. We first present our calibration of the model, paying particular attention to how the data discipline the media-reporting parameters. We then use the calibrated model to examine how the distribution of media coverage shapes the macroeconomic effects of the media. Because the model has no aggregate uncertainty, we focus on its steady state.

4.1. Calibration

We calibrate the quarterly model to a sample of publicly traded firms from 1990 to 2021. We set the discount factor to $\beta = 0.99$, corresponding to a 4% annual real interest rate; the exogenous exit probability to $\xi = \frac{7.7\%}{4}$; and aggregate productivity to $Z = 2.5\%$, with the

Table 3: Internally calibrated parameters

Parameter	Description	Value
<i>Cash flow dynamics</i>		
ρ_z	Idiosyncratic productivity, persistence	0.95
σ_z	—, innovation standard deviation	0.11
<i>Investment technology</i>		
δ	Depreciation rate	3.30%
θ	Return to scale of investment technology	0.81
<i>Life-cycle dynamics</i>		
$\mu_{\log z}^{\text{entrant}}$	Entrants, average log productivity	-0.175
$\mu_{\log k}^{\text{entrant}}$	—, average log size	-1.761
<i>Information and financial friction</i>		
σ_a	Dispersion of capital-quality shock	0.18
ϕ^e	Fixed cost to issuing equity	0.015%
<i>Selective media reporting (reporting probability function)</i>		
$(\lambda_\alpha, \lambda_p)$	Location parameters (scaled by 1000)	(4.7, 2.0)
λ_ξ	Curvature	1.80

Notes: This table reports the internally calibrated parameters. The fixed cost of issuing equity, ϕ^e , is normalized by the average net profit of the firm population to facilitate interpretation.

latter two parameters matching the average exit rate of 7.7% and the annual cash-flow rate of 10% in the Compustat sample, respectively. We then calibrate the parameters reported in Table 3 to match the empirical moments in Table 4. Appendix D.1 details the construction of these empirical moments.

The calibration is overidentified: we jointly choose eleven internally calibrated parameters to match fourteen target moments. Because the model is nonlinear, the mapping between parameters and moments is many-to-many rather than one-to-one. Nevertheless, their economic roles provide a natural way to organize both the parameters and the moments into five groups. The first three groups of moments capture the dynamics of idiosyncratic productivity, the level and dispersion of investment, and differences in growth and productivity between young and mature firms. These moments are primarily informative about the parameters governing cash-flow dynamics, investment technology, and firms' life-cycle dynamics. These mappings are standard and intuitive in the firm-dynamics literature, so we focus our discussion on the remaining two groups.

The remaining parameter groups govern the two financing frictions central to our mech-

Table 4: Model fit and target moments

Moment	Data	Model
<i>Cash flow dynamics (annual)</i>		
Idiosyncratic TFP, persistence	0.70	0.69
–, standard deviation	0.38	0.37
<i>Investment (annual, %)</i>		
Investment rate, mean	6.00	6.12
–, standard deviation	5.53	5.40
<i>Life-cycle dynamics (%)</i>		
Growth rate, young minus mature firms	10.55	10.49
Idiosyncratic TFP, –	-7.52	-7.50
<i>Financing (%)</i>		
Fraction of firms issuing equity, annual mean	17.87	17.97
Issuance-fee ratio, mean	2.17	2.21
Selling-concession ratio when issuing, mean	2.97	3.02
<i>Media-coverage distribution (share of reported firms, %)</i>		
Top 20% of market value	87.22	86.18
60–80% of market value	8.45	10.74
40–60% of market value	2.54	2.18
20–40% of market value	1.30	0.56
Bottom 20% of market value	0.50	0.35

Notes: This table reports targeted moments and their model counterparts. All moments are measured using firms in the Compustat sample from 1990 to 2021, and the definitions and measurements of the target moments are detailed in Appendix D.1. The investment rate refers to firms’ investment normalized by their capital stock. The issuance-fee ratio is defined as the fixed cost paid by issuing firms, normalized by their issuance quantity. The selling concession when issuing equity is measured as the log difference between a firm’s stock price before and after revealing its equity issuance decision. Both moments related to equity issuance costs are from Lee and Masulis (2009). The media-coverage distribution for each market-value quintile is defined as the share of reported firms in that quintile.

anism: the explicit cost of issuing equity and the implicit cost arising from asymmetric information. The dispersion of capital-quality shocks determines the severity of the information friction, while media coverage determines a firm’s exposure to it. The reporting-probability function governs both the overall coverage rate and the allocation of coverage across firms. Together, the financial-friction and media-reporting parameters determine the effective financing constraints faced by firms.

We use the firm financing and media report moments to discipline different dimensions of these linked mechanisms. The issuance-fee ratio primarily disciplines the explicit issuance cost, ϕ^e , while the selling-concession ratio is informative about the dispersion of capital-

quality shocks, σ_a , and hence the severity of asymmetric information among firms that issue equity.²⁶ The fraction of firms issuing equity captures the incidence of issuance and is informative about the average severity of financing constraints. Conditional on the two direct cost measures, this financing outcome also helps discipline the overall coverage rate because media coverage relaxes information frictions and thereby affects the incidence of issuance. Finally, the distribution of reported firms across market-value groups in Table 4 disciplines the cross-sectional selectivity of media coverage.²⁷

Overall, Table 4 shows that the calibration matches the fourteen target moments reasonably well. Beyond the targeted moments, we assess the model along two untargeted dimensions related to firms' financing and investment. First, the model predicts that equity-issuing firms finance 16.6% of their investment externally, close to the 15.3% in the data. Second, to assess the model's implications for the relationship between media reports and investment, we apply the local-projection specification with the binary media-coverage indicator from Appendix Figure A.4 to a panel simulated from the model. The model generates a peak cumulative investment response to media coverage of 7.7%, compared with 6.1% in the data (90% confidence interval: [2.2%, 10.1%]).

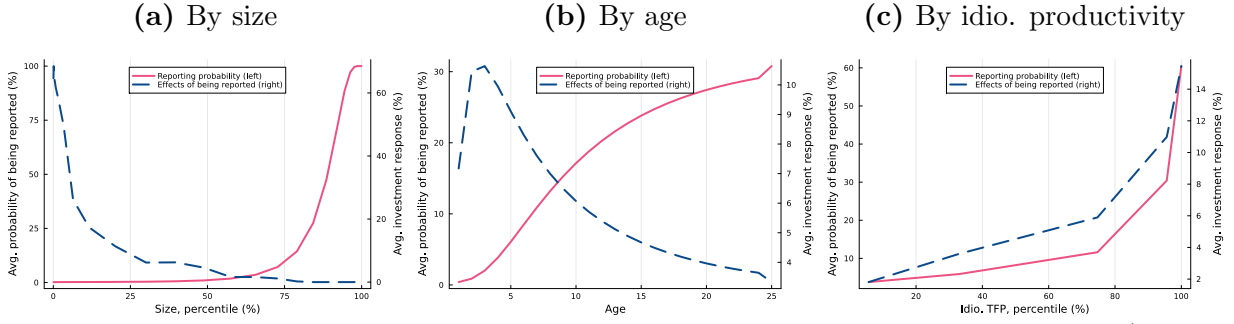
4.2. Distribution of media coverage and the effects of media reports

Figure 6 presents the probability of media coverage under our calibration (red solid lines) along three dimensions: size, age, and idiosyncratic productivity. Panels (a) and (b) indicate that larger and older firms are more likely to be reported, with the concentration being more pronounced along the size dimension. This pattern is consistent with the stylized facts documented in Figure A.2. The model also predicts a higher likelihood of media coverage for firms with greater idiosyncratic productivity (panel c). These qualitative relationships follow from equation (25). Newsworthiness scales with firm size and productivity because both are positively correlated with market value. Since firms, on average, grow in size and

²⁶The selling concession is the stock-price reduction that issuing firms offer investment banks in exchange for guaranteeing flotation. We measure it using the average stock-price decline following the announcement of an equity issuance. The average concession of 2.97% reported by Lee and Masulis (2009) lies within the 2–3% range of estimates in the literature. See Appendix D.1 for further details.

²⁷We use the distribution of reported firms rather than group-specific reporting frequencies because our data cover only three newspapers and therefore understate total media coverage. Appendix D.4.1 shows that the distribution of reported firms is invariant to the number of newspapers, which justifies using it as a target moment. We group firms by market value because it is observable and closely related to the model's newsworthiness measure.

Figure 6: Cross-sectional pattern of media reporting and its effects on firms’ investments



Notes: This figure plots the variation of the quarterly average probability of being reported (red solid lines) and the effects of being reported (blue dash lines) along size (capital stock), age, and idiosyncratic productivity. In this figure, the “effects of being reported” refers to the log-difference in a firm’s investment between when it is reported and when it is not. The last data point in Panel (b) represent firms aged ≥ 25 .

productivity over time, the probability of media coverage also rises with firm age.

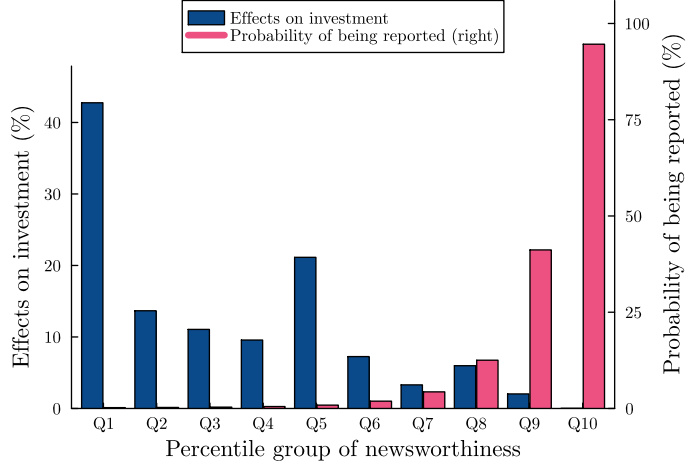
Alongside reporting probabilities, the blue dashed lines in Figure 6 plot the effect of reporting on investment across the same firm characteristics. For each firm state, we measure this effect as the log difference between investment when the firm is reported and when it is not. Because this comparison holds the firm’s state fixed, it isolates the causal effect of reporting from the endogenous selection of firms into coverage.

The effect of media coverage on investment decreases with firm size and age, but increases with productivity. This is because, as discussed in Section 3.1.1, media coverage can reduce information frictions related to equity issuance. The effects of media reports, therefore, depend on a firm’s reliance on external funding. Smaller and younger firms, with less internal financing, rely more heavily on external equity markets, making their investment more sensitive to media reports. Likewise, more productive firms have greater investment needs and therefore rely more heavily on external financing than less productive firms of similar size, making their investment more responsive to media coverage.

These patterns imply a misalignment between the distribution of media coverage and its impact on firm-level investment. To illustrate this misalignment more directly, Figure 7 plots the average investment responses to reporting alongside the probability of coverage across newsworthiness percentiles. Firms whose investment is most responsive to news reports are concentrated at the lower end of the newsworthiness distribution, receiving little coverage.²⁸

²⁸The investment response generally declines with newsworthiness, although the pattern is not monotonic across deciles. Firms in higher newsworthiness deciles tend to have both larger capital and higher idiosyncratic productivity, which have opposing implications for reporting responsiveness: Figure 6 shows that the effect on investment decreases with size but increases with productivity. Between Q4 and Q5, the effect of

Figure 7: Mis-allocation of media reporting



Notes: This figure reports the probability of being reported (red bars, right axis) and the effect of reporting on firm investment (blue bars). The effect on investment is the log difference in a firm’s investment between when it is reported and when it is not. Firms are sorted into deciles of newsworthiness, with “Q1” representing the 10% least newsworthy firms and “Q10” representing the 10% most newsworthy firms. For each decile, we report the average reporting probability and the percentage change in total investment when firms are reported rather than unreported.

In other words, the covariance term in the decomposition in equation (1) is strongly negative. This suggests that reallocating media coverage could generate a larger real effect on the economy. Building on this observation, the next subsection quantifies the role of allocation in determining the aggregate relevance of corporate news.

4.3. Aggregate relevance of media-report allocation

We now quantify how the allocation of media coverage affects the aggregate cost of asymmetric information. We measure this cost by comparing two steady states: a symmetric-information economy, in which investors observe firms’ capital quality, and an asymmetric-information economy without media reports.²⁹ As shown in Table 5, asymmetric information reduces aggregate investment by 6.7%, the capital stock by 4.7%, output by 5.3%, and aggregate firm value by 4.0%. We use the 4.0% loss in aggregate firm value as our welfare-relevant measure of the cost of asymmetric information.³⁰

a relatively large increase in average productivity appears to outweigh the lower responsiveness associated with the concurrent increase in capital. The empirical pattern in Figure 5 is weaker in comparison, likely due to attenuation: our coverage measure captures only a subset of total news coverage, and newsworthiness is unobserved empirically and proxied by firm size.

²⁹We keep all parameters unrelated to media reporting fixed at their baseline values when evaluating these two counterfactual steady states.

³⁰Appendix D.4 provides a detailed derivation of why aggregate firm value is a well-defined measure of welfare in the model.

Table 5: Aggregate losses from asymmetric information and the role of media reporting

Outcome	Loss relative to symmetric information (%)			Fraction of no-media loss eliminated (%)	
	No media	Baseline	Constrained- efficient	Baseline	Constrained- efficient
Investment	6.7	6.4	4.9×10^{-4}	4.8	99.99
Capital stock	4.7	4.5	2.6×10^{-4}	4.2	99.99
Output	5.3	5.0	4.5×10^{-4}	4.7	99.99
Firm value	4.0	3.8	6.8×10^{-4}	4.5	99.98

Notes: Aggregate loss is the percentage shortfall relative to the symmetric-information equilibrium. For each outcome, the fraction of no-media loss eliminated is the share of the no-media loss removed by media reporting. The baseline and constrained-efficient allocations have the same reporting capacity and differ only in how reports are allocated across firms.

We next evaluate how much of this cost is eliminated by the baseline media allocation, which is calibrated to match the distribution of media coverage in the data. Media outlets report on 15.7% of firms in the baseline economy. Since 18.2% of firms would increase investment and firm value if reported, the baseline reporting capacity could, in principle, cover 86% of the firms that would benefit from coverage. Despite this, the baseline allocation of media generates only small aggregate gains. Table 5 shows that, relative to an economy without media, it raises aggregate output by just 0.3% and aggregate firm value by 0.2%. That is, media eliminates only 4.5% of the firm-value cost of asymmetric information, with similarly modest effects of 4.2–4.8% across the other aggregate outcomes.

Our key hypothesis for why these effects are small concerns the way media is allocated across firms: the firms that are covered are not those that need it most. To isolate this force, we construct a constrained-efficient allocation that holds reporting capacity fixed at its baseline level. Rather than allocating reports via newsworthiness, the planner ranks each firm state, (k, z, a) , by the increase in firm value generated by reporting, $\Delta V(k, z, a) \equiv V(k, z, a, 1) - V(k, z, a, 0)$, and reports the states with the largest gains. This reallocation eliminates almost all aggregate loss from asymmetric information.

The size of this effect is striking, because the planner has the same reporting capacity as the outlets in the baseline economy, which in that case was not enough to cover all constrained firms. The reason for the large effect is that, beyond the direct ‘treatment effects’ of reporting a given firm, reallocating media coverage also changes the *distribution* of firm states. Specifically, the constrained-efficient allocation targets financially constrained firms,

who tend to be smaller. Relaxing their constraints allows them to grow faster, reshaping the steady-state firm distribution so that fewer firms remain financially constrained. As a result, the same reporting capacity can cover all remaining constrained firms and eliminate almost all of the loss from asymmetric information.³¹

Compared with 4.5% under the baseline allocation, the aggregate role of media is approximately 22 times as large when reports are efficiently allocated. Because the reporting capacity is unchanged, this difference is entirely attributable to the *allocation* of reports across firms. This result highlights the paper’s central quantitative message: conditional on the amount of media coverage, the allocation of coverage across firms is the dominant determinant of its aggregate effects.

5. Conclusion

News outlets provide valuable information to their readers, but constraints on space and journalistic resources mean they have to make judgments on which firms are most newsworthy. We find that these judgments overwhelmingly favor reporting on the largest firms in the economy—firms that benefit the least from media coverage. This selectivity has important effects on firm dynamics and aggregate investment. Reallocating media reports can substantially increase media’s role in alleviating the economic cost of information frictions.

Both cross-time and cross-sectional variations in the media’s production of firm news are endogenously determined, and the way the media reports firms’ news has important implications for the macroeconomy. Our findings suggest that large firms that dominate their markets, or “superstar” firms (e.g., [Covarrubias et al., 2020](#); [De Loecker et al., 2020](#); [Autor et al., 2020](#)), do so not only with better information technology ([Kwon et al., 2024](#)) but also with superior media coverage. The interaction between information provision and corporate finance highlights an intangible form of market power enjoyed by these large firms, which warrants closer investigation in future studies.

³¹The near-complete elimination is a quantitative result of our calibrated model, rather than a theoretical implication of the framework. Figure D.1 in Appendix D.4.2 shows how the share of the loss from information frictions that can be eliminated under constrained-efficient allocation varies with reporting capacity.

References

- Alves, Felipe, Greg Kaplan, Benjamin Moll, and Giovanni L. Violante, “A Further Look at the Propagation of Monetary Policy Shocks in HANK,” *Journal of Money, Credit and Banking*, 2020, 52 (S2), 521–559.
- AMF, “The stock market attracted a record number of retail investors in 2025,” 2026.
- Angeletos, George-Marios and Karthik A. Sastry, “Inattentive Economies,” *Journal of Political Economy*, July 2025, 133 (7), 2265–2319.
- Araci, Dogu, “FinBERT: Financial sentiment analysis with pre-trained language models,” *arXiv preprint arXiv:1908.10063*, 2019.
- Armona, Luis, Matthew Gentzkow, Emir Kamenica, and Jesse M. Shapiro, “What Is Newsworthy? Theory and Evidence,” *American Economic Review: Insights*, September 2026, 8 (3), 354–73.
- Asquith, Paul and David W. Mullins, “Equity Issues and Offering Dilution,” *Journal of Financial Economics*, January 1986, 15 (1), 61–89.
- Asriyan, Vladimir, “Balance Sheet Channel with Information-Trading Frictions in Secondary Markets,” *The Review of Economic Studies*, January 2021, 88 (1), 44–90.
- Auclert, Adrien, “Monetary Policy and the Redistribution Channel,” *American Economic Review*, June 2019, 109 (6), 2333–67.
- , Matthew Rognlie, and Ludwig Straub, “The Intertemporal Keynesian Cross,” *Journal of Political Economy*, 2024, 132 (12), 4068–4121.
- Autor, David, David Dorn, Lawrence F Katz, Christina Patterson, and John Van Reenen, “The fall of the labor share and the rise of superstar firms,” *The Quarterly Journal of Economics*, 2020, 135 (2), 645–709.
- Banks, Jeffrey S. and Joel Sobel, “Equilibrium Selection in Signaling Games,” *Econometrica*, 1987, 55 (3), 647–661.
- Beaudry, Paul and Franck Portier, “News-Driven Business Cycles: Insights and Challenges,” *Journal of Economic Literature*, December 2014, 52 (4), 993–1074.
- Bigio, Saki, “Endogenous Liquidity and the Business Cycle,” *American Economic Review*, June 2015, 105 (6), 1883–1927.
- Brunnermeier, Markus K, Thomas M Eisenbach, and Yuliy Sannikov, “Macroeconomics with financial frictions: A survey,” *National Bureau of Economic Research*, 2012.
- Bui, Ha, Zhen Huo, Andrei A Levchenko, and Nitya Pandalai-Nayar, “Information Frictions and News Media in Global Value Chains,” *Manuscript*, 2022.
- Bybee, Leland, Bryan T Kelly, Asaf Manela, and Dacheng Xiu, “The Structure of Economic News,” *Manuscript*, 2020.
- Caramp, Nicolas, “Sowing the seeds of financial crises: Endogenous asset creation and adverse selection,” *Review of Economic Studies*, 2024, p. rdae092.
- Chahrour, Ryan, Kristoffer Nimark, and Stefan Pitschner, “Sectoral Media Focus and Aggregate Fluctuations,” *American Economic Review*, 2021, 111 (12), 3872–3922.
- Chan, Wesley S, “Stock price reaction to news and no-news: drift and reversal after headlines,” *Journal of financial economics*, 2003, 70 (2), 223–260.

- Chiang, Yu-Ting**, “Media Competition for Attention,” *Manuscript*, 2020.
- Cho, In-Koo and David M. Kreps**, “Signaling Games and Stable Equilibria,” *The Quarterly Journal of Economics*, 1987, *102* (2), 179–221.
- Choe, Hyuk, Ronald W. Masulis, and Vikram Nanda**, “Common stock offerings across the business cycle: Theory and evidence,” *Journal of Empirical Finance*, 1993, *1* (1), 3–31.
- Cloyne, James, Clodomiro Ferreira, Maren Froemel, and Paolo Surico**, “Monetary Policy, Corporate Finance, and Investment,” *Journal of the European Economic Association*, 03 2023, *21* (6), 2586–2634.
- Coibion, O, Y Gorodnichenko, and T Ropele**, “Inflation Expectations and Misallocation of Resources: Evidence from Italy,” *Manuscript*, 2023.
- Coibion, Olivier, Yuriy Gorodnichenko, and Tiziano Ropele**, “Inflation Expectations and Firm Decisions: New Causal Evidence,” *The Quarterly Journal of Economics*, 2020, *135* (1), 165–219.
- Cooley, Thomas F and Vincenzo Quadrini**, “Financial Markets and Firm Dynamics,” *American Economic Review*, 2001, *91* (5), 1286–1310.
- Covarrubias, Matias, Germán Gutiérrez, and Thomas Philippon**, “From Good to Bad Concentration? US Industries over the Past 30 Years,” *NBER Macroeconomics Annual*, 2020, *34* (1), 1–46.
- Cutler, David M, James M Poterba, and Lawrence H Summers**, *What Moves Stock Prices?*, Vol. 487, National Bureau of Economic Research Cambridge, Massachusetts, 1988.
- David, Joel M. and Venky Venkateswaran**, “The Sources of Capital Misallocation,” *American Economic Review*, July 2019, *109* (7), 2531–2567.
- , **Hugo A. Hopenhayn, and Venky Venkateswaran**, “Information, Misallocation, and Aggregate Productivity,” *The Quarterly Journal of Economics*, May 2016, *131* (2), 943–1005.
- Denti, Tommaso and Kristoffer Nimark**, “Attention Costs, Economies of Scale and Markets for Information,” *Manuscript*, 2022.
- Doms, Mark and Timothy Dunne**, “Capital adjustment patterns in manufacturing plants,” *Review of Economic Dynamics*, 1998, *1* (2), 409–429.
- Dougal, Casey, Joseph Engelberg, Diego García, and Christopher A. Parsons**, “Journalists and the Stock Market,” *The Review of Financial Studies*, March 2012, *25* (3), 639–679.
- Eisfeldt, Andrea L.**, “Endogenous Liquidity in Asset Markets,” *The Journal of Finance*, 2004, *59* (1), 1–30.
- Elder, Bryce**, “Tonight We’re Going to Stonk-Punt like It’s 2021,” *Financial Times*, February 2025.
- Eliaz, Kfir and Ran Spiegler**, “News Media as Suppliers of Narratives (and Information),” *Manuscript*, April 2024.
- Engelberg, Joseph E. and Christopher A. Parsons**, “The Causal Impact of Media in Financial Markets,” *The Journal of Finance*, 2011, *66* (1), 67–97.

- Farhi, Emmanuel and Xavier Gabaix**, “Optimal Taxation with Behavioral Agents,” *American Economic Review*, January 2020, *110* (1), 298–336.
- Gabaix, Xavier**, “The granular origins of aggregate fluctuations,” *Econometrica*, 2011, *79* (3), 733–772.
- , “A Sparsity-Based Model of Bounded Rationality,” *The Quarterly Journal of Economics*, November 2014, *129* (4), 1661–1710.
- , “A Behavioral New Keynesian Model,” *American Economic Review*, 2020, *110* (8), 2271–2327.
- Gentzkow, Matthew and Jesse M Shapiro**, “Competition and Truth in the Market for News,” *Journal of Economic perspectives*, 2008, *22* (2), 133–154.
- **and** – , “What Drives Media Slant? Evidence from US Daily Newspapers,” *Econometrica*, 2010, *78* (1), 35–71.
- Gertler, Mark and Simon Gilchrist**, “The Role of Credit Market Imperfections in the Monetary Transmission Mechanism: Arguments and Evidence,” *The Scandinavian Journal of Economics*, 1993, pp. 43–64.
- , **Nobuhiro Kiyotaki**, and **Andrea Prestipino**, “A Macroeconomic Model with Financial Panics,” *The Review of Economic Studies*, 05 2019, *87* (1), 240–288.
- Gorton, Gary and George Pennacchi**, “Financial Intermediaries and Liquidity Creation,” *The Journal of Finance*, 1990, *45* (1), 49–71.
- **and Guillermo Ordonez**, “Collateral Crises,” *American Economic Review*, 2014, *104* (2), 343–378.
- Greenwood, Robin, Toomas Laarits, and Jeffrey Wurgler**, “Stock Market Stimulus,” *The Review of Financial Studies*, October 2023, *36* (10), 4082–4112.
- Grossman, Sanford J. and Joseph E. Stiglitz**, “On the Impossibility of Informationally Efficient Markets,” *The American Economic Review*, 1980, *70* (3), 393–408.
- Guo, Xing, Pablo Ottonello, Thomas Winberry, and Toni Whited**, “Firm Heterogeneity and Adverse Selection in External Finance: Micro Evidence and Macro Implications,” jul 2025. Revise and Resubmit at the Review of Economic Studies.
- Hoberg, Gerard and Vojislav Maksimovic**, “Redefining financial constraints: A text-based analysis,” *The Review of Financial Studies*, 2015, *28* (5), 1312–1352.
- Hu, Allen**, “Financial News Production,” *Manuscript*, 2024.
- Huberman, Gur and Tomer Regev**, “Contagious Speculation and a Cure for Cancer: A Nonevent That Made Stock Prices Soar,” *The Journal of Finance*, 2001, *56* (1), 387–396.
- İmrohoroglu, Ayşe and Şelale Tüzel**, “Firm-Level Productivity, Risk, and Return,” *Management science*, 2014, *60* (8), 2073–2090.
- Jordà, Òscar**, “Estimation and inference of impulse responses by local projections,” *American economic review*, 2005, *95* (1), 161–182.
- Kahle, Kathleen M and René M Stulz**, “Is the US Public Corporation in Trouble?,” *Journal of Economic Perspectives*, 2017, *31* (3), 67–88.

- Korajczyk, Robert A., Deborah J. Lucas, and Robert L. McDonald**, “The Effect of Information Releases on the Pricing and Timing of Equity Issues,” *The Review of Financial Studies*, 1991, 4 (4), 685–708.
- Kumar, Alok and Charles M.c. Lee**, “Retail Investor Sentiment and Return Comovements,” *The Journal of Finance*, 2006, 61 (5), 2451–2486.
- Kurlat, Pablo**, “Lemons Markets and the Transmission of Aggregate Shocks,” *American Economic Review*, June 2013, 103 (4), 1463–1489.
- Kwon, Spencer Y, Yueran Ma, and Kaspar Zimmermann**, “100 Years of Rising Corporate Concentration,” *American Economic Review*, 2024, 114 (7), 2111–2140.
- Larsen, Vegard H., Leif Anders Thorsrud, and Julia Zhulanova**, “News-Driven Inflation Expectations and Information Rigidities,” *Journal of Monetary Economics*, 2021, 117, 507–520.
- Lee, Gemma and Ronald W. Masulis**, “Seasoned Equity Offerings: Quality of Accounting Information and Expected Flotation Costs,” *Journal of Financial Economics*, 2009, 92 (3), 443–469.
- Levenshtein, VI**, “Binary Codes Capable of Correcting Deletions, Insertions, and Reversals,” *Proceedings of the Soviet Physics Doklady*, 1966, 10 (8), 707–710.
- Loecker, Jan De, Jan Eeckhout, and Gabriel Unger**, “The Rise of Market Power and the Macroeconomic Implications,” *The Quarterly Journal of Economics*, 2020, 135 (2), 561–644.
- Loughran, Tim and Bill McDonald**, “When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks,” *The Journal of finance*, 2011, 66 (1), 35–65.
- Macaulay, Alistair and Wenting Song**, “Narrative-Driven Fluctuations in Sentiment: Evidence Linking Traditional and Social Media,” *Manuscript*, 2022.
- Maćkowiak, Bartosz, Filip Matějka, and Mirko Wiederholt**, “Rational Inattention: A Review,” *Journal of Economic Literature*, March 2023, 61 (1), 226–273.
- Marinovic, Iván, Marco Ottaviani, and Peter Sorensen**, “Forecasters’ Objectives and Strategies,” in Graham Elliott and Allan Timmermann, eds., *Handbook of Economic Forecasting*, Vol. 2, Elsevier, January 2013, pp. 690–720.
- Martineau, Charles and Jordi Mondria**, “News Selection and Asset Pricing Implications,” *Manuscript*, 2022.
- McKeon, Stephen B.**, “Employee Option Exercise and Equity Issuance Motives,” *Manuscript*, 2015.
- Myers, Stewart C. and Nicholas S. Majluf**, “Corporate Financing and Investment Decisions When Firms Have Information That Investors Do Not Have,” *Journal of Financial Economics*, June 1984, 13 (2), 187–221.
- Nieuwerburgh, Stijn Van and Laura Veldkamp**, “Information Acquisition and Under-Diversification,” *Review of Economic Studies*, April 2010, 77 (2), 779–805.
- Nimark, Kristoffer P**, “Man-Bites-Dog Business Cycles,” *American Economic Review*, 2014, 104 (8), 2320–67.
- **and Stefan Pitschner**, “News Media and Delegated Information Choice,” *Journal of Economic Theory*, 2019, 181, 160–196.

- Ottaviani, Marco and Peter Norman Sørensen**, “The Strategy of Professional Forecasting,” *Journal of Financial Economics*, August 2006, *81* (2), 441–466.
- Ottonello, Pablo and Thomas Winberry**, “Financial Heterogeneity and the Investment Channel of Monetary Policy,” *Econometrica*, 2020, *88* (6), 2473–2502.
- Perego, Jacopo and Sevgi Yuksel**, “Media Competition and Social Disagreement,” *Econometrica*, 2022, *90* (1), 223–265.
- Peress, Joel**, “The Media and the Diffusion of Information in Financial Markets: Evidence from Newspaper Strikes,” *the Journal of Finance*, 2014, *69* (5), 2007–2043.
- Sterk, Vincent, Petr Sedláček, and Benjamin Pugsley**, “The Nature of Firm Growth,” *American Economic Review*, February 2021, *111* (2), 547–79.
- Tetlock, Paul C.**, “Does Public Financial News Resolve Asymmetric Information?,” *The Review of Financial Studies*, September 2010, *23* (9), 3520–3557.
- Veldkamp, Laura**, *Information Choice in Macroeconomics and Finance*, Princeton University Press, 2023.
- Ye, Junjie, Xuanting Chen, Nuo Xu, Can Zu, Zekai Shao, Shichun Liu, Yuhan Cui, Zeyang Zhou, Chao Gong, Yang Shen et al.**, “A comprehensive capability analysis of gpt-3 and gpt-3.5 series models,” *arXiv preprint arXiv:2303.10420*, 2023.

ONLINE APPENDICES

A. Additional Details for the Empirical Analysis

A.1. Data construction

This section describes the firm-level variables used in the empirical analysis of the paper, based on quarterly Compustat data. The definition follows standard practices in the firm dynamics literature (e.g., [Kahle and Stulz, 2017](#); [Ottonello and Winberry, 2020](#)).

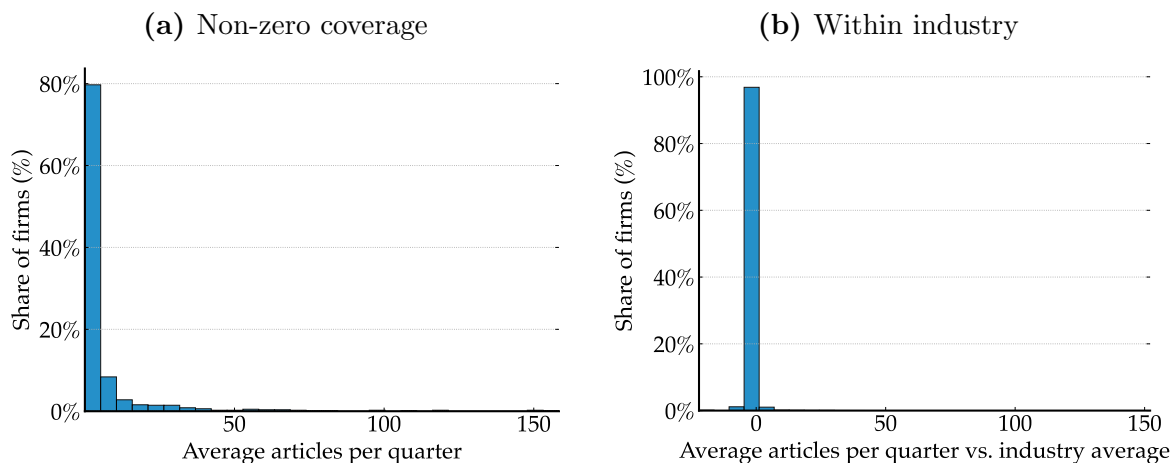
Variable definitions

1. *Size*: the log of total real assets (`atq`), deflated using the BLS implicit price deflator.
2. *Age*: the number of years since the CRSP listing.
3. *Leverage*: the sum of the debt in current liabilities and the long-term debt (`dlttq+dlc`) over the sum of total assets and the market valuation minus the common equity (`atq-ceqq+cshoq*prccq`)
4. *Investment*: defined as $\Delta \log k_{it}$, where k_{it} denotes the capital stock of firm i at the end of quarter t . Following [Ottonello and Winberry \(2020\)](#), for each firm, we set the first value of k_{it} to be gross plant, property, and equipment (`ppegtq`) in the first period in which this variable is reported in Compustat and the subsequent value of k_{it} to be the changes in the net value of the plant, property, and equipment (`ppentq`). If a firm has a missing observation for `ppentq` located between two periods with non-missing observations, we estimate its value using a linear interpolation with the values of `ppentq`; if two or more consecutive observations are missing, we do not do any imputation.
5. *Equity issuance*: defined as the sale of common and preferred stock (`sstky` in the first fiscal quarter and changes in the `sstky` for the second to fourth fiscal quarters). Following [McKeon \(2015\)](#), we classify equity issuances that are smaller than 3% of a firm's market capitalization as zero issuance.

6. *Cumulative equity issuance probability*: an indicator variable that takes the value of one if a firm has issued new equity between quarters t and $t + h$ (i.e., the cumulative equity issuance probability $E_{it+h} = 1$ if $e_{it-1} = 0$ and $\sum_{\tau=0,\dots,h} e_{it+\tau} > 0$, where e_{it} denotes firm i 's equity issuance in quarter t); and zero otherwise.

A.2. Additional tables and figures

Figure A.1: Distribution of corporate news coverage: Non-zero coverage and within industry



Notes: Panel (a) reports the distribution of average articles per quarter for firms that receive non-zero coverage over the sample period. Panel (b) reports the distribution of average articles per quarter compared to industry average, which is equivalent to the residual from regressing average articles per quarter on 4-digit NAICS fixed effects.

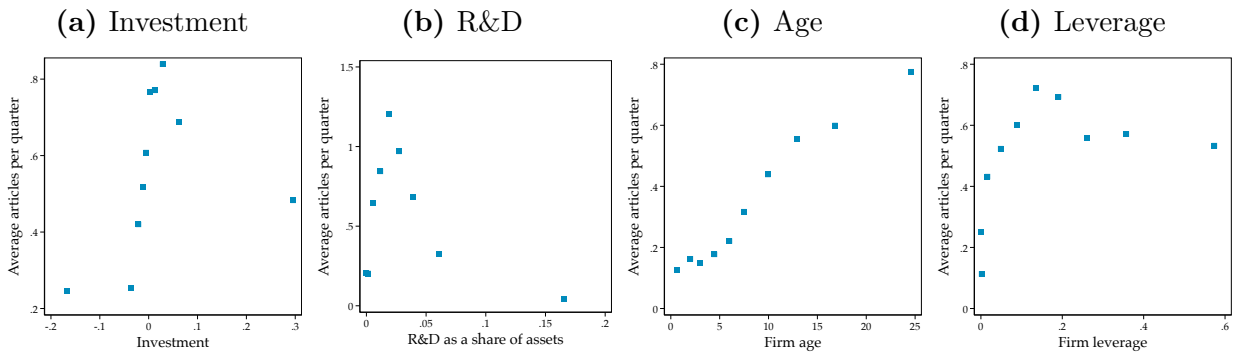
Table A.1: Top 10 firms ranked by total media coverage in the sample period

Rank	Firm	Articles
1	BlackRock	36,455
2	AT&T	22,775
3	General Motors	18,380
4	Microsoft	15,314
5	Apple	13,995
6	Time Warner	11,708
7	Alphabet	10,402
8	Citigroup	9,844
9	Merrill Lynch	9,070
10	Boeing	8,965
Total articles on top 10 firms		156,908
Total articles on remaining firms		228,790

Table A.2: Variance decomposition of firms' media coverage

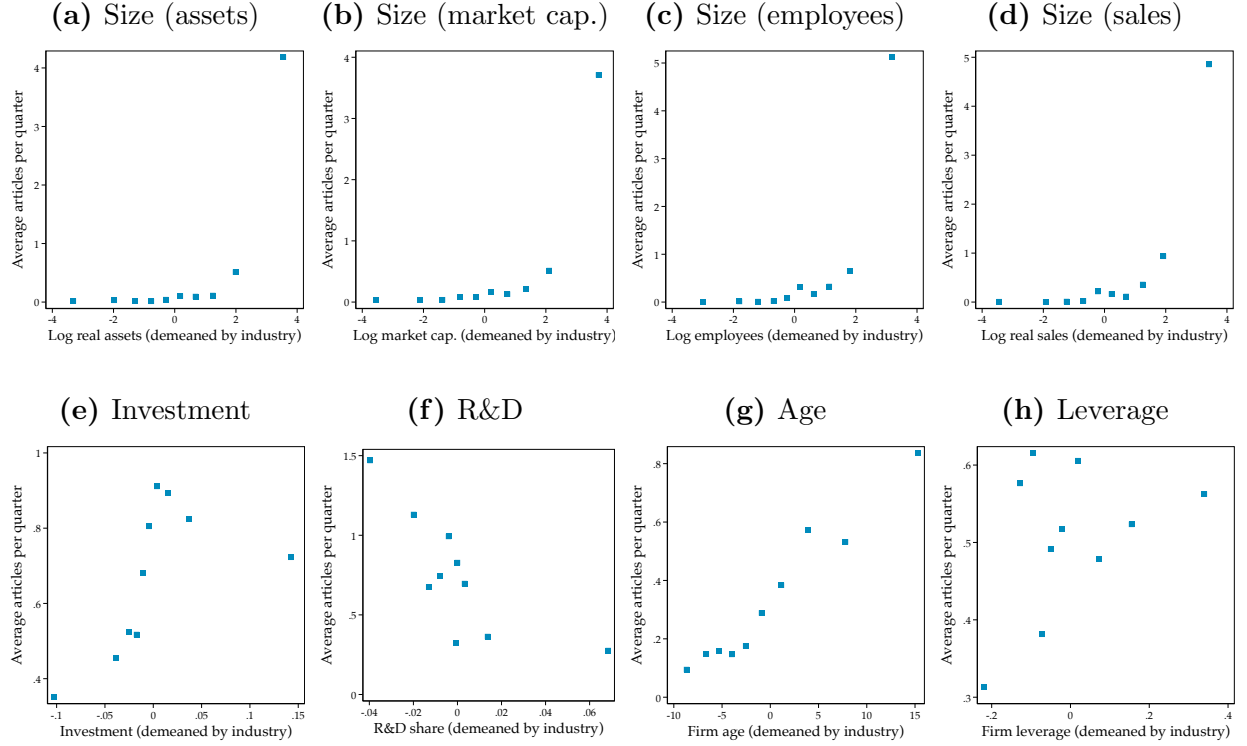
	Mean	SD	R ²		Mean	SD	R ²
Articles per quarter	0.528	6.623	0.000	Probability of media coverage	1.514%	0.122	0.000
Time		6.622	0.000			0.122	0.002
Industry		6.489	0.040			0.119	0.045
Firm		4.057	0.625			0.085	0.515
Industry $\times$ Time + Firm		3.902	0.656			0.081	0.563

Notes: This table reports the variance decomposition of media coverage. The left-hand panel reports the standard deviation of ε_{it} and the R^2 from estimating $h_{it} = \alpha_{st} + \alpha_i + \varepsilon_{it}$, where h_{it} is the article counts containing firm i in major newspapers in quarter t , α_{st} is the sector-by-time fixed effect, and α_i is the firm fixed effect. The right-hand panel reports the standard deviation of ε_{it} and the R^2 from estimating $\mathbb{1}(h_{it} > 0) = \alpha_{st} + \alpha_i + \varepsilon_{it}$, where the dependent variable is an indicator variable for media coverage, $\mathbb{1}(h_{it} > 0)$.

Figure A.2: Other firm characteristics and media coverage

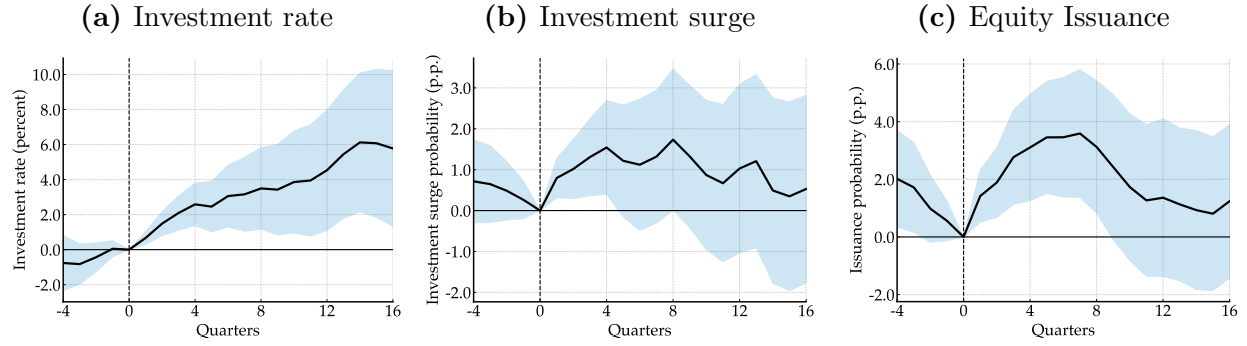
Notes: This figure reports binned scatterplots of average news articles per quarter. Each bin represents a decile of firms, sorted from the smallest to largest. Firm characteristics include investment rate in panel (a), and research and development expenses as a share of assets in panel (b), firm age (years since the IPO) in panel (c), market leverage in panel (d).

Figure A.3: Media coverage and within-industry firm characteristics



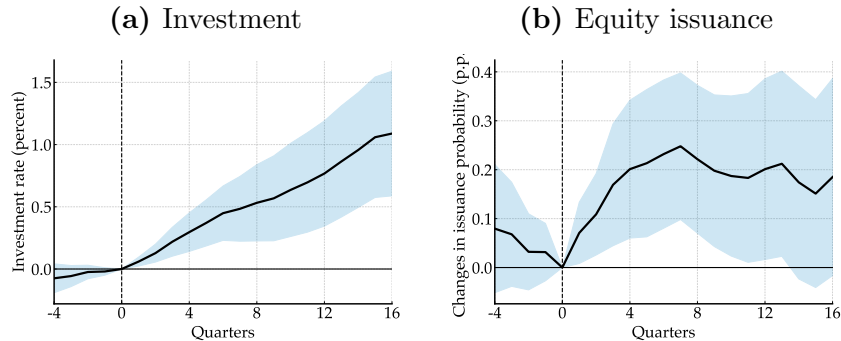
Notes: This figure reports binned scatterplots of average news articles per quarter. Each dot represents a decile of firms by firm characteristics relative to the industry (4-digit NAICS) average, sorted from the smallest to largest. The top panel reports the relationship between news coverage and firm size, measured by log real assets in panel (a), market capitalization (prices per share times shares outstanding) in panel (b), log employee numbers in panel (c), and log real sales in panel (d). The bottom panel reports the relationship between news coverage and other firm characteristics, including investment rate in panel (e), and research and development expenses as a share of assets in panel (f), firm age (years since the IPO) in panel (g), and market leverage in panel (h).

Figure A.4: Alternative media coverage measure: coverage indicator $\mathbb{1}(\nu_{it} > 0)$



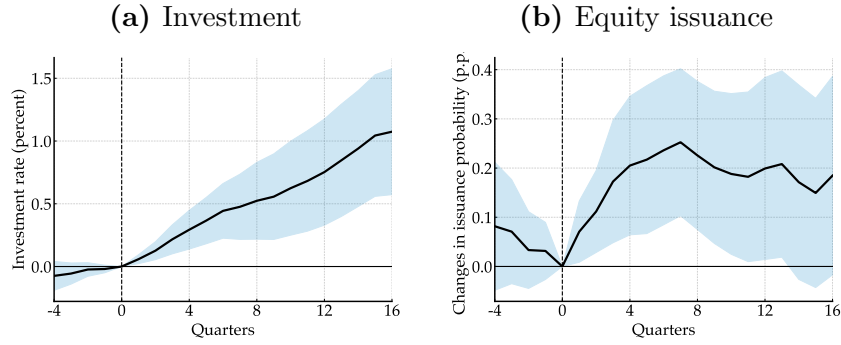
Notes: This figure reports variants of the baseline local projections in equation (2) using an alternative media coverage measure, for quarters $-4 \leq h \leq 16$: $y_{it+h} - y_{it} = \alpha_{st} + \alpha_i + \beta_h \mathbb{1}(\nu_{it} > 0) + \Gamma' Z_{it-1} + u_{ith}$, where α_{st} denotes a sector-by-quarter fixed effect (with sectors defined at the 4-digit NAICS level); α_i denotes a firm fixed effect; $\mathbb{1}(\nu_{it} > 0)$ is an indicator variable which takes the value of 1 if firm i has any coverage in major U.S. newspapers in quarter t , and 0 otherwise; and Z_{it-1} is a vector of firm controls including size, age, and real sales growth. The dependent variable y_{it} includes (a) the investment rate, defined as the log change in the book value of the firm's tangible capital stock; (b) the cumulative probability of investment surge, defined as an indicator variable that takes the value of 1 if a firm has quarter investment rate higher than 20% between quarters t and $t+h$, and zero otherwise; and (c) the cumulative probability of equity issuance, defined as an indicator variable that takes the value of 1 if a firm issues new equity between quarters t and $t+h$, and zero otherwise. Standard errors are double clustered by firm and quarter. 90% confidence intervals are reported.

Figure A.5: Excluding firms that share common ownership with media outlets



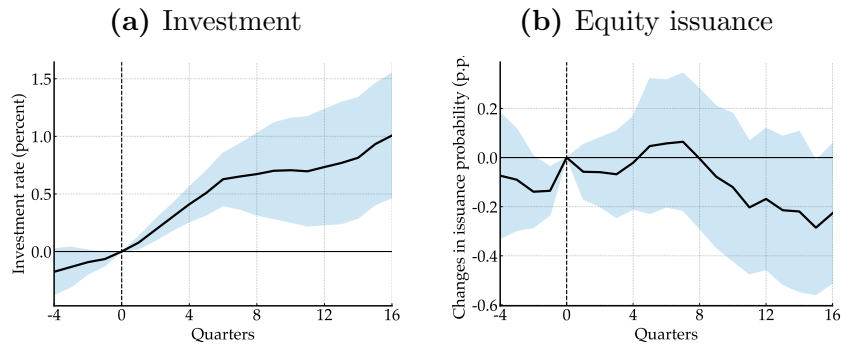
Notes: This figure reports variants of the baseline local projections in equation (2), where firms that share common owners with the media outlets in our sample (The Wall Street Journal, New York Times, and USA Today) are excluded.

Figure A.6: Excluding firms in the media sector



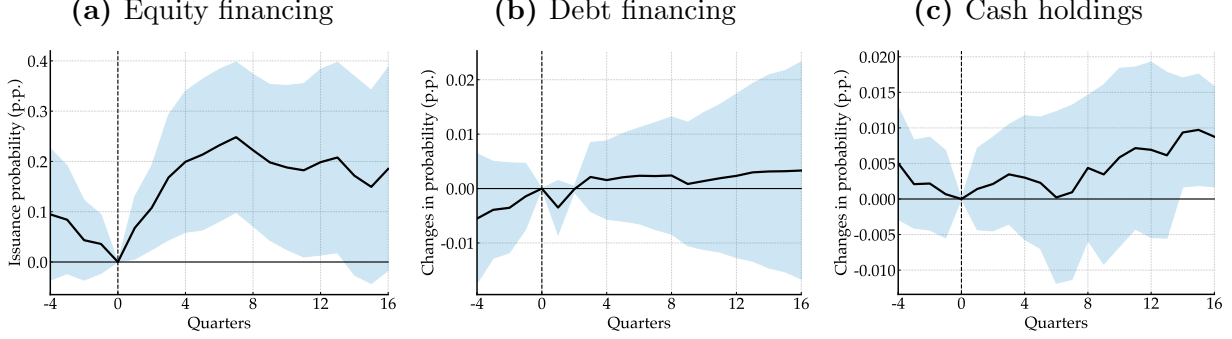
Notes: This figure reports variants of the baseline local projections in equation (2), where firms in the media sector (NAICS 5418) are excluded.

Figure A.7: Post-2005 sample



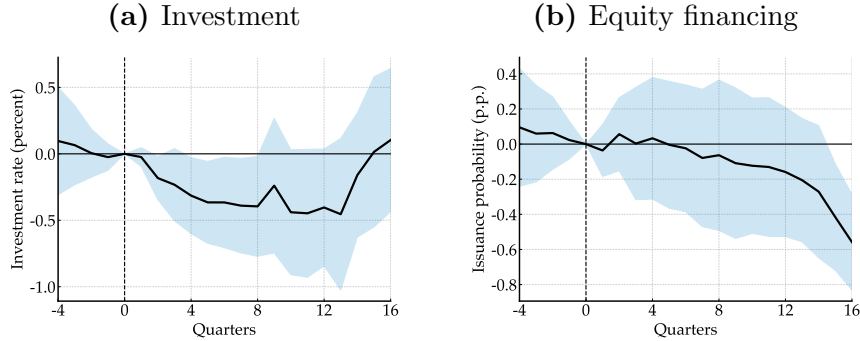
Notes: This figure reports variants of the baseline local projections in equation (2), where the sample is restricted to 2005 and later, corresponding to the creation of Twitter.

Figure A.8: The effects of media coverage on different forms of firm financing



Notes: This figure reports variants of the baseline local projections in equation (2): $y_{it+h} - y_{it} = \alpha_{st} + \alpha_i + \beta_h(\nu_{it} - \mathbb{E}_i \nu_{it}) + \Gamma' Z_{it-1} + u_{ith}$, where α_{st} denotes a sector-by-quarter fixed effect; α_i denotes a firm fixed effect; ν_{it} denotes the number of mentions of firm i in major U.S. newspapers quarter t ; $\mathbb{E}_i \nu_{it}$ denotes the average number of mentions of the firm over the sample; and Z_{it-1} is a vector of firm controls including size, age, and real sales growth. The dependent variable y_{it} includes (a) the cumulative probability of equity financing, repeated from the baseline estimates in Figure 3; (b) the cumulative probability of debt financing, defined as an indicator variable that takes the value of 1 if a firm increases debt financing between quarters t and $t+h$, and zero otherwise; and (c) the cumulative probability of increasing cash holdings, defined as an indicator variable that takes the value of 1 if a firm increases cash holdings between quarters t and $t+h$, and zero otherwise. We have standardized $(\nu_{it} - \mathbb{E}_i \nu_{it})$ across the entire sample. Standard errors are double clustered by firm and quarter. 90% confidence intervals are reported.

Figure A.9: Effects of Twitter coverage



Notes: This figure reports the results from estimating the local projections in equation (2) for quarters $-4 \leq h \leq 16$: $y_{it+h} - y_{it} = \alpha_{st} + \alpha_i + \beta_h(\tau_{it} - \mathbb{E}_i \tau_{it}) + \Gamma' Z_{it} + u_{ith}$, where the dependent variables y_{it} include (a) the cumulative investment rate and (b) the cumulative probability of equity financing, defined in the main text; α_{st} denotes the sector-by-quarter fixed effect; α_i denotes the firm fixed effect; τ_{it} denotes the number of firm mentions on Twitter in quarter t ; $\mathbb{E}_i \tau_{it}$ denotes the average number of firm mentions on Twitter over the sample; and Z_{it} is a vector of firm controls including size, age, and real sales growth. We have standardized $(\tau_{it} - \mathbb{E}_i \tau_{it})$ over the entire sample. Standard errors are double clustered by firm and quarter. 90% confidence intervals are reported.

A.3. Topics in news articles

To analyze the content of the news coverage, we employ latent Dirichlet allocation (LDA) to extract 20 distinct “topics” that represent the text of these newspaper articles. LDA, a generative probabilistic model from natural language processing, assumes that each article is a mixture of topics and each topic is a mixture of words. By analyzing the co-occurrence patterns of words within the articles, the model identifies the underlying topics that best represent the content.

We use the **gensim** library in Python to estimate the LDA model. The pre-processing of texts includes removing stop words, numbers, cases, and single-letter words, and then lemmatizing. We specify uniform Dirichlet priors. To select the model hyperparameter that governs the number of topics, we take a data-driven approach and perform a grid search of 200 topics in increments of 20. Through this procedure, we choose 20 as the number of topics as this generates the highest topic coherence.

The estimated topics are reported in Appendix Figure A.10. To construct the measure of media coverage purged of coverage on investment or financing, we compute the share of a firm’s quarterly news coverage containing any articles that have any positive loadings on topics related to firm investment and financing—topic 5 (“investment”), topic 16 (“financing”), and topic 8 (“financing from international markets”) in Appendix Figure A.10—and exclude it from the baseline coverage measure.

We use this purged coverage measure to estimate a variant of the baseline local projection in (2):

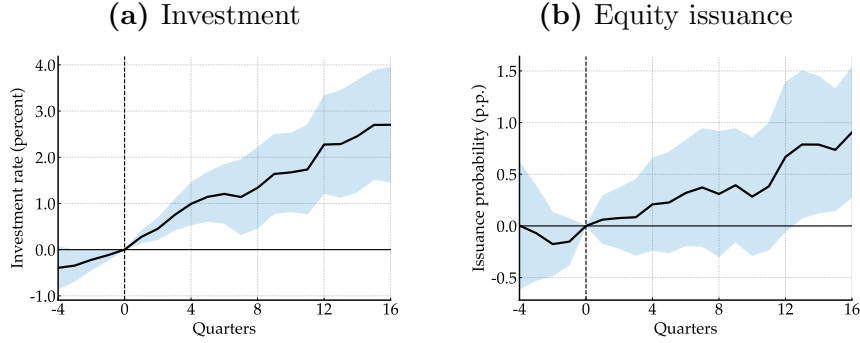
$$y_{it+h} - y_{it} = \alpha_{st} + \alpha_i + \beta_h(\nu_{it}^{\text{purged}} - \mathbb{E}_i \nu_{it}^{\text{purged}}) + \Gamma' Z_{it-1} + u_{ith},$$

where the dependent variables y_{it} include (a) the cumulative investment rate and (b) the cumulative probability of equity financing, defined in the main text; $\{\alpha_{st}, \alpha_i\}$ denote the sector-by-quarter and firm fixed effects; ν_{it}^{purged} denotes the number of mentions of firm i in major U.S. newspapers quarter t excluding any articles with positive LDA loadings on topics related to investment and equity financing; $\mathbb{E}_i \nu_{it}^{\text{purged}}$ denotes the average of the firm’s ν_{it}^{purged} over the sample; and Z_{it-1} is a vector of firm controls including size, age, and real sales growth. Estimates are reported in Appendix Figure A.11.

Figure A.10: Topics in firm news coverage



Figure A.11: News coverage, firm investment, and financing: Excluding the coverage in investment and financing articles



Notes: This figure reports variants of the baseline local projections in equation (2). We estimate $y_{it+h} - y_{it} = \alpha_{st} + \alpha_i + \beta_h(\nu_{it}^{\text{purged}} - \mathbb{E}_i \nu_{it}^{\text{purged}}) + \Gamma' Z_{it-1} + u_{ith}$, where the dependent variables y_{it} include (a) the cumulative investment rate and (b) the cumulative probability of equity financing, defined in the main text; $\{\alpha_{st}, \alpha_i\}$ denote the sector-by-quarter and firm fixed effects; ν_{it}^{purged} denotes the number of mentions of firm i in major U.S. newspapers quarter t excluding any articles with positive LDA loadings on topics related to investment and equity financing; $\mathbb{E}_i \nu_{it}^{\text{purged}}$ denotes the average of the firm's ν_{it}^{purged} over the sample; and Z_{it-1} is a vector of firm controls including size, age, and real sales growth. Standard errors are double clustered by firm and quarter. 90% confidence intervals are reported.

A.4. Sentiment of news articles

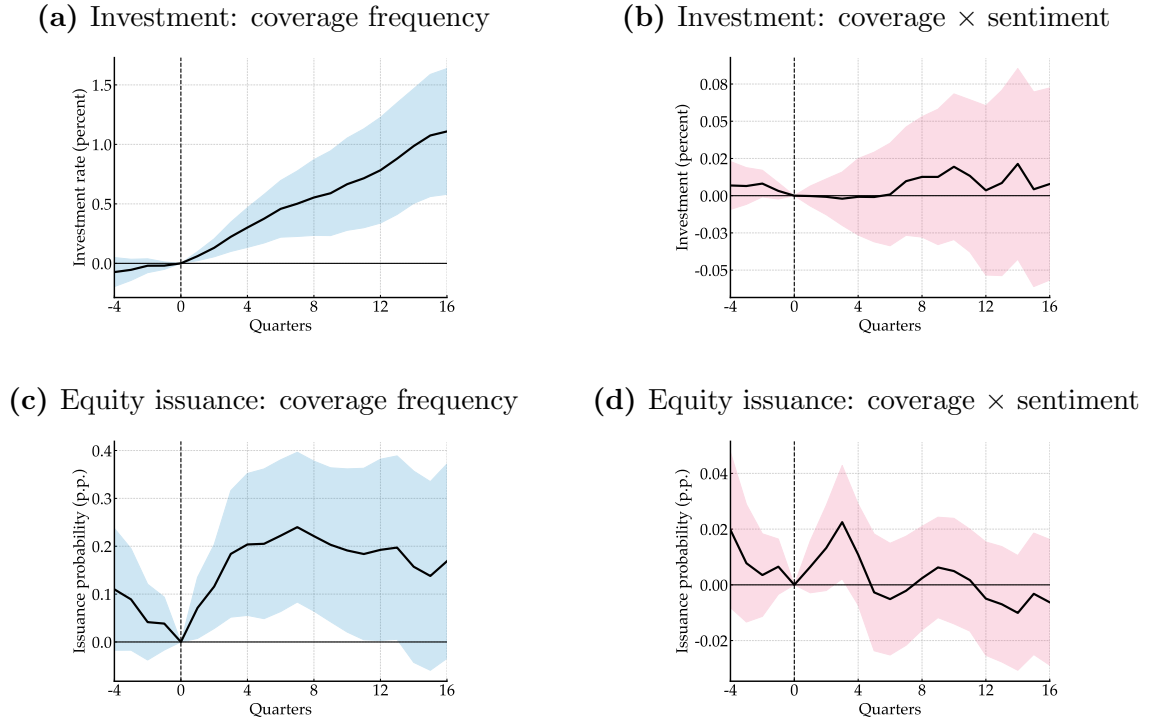
This section describes the construction of the news sentiment measure. For each news article k , we compute three sentiment measures:

1. Keyword-based measure, $\text{sent}_k^{\text{LM}} = (N_k^+ - N_k^-)/N_k$, where N_k^+ , N_k^- , and N_k denote the number of positive, negative, and total words in article k , respectively, according to the dictionary constructed by [Loughran and McDonald \(2011\)](#).
2. FinBERT-based measure, $\text{sent}_k^{\text{BERT}} = (N_k^+ - N_k^-)/N_k$, where N_k^+ , N_k^- , and N_k denote the number of positive, negative, and total sentences in article k , respectively. Sentiment is classified at the sentence level using FinBERT ([Araci, 2019](#)), a version of BERT pre-trained on financial texts.
3. GPT-based measure, $\text{sent}_k^{\text{GPT}} \in \{-1, 0, 1\}$. We query GPT 3.5 ([Ye et al., 2023](#)) for the sentiment of an article and classify an article as negative (-1), neutral (0), or positive (1). The prompt used is: “Classify the sentiment of this news article. Return ‘positive’ if the sentiment of the article is positive, return ‘negative’ if the sentiment of the article is negative, and return ‘neutral’ if the sentiment of the article is neutral.”

To match the frequency of the empirical analysis, we compute the sentiment of firm i ’s coverage in quarter t as the average sentiment of the firm’s news articles within the quarter. Since each sentiment measure has its strengths and limitations, we take the first principal component across the three measures as our sentiment measure, sent_{it} .

Figure [A.12](#) reports the local projection results for which the frequency of news coverage is interacted with the sentiment of the coverage.

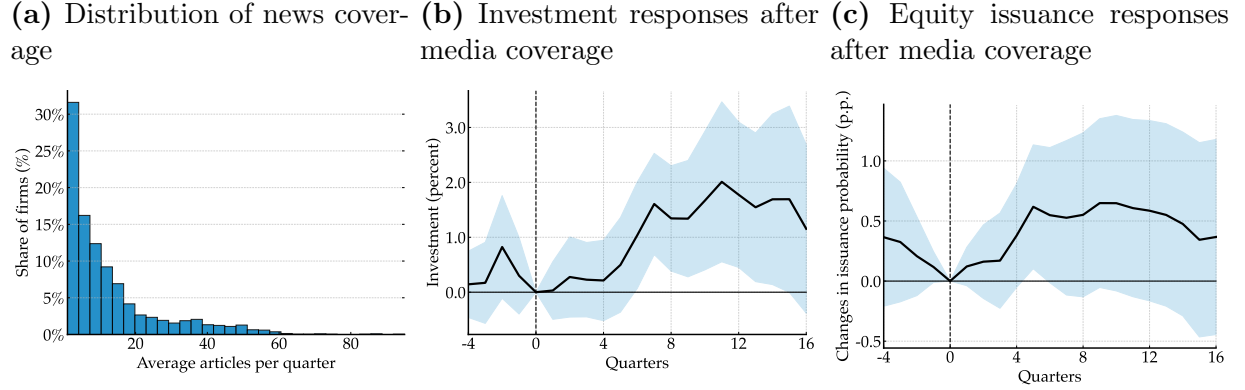
Figure A.12: News coverage sentiment



Notes: This figure reports results from estimating $\Delta_h y_{it} = \alpha_{st} + \alpha_i + \beta_{vh}(\nu_{it} - \mathbb{E}_i \nu_{it}) + \beta_{sh} \text{sent}_{it} + \beta_{vsh}(\nu_{it} - \mathbb{E}_i \nu_{it}) \times \text{sent}_{it} + \Gamma' Z_{it} + u_{it}$, where sent_{it} denotes the average sentiment of firm i 's media coverage in quarter t , and the sentiment measure is defined in Appendix Section A.4. Panels (a) and (c) report estimated β_{vh} , and panels (b) and (d) report estimated β_{vsh} . 90% confidence bands are reported.

A.5. News coverage in France

Figure A.13: Media coverage and firm outcomes in France



Notes: Panel (a) reports the distribution of corporate news coverage in major French newspapers from 2005 to 2022, including *Les Echos*, *Le Monde*, *La Tribune*, and *Le Figaro*. Panels (b) and (c) report estimates from the local projections in equation (2) for quarters $-4 \leq h \leq 16$: $y_{it+h} - y_{it} = \alpha_{st} + \alpha_i + \beta_h \nu_{it} + \Gamma' Z_{it} + u_{ith}$, where $\{\alpha_{st}, \alpha_i\}$ denote the sector-by-quarter and firm fixed effects; ν_{it} denotes the news coverage of firm i in major French newspapers in quarter t , demeaned at the firm level and standardized; and Z_{it} is a vector of firm controls including size, age, and real sales growth. The dependent variable y_{it} includes the investment rate ($\Delta \log k_{it}$) in panel (b), defined as the log change in capital expenditures, and the cumulative probability of equity issuance (E_{it}) in panel (c), defined as an indicator variable that takes the value of 1 if a firm issues new equity between quarters t and $t + h$, and zero otherwise. Standard errors are double clustered by firm and quarter. 90% confidence intervals are reported.

Table A.3: National media strikes in France

Quarter	Date	Description
2005Q4	October 4, 2005	Unions of journalists and technicians in public broadcasting went on strike as part of a national day of action.
	October 20, 2005	The Agence France-Presse journalists' unions went on strike to oppose the announced closure of a regional office.
2008Q1	February 13, 2008	Workers in public broadcasting went on strike to protest President Nicolas Sarkozy's media reform.
2008Q4	November 25, 2008	Public broadcaster workers went on strike to protest bill passed reforming public broadcasting by President Sarkozy.
2013Q1	February 1, 2013	The Agence France Presse journalists' unions went on strike to call for the withdrawal of the "France Region" project.
2018Q2	April 1, 2018	National strikes, including by broadcasters, against President Emmanuel Macron's reforms to the public sector.

Notes: Information on national media strikes in France from 2005 to 2021 was found by searching Factiva using the search words "strike or grève" and "journalist or journaliste" or "strike or grève" and "broadcaster or diffuseur" and restricting the region to France, the industry to Media/Entertainment, the subject to Labor Dispute, and by excluding strikes against individual newspapers.

Table A.4: Descriptive statistics of strikes and nonstrikes

	Media strikes	Nonstrikes	Total
Number of quarters	5	9	68
Avg number of firm news	2,814	3,155	2,337
Avg share of firm news	8.87%	10.12%	8.25%

Notes: “Media strikes” includes 2005Q4, 2008Q1, 2008Q4, 2013Q1, and 2018Q2. “Nonstrikes” refers to quarters adjacent to strikes which do not include a media strike. “Total” refers to all quarters in the French sample from 2005 to 2021.

Table A.5: Equity issuance and media strikes

	(1)	(2)	(3)
	Equity issuance probability		
Strike indicator	0.005 (0.018)	-0.002 (0.016)	-0.002 (0.017)
Size		0.002** (0.001)	0.002** (0.001)
Age		-0.013*** (0.004)	-0.014*** (0.004)
Real sales growth		-0.000*** (0.000)	-0.000*** (0.000)
rGDP growth			-0.000 (0.001)
Aggregate investment			-0.000 (0.000)
Observations	36454	33348	28493
R^2	0.019	0.025	0.026
Industry FE	yes	yes	yes
Double-clustered SE	yes	yes	yes

Notes: This table reports estimates from $E_{it} = \alpha_{st} + \beta S_t + \Gamma' Z_{it} + u_{it}$, where E_{it} is an indicator variable that equals 1 if firm i issues equity in quarter t ; α_{st} is a sector-by-fiscal-quarter fixed effect; S_t is an indicator variable which equals 1 if there is a media strike in quarter t ; and Z_{it} is a vector of controls including firm size, age, real sales growth, real GDP growth, inflation, and aggregate investment. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table A.6: Probability and size of issuance following media strikes

	Equity issuance probability				Log equity issuance			
	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 1$	$h = 2$	$h = 3$	$h = 4$
Strike indicator	-0.019 (0.018)	-0.034** (0.014)	-0.044** (0.018)	0.028 (0.038)	-0.169 (0.135)	-0.272 (0.220)	-0.048 (0.084)	0.197 (0.136)
Observations	32472	31650	30863	30141	11272	11048	10866	10657
R^2	0.049	0.070	0.082	0.083	0.518	0.524	0.528	0.536
Firm FE	yes	yes	yes	yes	yes	yes	yes	yes
Double-clustered SE	yes	yes	yes	yes	yes	yes	yes	yes
Firm controls	yes	yes	yes	yes	yes	yes	yes	yes

Notes: The table reports estimates from $y_{it+h} = \alpha_{st} + \beta S_t + \Gamma' Z_{it} + u_{it}$ for horizon h , where $y_{it+h} \in \{E_{it+h}, \log e_{it+h}\}$ is either an indicator variable for whether a firm issues equity or the log of equity issuance; α_{st} is a sector-by-fiscal-quarter fixed effect; S_t is an indicator variable which equals 1 if there is a media strike in quarter t ; and Z_{it} is a vector of firm controls including firm size, age, and real sales growth. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

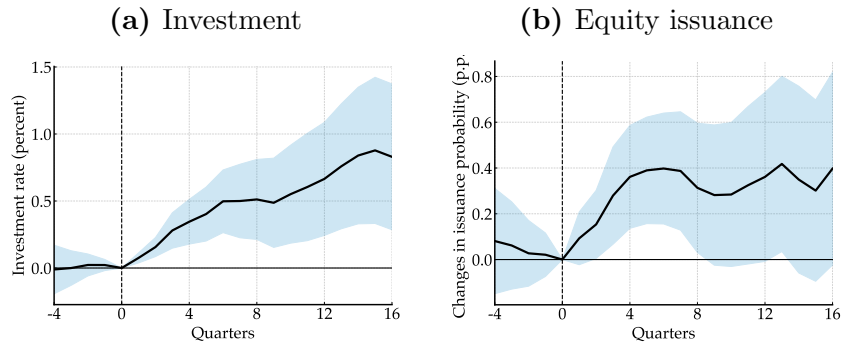
A.6. Excluding the largest firms

A potential interpretation of our empirical findings is that because large firms are *granular* and have disproportionately large effects on the macroeconomy (Gabaix, 2011), media outlets may devote outsized attention to them because their news is informative about aggregate outcomes. Under this interpretation, the observed media allocation is not necessarily misallocated.

To test whether our findings are driven by the media’s focus on granular firms, we re-estimate the local projections (Appendix Figure A.14), the covariance between firm size and responsiveness to coverage (Appendix Figure A.15), and firm responsiveness by size decile (Appendix Figure A.16) after excluding the largest 1% firms in our sample. News about these firms may be viewed as especially informative because of their outsized influence on aggregate outcomes. By excluding them from the sample, we test whether media outlets continue to devote greater attention to large firms that are less likely to have granular influence on the macroeconomy.

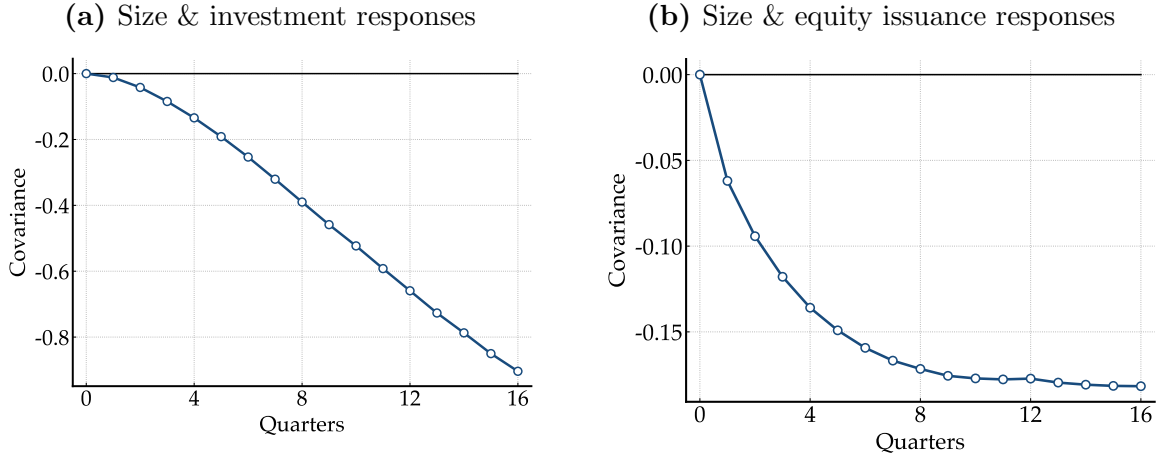
Our empirical results remain both qualitatively and quantitatively unchanged after excluding the largest firms, suggesting that the media’s preference for large firms is not entirely driven by firm granularity.

Figure A.14: Local projection: excluding largest firms



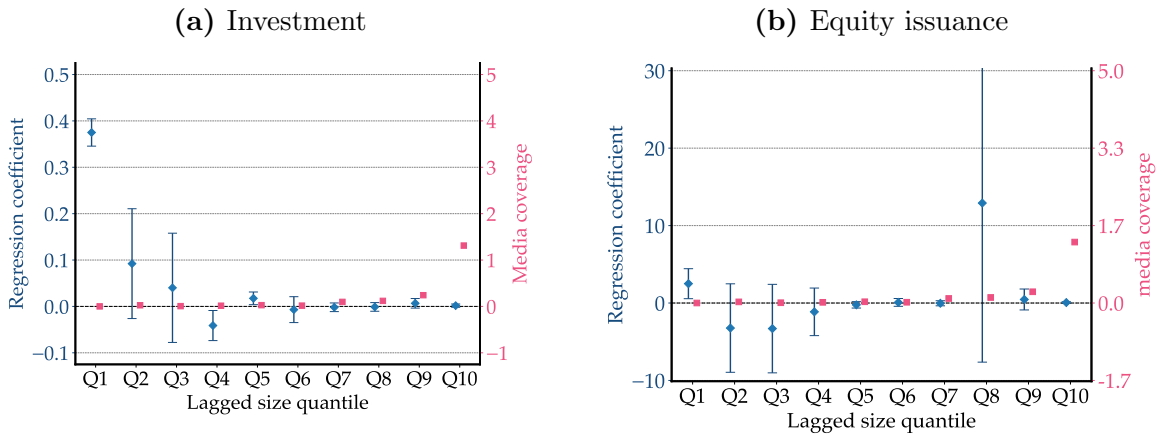
Notes: This figure estimates the local projections in equation (2) described in the main text, excluding firms with real assets above the 99th percentile. 90% confidence intervals are reported.

Figure A.15: Covariance: firm size and responsiveness to coverage, excluding largest firms



Notes: This figure reports the covariance, $\text{Cov}(s_{it}, \widehat{\Delta_h y_{it}})$, between firm size, s_{it} , for firm i in quarter t and its cumulative responses to media coverage, $\widehat{\Delta_h y_{it}}$, measured by the fitted value implied by the baseline local-projection specification in equation (2) for horizon h based on a sample that excludes the largest 1% firms by real assets. In panel (a), $\Delta_h y_{it}$ denotes the investment rate ($\Delta_h \log k_{it}$), defined as the log change in the book value of the firm's tangible capital stock. In panel (b), $\Delta_h y_{it}$ denotes the cumulative probability of equity issuance (E_{it}), defined as an indicator variable that takes the value of 1 if a firm issues new equity between quarters t and $t + h$ and 0 otherwise.

Figure A.16: Media coverage and firm responsiveness by size decile, excluding largest firms



Notes: This figure reports the results from estimating equation (6) in the main text, excluding firms with real assets above the 99th percentile in each quarter. Both panels are estimated on the same sample of firms. Standard errors are double clustered by firm and quarter. 90% confidence intervals are reported.

B. Additional Details for the Theoretical Analysis

B.1. Proof of Proposition 1

Proof. We show that there is a unique reporting policy that can be sustained as a symmetric equilibrium. To find this, we begin by considering an arbitrary candidate reporting policy. We then show that there is a unique candidate reporting policy from which no outlet would find it optimal to deviate.

The candidate reporting policy is characterized by a vector of reporting choices $\mathbf{m}_t = \{m_{jt}\}_{j=0}^1$, which satisfies the space constraint (16). Without loss of generality, assume that $\mathbf{m}_t$ involves all outlets reporting on firm j and all outlets not reporting on firm j' .

Optimal forecasts For a firm j that is reported on by outlet i , forecaster i can forecast the market value precisely as they observe all of that firm's state variables

$$\text{Pred}_t(k_{jt}, z_{jt}, 1, \mathcal{I}_{it}^{\text{news}} | m_{ijt}^o = 1) = \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1). \quad (29)$$

Substituting this forecast into equation (22) reveals that when $m_{ijt}^o = 1$, forecaster i makes no forecast errors.

For a firm that is not reported by outlet i , the optimal forecast by forecaster i is given by

$$\text{Pred}_t(k_{jt}, z_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}} | m_{ijt}^o = 0) = \mathbb{E}[\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}) | k_{jt}, z_{jt}, m_{jt}]. \quad (30)$$

The forecast in (30) is uncertain because the forecaster is uncertain about a_{jt} . In general, this will lead forecaster i to make non-zero forecast errors when $m_{ijt}^o = 0$.³²

Forecaster utility at equilibrium Since we consider a symmetric reporting policy, all outlets make the same reporting decisions. This means all forecasters have the same infor-

³²If $m_{jt} = 1$, forecaster i is aware that investors know a_{jt} , but they do not themselves observe this as their outlet did not report it. If instead $m_{jt} = 0$, forecaster i knows that no outlet reported a_{jt} and so it is not part of the investors' information set. However, there is still uncertainty in that case, because the realized a_{jt} can still affect the realized market value, indirectly, through firm j 's equity issuance (see Section 3.1.1). Since forecasters make predictions before equity markets open, they cannot observe equity issuance. They do not therefore know what the investors' posteriors will be and so cannot be certain about the realization of the market value.

mation set and make the same forecast errors. As a result,

$$\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) = \overline{\text{FE}}_{-it}(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{-it}^{\text{news}}), \quad (31)$$

and thus $U_{it} = 0$.

Outlet deviations A minimal deviation from $\mathbf{m}_t$ consists of an outlet i ceasing to report on firm j and, instead, reporting on firm j' . $\mathbf{m}_t$ can only be sustained in equilibrium if no outlet finds it optimal to deviate in this way. Since in the absence of any deviation we have obtained that $U_{it} = 0$ with certainty, a sufficient condition for $\mathbf{m}_t$ to be an equilibrium is that

$$\mathbb{E} \hat{U}_{it}(j, j') + \kappa_{jt} - \kappa_{j't} \leq 0, \quad (32)$$

where $\hat{U}_{it}(j, j')$ is the utility of forecaster i if outlet i deviates. If this condition holds for all pairs of the reported and unreported firms, j, j' , the news outlets never deviate and $\mathbf{m}_t$ is an equilibrium.

We now proceed to find an expression for $\mathbb{E} \hat{U}_{it}(j, j')$. First, notice that the deviation would have no effect on firms other than j and j' . From the definition of the forecaster's utility in equation (21), we therefore have

$$\begin{aligned} \mathbb{E} \hat{U}_{it}(j, j') &= -\mathbb{E} [\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) - \overline{\text{FE}}_{-it}(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{-it}^{\text{news}})] \\ &\quad -\mathbb{E} [\text{FE}_t(k_{j't}, z_{j't}, a_{j't}, m_{j't}, \mathcal{I}_{it}^{\text{news}}) - \overline{\text{FE}}_{-it}(k_{j't}, z_{j't}, a_{j't}, m_{j't}, \mathcal{I}_{-it}^{\text{news}})]. \end{aligned} \quad (33)$$

The first two terms give the utility change from no longer reporting on firm j . Other forecasters are still reporting on j and so it remains the case that $m_{jt} = 1$ and the realized market value of firm j is unchanged. The average forecast error of the other forecasters $\overline{\text{FE}}_{-i}(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{-it}^{\text{news}})$, therefore, remains unchanged at 0. However, forecaster i 's forecast does change, as their information set no longer contains a_{jt} . Specifically,

$$\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) = [\text{Pred}_t(k_{jt}, z_{jt}, 1, \mathcal{I}_{it}^{\text{news}} | m_{ijt}^o = 0) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2. \quad (34)$$

Substituting out for the optimal forecast using

$$\text{Pred}_t(k_{jt}, z_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) = \mathbb{E}[\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}) | k_{jt}, z_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}]. \quad (35)$$

and taking the expectations, we obtain

$$\begin{aligned} \mathbb{E}[\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}})] &= \mathbb{E} \left[\mathbb{E}(\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) | k_{jt}, z_{jt}, m_{jt} = 1) \right. \\ &\quad \left. - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) \right]^2, \end{aligned} \quad (36)$$

$$= \mathbb{V}[\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) | k_{jt}, z_{jt}], \quad (37)$$

where $\mathbb{V}[\cdot]$ denotes the variance with respect to a_{jt} .

The second two terms of equation (33) give the utility change due to reporting on firm j' . Recall that investors observe a firm's asset quality if at least one outlet reports it (equation (20)). Since outlet i has reported on firm j' , that firm's asset quality $a_{j't}$ is transmitted to investors and so $m_{j't} = 1$. As a result, forecaster i observes all of the determinants of firm j' 's market value and is able to make an accurate forecast (equation (29)). Forecaster i therefore makes no forecast error about firm j' .

However, although forecaster i makes no forecast error about j' under this deviation, the same is not true of the other forecasters. Their outlets have not reported on j' ($m_{i'j't}^o = 0$) and so they do not have sufficient information to precisely infer that firm's market value. This generates a forecast error, given by

$$\overline{\text{FE}}_{-it}(k_{j't}, z_{j't}, a_{j't}, 1, \mathcal{I}_{-it}^{\text{news}}) = \int_{i' \neq i} [\text{Pred}_t(k_{j't}, z_{j't}, \mathcal{I}_{i't}^{\text{news}} | m_{i'j't}^o = 0) - \text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1)]^2 di' \quad (38)$$

All outlets, i' , are identical, so when using the same steps as those used to derive equation (37), the expectation of this average forecast error becomes

$$\mathbb{E} [\overline{\text{FE}}_{-it}(k_{j't}, z_{j't}, a_{j't}, 1, \mathcal{I}_{-it}^{\text{news}})] = \mathbb{V}[\text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1) | k_{j't}, z_{j't}]. \quad (39)$$

Substituting these results into equation (33), the utility of deviating in this way is

$$\mathbb{E} \hat{U}_{it}(j, j') = \mathbb{V}[\text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1) | k_{j't}, z_{j't}] - \mathbb{V}[\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) | k_{jt}, z_{jt}]. \quad (40)$$

Condition (32) is therefore satisfied, and the candidate, $\mathbf{m}_t$, can be sustained as a symmetric equilibrium, if and only if

$$\mathbb{V}[\text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1) | k_{j't}, z_{j't}] - \kappa_{j't} \leq \mathbb{V}[\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) | k_{jt}, z_{jt}] - \kappa_{jt} \quad (41)$$

for all pairs of reported and unreported firms j, j' . The reporting policy for which this condition holds is the one described in equations (24)-(25), with the threshold N_t^* determined by:

$$\int_0^1 \mathbb{1}(N_t(k_{jt}, z_{jt}, a_{jt}) - \kappa_{jt} \geq N_t^*) dj = r. \quad (42)$$

□

B.2. Alternative assumptions on outlets and forecasters

We consider two plausible alternative assumptions to the media-reporting policy in Section 3. Even though the resulting newsworthiness function differs slightly from equation (25), the qualitative properties remain unchanged.

B.2.1. Outlet objective function

In Section 3, we assumed that the media outlets maximize the expected utility of their forecasters. However, the media outlets observe all realizations of a_{jt} and the reporting decisions of the other outlets. These outlets are therefore able to predict the *realized* utility of their forecasters when they make their reporting decisions. If we allow the outlet to maximize this realized utility, their problem is as in Section 3.1.2, except that the objective function changes to

$$\max_{m_{ijt}^o} - \int_0^1 [\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) - \overline{\text{FE}}_{-it}(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{-it}^{\text{news}})] dj - \int_0^1 \kappa_{jt} m_{ijt}^o dj \quad (43)$$

subject to equations (18)–(19) and optimal predictions (35).

In this case, a vector $\mathbf{m}_t$ can be sustained as a symmetric reporting equilibrium in pure strategies if and only if

$$\hat{U}_{it}(j, j') + \kappa_{jt} - \kappa_{j't} \leq 0 \quad (44)$$

for all pairs of reported and unreported firms j, j' . This differs from equation (32) in that there is no longer an expectation operator present.

The results on the realized forecast errors derived for the baseline model continue to hold, as nothing has changed in the forecaster's problem. $\hat{U}_{it}(j, j')$ is therefore given by

$$\begin{aligned} \hat{U}_{it}(j, j') = & \left[\mathbb{E}(\text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1) | k_{j't}, z_{j't}, m_{j't} = 1) - \text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1) \right]^2 \\ & - \left[\mathbb{E}(\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) | k_{jt}, z_{jt}, m_{jt} = 1) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) \right]^2. \end{aligned} \quad (45)$$

The unique symmetric pure strategy reporting equilibrium is therefore as in Section 3.2, except that the newsworthiness function is modified to

$$N_t(k_{jt}, z_{jt}, a_{jt}) = \left[\mathbb{E}(\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) | k_{jt}, z_{jt}, m_{jt} = 1) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) \right]^2. \quad (46)$$

Like the form in equation (25), this is increasing in the firm size. The key difference is that the newsworthiness function now also depends on the realized a_{jt} .

B.2.2. Forecaster information

In Section 3.1, we assumed that forecasters can observe the reporting decisions of outlets other than their own. This allowed for a simple characterization of the equilibrium reporting policy, but it is not essential for our results. In this subsection, we derive the equilibrium reporting policy under the alternative assumption that forecaster i does not observe the reporting decisions of other outlets, as in, for example, Nimark and Pitschner (2019). We continue to assume, as in the previous derivation, that the outlet maximizes the realized utility of their forecaster. The outlet problem is therefore unchanged: their objective is as in equation (43), and the constraints are as in equations (18)–(19). A vector $\mathbf{m}_t$ can be sustained as a symmetric pure strategy equilibrium if and only if condition (44) holds for all

pairs of reported and unreported firms j, j' . The key way this alternative assumption changes the model is that forecasters no longer necessarily observe the aggregate media indicator m_{jt} .

As in Section 3.1, if outlet i reports on a firm, j , then $m_{jt} = 1$. Moreover, forecaster i can infer that $m_{jt} = 1$ for certain: they see that their outlet has reported on firm j , which is sufficient to imply $m_{jt} = 1$ (equation (20)). As in Appendix B.1, if an outlet deviates to report on firm j' , while other outlets report on j , we therefore have

$$\text{FE}_t(k_{j't}, z_{j't}, a_{j't}, m_{j't} = 1, \mathcal{I}_{it}^{\text{news}}) = \overline{\text{FE}}_{t-i}(k_{jt}, z_{jt}, a_{jt}, m_{jt} = 1, \mathcal{I}_{-it}^{\text{news}}) = 0 \quad (47)$$

The utility change from deviating therefore reduces to

$$\begin{aligned} \hat{U}_{it}(j, j') = & \left[\mathbb{E}(\text{MV}_t(k_{j't}, z_{j't}, a_{j't}, m_{j't}) | k_{j't}, z_{j't}, m_{i'j't}^o = 0) - \text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1) \right]^2 \\ & - \left[\mathbb{E}(\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}) | k_{jt}, z_{jt}, m_{ijt}^o = 0) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) \right]^2. \end{aligned} \quad (48)$$

This differs from equation (45) because the expectations are formed without the knowledge of the true $m_{jt}, m_{j't}$. In both cases, all the forecasters know is what their own outlets have published. In the first of the expectations, this is the expected market value of firm j' formed by forecasters other than forecaster i , whose outlet did not report on j' ($m_{i'j't}^o = 0$). In the second expectation, the expected market value of firm j is formed by forecaster i , whose outlet has deviated and is not reporting on firm j ($m_{ijt}^o = 0$). In both cases, the true aggregate reporting indicator is $m_{jt} = m_{j't} = 1$: outlet i reports on firm j' , and all other outlets report on firm j . As the outlets can still observe each other's reporting choices, each outlet is aware of this fact; only the forecasters are not.

The law of iterated expectations implies

$$\begin{aligned} & \mathbb{E}(\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}) | k_{jt}, z_{jt}, m_{ijt}^o = 0) \\ &= \Pr(m_{jt} = 1 | k_{jt}, z_{jt}, m_{ijt}^o = 0) \mathbb{E}(\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) | k_{jt}, z_{jt}, m_{ijt}^o = 0, m_{jt} = 1) \\ &+ (1 - \Pr(m_{jt} = 1 | k_{jt}, z_{jt}, m_{ijt}^o = 0)) \mathbb{E}(\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 0) | k_{jt}, z_{jt}, m_{ijt}^o = 0, m_{jt} = 0), \end{aligned} \quad (49)$$

where $\Pr(m_{jt} = 1 | k_{jt}, z_{jt}, m_{ijt}^o = 0)$ is the perceived probability that forecaster i attaches to $m_{jt} = 1$, conditional on their observations.

The term $\Pr(m_{jt} = 1 | k_{jt}, z_{jt}, m_{ijt}^o = 0)$ represents, for a forecaster observing that their

outlet *did not* report on a firm, the probability that some other outlet *did* report on that firm during the period. Under rational expectations, although forecaster i does not observe the reporting decisions of the outlet belonging to forecaster i' (and vice versa), they are able to understand the policy function that is driving that other outlet's decisions and, thus, the process for determining m_{jt} .

Under rational expectations, forecasters also understand that they are in a symmetric media equilibrium, which implies

$$\Pr(m_{jt} = 1 | k_{jt}, z_{jt}, m_{ijt}^o = 0) = 0. \quad (50)$$

That is, when forecasters observe that their own outlet has not reported on a particular firm, they infer that no outlet has done so.

There is one nuance here that is noteworthy. Forecasters infer that $m_{jt} = m_{ijt}^o$ because they have rational expectations, so they have full knowledge of the equilibrium. In equilibrium, their inference on m_{jt} is therefore correct. However, in equation (48) we are considering a deviation from that equilibrium, implicitly assuming that if such a deviation were to happen, forecasters would not be able to identify that it had happened. As a result, they would continue to forecast $m_{jt} = m_{ijt}^o$ with certainty, even though this would have been incorrect under the deviation. This assumption is consistent with rational expectations: In any equilibrium, such a deviation is a probability-zero event, so it is rational to attach no weight to it. All the forecaster observes is k_{jt} , z_{jt} and m_{ijt}^o , and none of this reveals that a deviation is occurring. This is one key reason why deviations create forecast errors, as they lead forecasters to make errors about m_{jt} . The utility change from deviating in equation (48), therefore, becomes

$$\begin{aligned} \hat{U}_{it}(j, j') = & \left[\mathbb{E}(\text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 0) | k_{j't}, z_{j't}) - \text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1) \right]^2 \\ & - \left[\mathbb{E}(\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 0) | k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) \right]^2 \end{aligned} \quad (51)$$

The unique symmetric pure strategy reporting equilibrium is the same as in Section 3.2, except that the newsworthiness function is modified to

$$N_t(k_{jt}, z_{jt}, a_{jt}) = \left[\mathbb{E}(\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 0) | k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) \right]^2. \quad (52)$$

A firm is more newsworthy if its news coverage would substantially alter the beliefs of forecasters and investors and would lead to a large change in market valuations. Newsworthiness is increasing in firm size (as in equation (25)) and depends on realizing a_{jt} (as in equation (46)).

B.3. Equity market equilibrium under asymmetric information

This appendix formally characterizes the equity-market allocation used in Section 3.1.1. For a firm reported by the media, investors observe capital quality and the firm solves

$$V_t(k, z, a, 1) = \max_{e \geq 0} \{W_t(k, z, a, e) - e\}. \quad (53)$$

For an unreported firm, the equity price depends on issuance rather than directly on capital quality. We therefore write it as $P_t(k, z, e)$.

Fix a public state (k, z) . Capital quality has ordered support $a_1 < a_2 < \dots < a_N$ with prior $\mathcal{G}(a)$. Conditional on (k, z) , an equity-market equilibrium consists of an issuance policy $\mathbf{e}_t(k, z, a)$, investor beliefs $\mathcal{B}_t(a \mid k, z, e)$, and an equity-price schedule $P_t(k, z, e)$ such that firms optimize, on-path beliefs satisfy Bayes' rule, off-path beliefs satisfy the Divinity Criterion, and prices satisfy investor break-even. In the remainder of this appendix, we suppress the time subscript.

We apply the characterization in Guo et al. (2025). The relevant properties of the post-issuance value function are

$$\frac{\partial W(k, z, a, e)}{\partial e} > 0, \quad \frac{\partial^2 W(k, z, a, e)}{\partial e^2} < 0, \quad \frac{\partial^2 W(k, z, a, e)}{\partial e \partial a} \leq 0. \quad (54)$$

Under these properties, positive issuance choices separate across capital qualities, while firms that do not issue may pool at zero. The equilibrium allocation, supporting beliefs, and prices are characterized jointly below.

Proposition 2. *For firms with common public states (k, z) , define full-information issuance as*

$$\mathbf{e}^*(k, z, a) \equiv \arg \max_{e \geq 0} \{W(k, z, a, e) - e\}. \quad (55)$$

The equilibrium issuance policy is constructed recursively:

1. The lowest-quality type chooses

$$\mathbf{e}(k, z, a_1) = \mathbf{e}^*(k, z, a_1). \quad (56)$$

2. Given $\mathbf{e}(k, z, a_\iota) > 0$, the no-mimic bound $\bar{e}_{\iota+1}$ for type $a_{\iota+1}$ solves

$$W(k, z, a_\iota, \mathbf{e}(k, z, a_\iota)) - \mathbf{e}(k, z, a_\iota) \quad (57)$$

$$= W(k, z, a_\iota, \bar{e}_{\iota+1}) - \bar{e}_{\iota+1} \frac{W(k, z, a_\iota, \bar{e}_{\iota+1})}{W(k, z, a_{\iota+1}, \bar{e}_{\iota+1})}. \quad (58)$$

Type $a_{\iota+1}$ then chooses

$$\mathbf{e}(k, z, a_{\iota+1}) = \begin{cases} \min\{\mathbf{e}^*(k, z, a_{\iota+1}), \bar{e}_{\iota+1}\}, & W(k, z, a_{\iota+1}, \bar{e}_{\iota+1}) > W(k, z, a_{\iota+1}, 0), \\ 0, & \text{otherwise.} \end{cases} \quad (59)$$

3. If $\mathbf{e}(k, z, a_\iota) = 0$, then $\mathbf{e}(k, z, a) = 0$ for all $a > a_\iota$.

Let $a_{\bar{\iota}}$ denote the highest-quality type that issues positive equity. The belief supporting this allocation is

$$\mathcal{B}(a \mid k, z, e) = \begin{cases} \mathbb{1}_{a=a_1}, & e > \mathbf{e}(k, z, a_2), \\ \mathbb{1}_{a=a_{\iota-1}}, & e \in (\mathbf{e}(k, z, a_\iota), \mathbf{e}(k, z, a_{\iota-1})], \\ \frac{\mathcal{G}(a) \mathbb{1}_{\mathbf{e}(k, z, a)=0}}{\int_{\tilde{a}: \mathbf{e}(k, z, \tilde{a})=0} \mathcal{G}(\tilde{a}) d\tilde{a}}, & e \leq \mathbf{e}(k, z, a_{\bar{\iota}}). \end{cases} \quad (60)$$

Thus, positive on-path issuance identifies the issuing type, off-path issuance between adjacent equilibrium choices is attributed to the type selected by the Divinity Criterion, and zero issuance is priced using the prior conditional on the zero-issuance pool.

Investor break-even under these beliefs implies the price schedule

$$P(k, z, e) = \begin{cases} W(k, z, a_1, e) - e, & e > \mathbf{e}(k, z, a_2), \\ W(k, z, a_{\ell-1}, e) - e, & e \in (\mathbf{e}(k, z, a_{\ell}), \mathbf{e}(k, z, a_{\ell-1})], \\ \frac{\int_{\tilde{a}:\mathbf{e}(k,z,\tilde{a})=0} W(k, z, \tilde{a}, e) \mathcal{G}(\tilde{a}) d\tilde{a}}{\int_{\tilde{a}:\mathbf{e}(k,z,\tilde{a})=0} \mathcal{G}(\tilde{a}) d\tilde{a}} - e, & e \leq \mathbf{e}(k, z, a_{\ell}). \end{cases} \quad (61)$$

Proposition 2 is an application of Guo et al. (2025). Their characterization establishes the global optimality of the recursive issuance policy and shows that the stated off-path beliefs satisfy the Divinity Criterion. We therefore omit the general proof and use the proposition to summarize the allocation, beliefs, and prices required by the quantitative model.

B.4. Reporting cost distribution

To obtain the reporting probability function (28), we let c_{jt} denote the net threshold cost $\kappa_{jt} + N_t^*$, such that (27) implies

$$R_t(k, z) = \Pr(c_{jt} \leq N_t(k, z)) = F_c(N_t(k, z)) \quad (62)$$

where F_c is the CDF of c_{jt} . The second equality uses the assumption (stated above) that κ_{jt} (and thus c_{jt}) is independent of firm fundamentals. We then specify the distribution of c_{jt} directly, as

$$c_{jt} = \begin{cases} 0 & \text{with probability } \lambda_p \\ W_{jt} \sim \text{Weibull}(\lambda_\alpha, \lambda_\xi) & \text{with probability } 1 - \lambda_p \end{cases}. \quad (63)$$

This is a standard zero-inflated Weibull distribution, fully parameterized by the mixture parameter λ_p and the standard Weibull scale and shape parameters $\lambda_\alpha, \lambda_\xi$. This is a general parametric form that allows us to match the quantitative features of observed reporting patterns in our data with a minimal number of parameters. Since the distribution in (63) is specified for c_{jt} , the distribution of raw reporting costs κ_{jt} is a three-parameter zero-inflated

Weibull with shifted location $-N_t^*$.

Note that by using this location-shifted distribution for κ_{jt} , we also do not have to specify the reporting limit r independently of the other parameters $\lambda_p, \lambda_\alpha, \lambda_\xi$. Integrating $R_t(k, z)$ over firms pins down r as a function of those three distribution parameters. The calibration of the media's policy therefore only involves calibrating $\lambda_p, \lambda_\alpha, \lambda_\xi$.

B.5. Aggregate welfare function

Let $\mathcal{M}$ denote a stationary media-reporting policy, and let $\mathcal{F}^\mathcal{M}(k, z)$ denote the associated stationary distribution of firms over capital and productivity. Because capital quality and the reporting-cost shock are i.i.d., the full state distribution is $\mathcal{F}^\mathcal{M}(k, z)\mathcal{G}(a)\mathcal{H}(\kappa)$. Aggregate firm value under policy $\mathcal{M}$ is

$$\bar{V}(\mathcal{M}) \equiv \int V^\mathcal{M}(k, z, a, \mathcal{M}(k, z, a, \kappa)) \mathcal{F}^\mathcal{M}(k, z) \mathcal{G}(a) \mathcal{H}(\kappa) dk dz da d\kappa. \quad (64)$$

The value function $V^\mathcal{M}$ is the present discounted value accruing to a firm's existing shareholders and already incorporates the effects of the reporting policy on current financing and investment decisions, continuation values, and the stationary firm distribution.

To relate (64) to welfare, we take as our criterion the sum of shareholder value, the net surplus of external investors, and forecasters' utility:

$$\mathcal{W}(\mathcal{M}) = \bar{V}(\mathcal{M}) + \Pi^I(\mathcal{M}) + \bar{U}(\mathcal{M}). \quad (65)$$

The net surplus of external investors is zero. For any positive equity issuance e , investors pay e for a claim to the issuing firm's future payouts. The investor break-even condition implies that the expected value of this claim, conditional on investors' information, is also e . This equality holds state by state for reported firms and in expectation across observationally equivalent unreported firms. Hence equity issuance reallocates firm ownership but does not create net investor surplus:

$$\Pi^I(\mathcal{M}) = 0. \quad (66)$$

Forecasters also make no contribution to aggregate welfare under the relative-performance objective in equation (21). Integrating their utilities over the unit mass of forecasters yields

$$\begin{aligned}\bar{U}(\mathcal{M}) &= \int_0^1 U_i(\mathcal{M}) di \\ &= - \int_0^1 \int_0^1 [\text{FE}_{ij} - \overline{\text{FE}}_{-i,j}] dj di = 0,\end{aligned}\tag{67}$$

because one forecaster’s performance relative to the average is offset by the corresponding relative performance of the other forecasters. Aggregate welfare therefore reduces to $\mathcal{W}(\mathcal{M}) = \bar{V}(\mathcal{M})$. Thus, conditional on reporting capacity, maximizing aggregate welfare is equivalent to maximizing aggregate firm value. This motivates the constrained-efficient allocation in Section 4.3, which directs reports toward the firm states with the largest gains $V(k, z, a, 1) - V(k, z, a, 0)$.³³

The one qualification to this analysis is that while firm value precisely captures the welfare of firms, investors, and forecasters, it abstracts away from the cognitive or effort-based reporting costs κ . This is consistent with the misallocation literature (e.g. David et al., 2016; David and Venkateswaran, 2019), in which it is common to evaluate gains from reallocating resources while abstracting from informational costs of achieving such a reallocation. Similarly, studies of welfare in behavioral models often take the same approach (e.g. Gabaix, 2014, 2020; Farhi and Gabaix, 2020).³⁴

As well as consistency with prior literature, there is a practical reason for this abstraction. Firm value is defined via the firm problem, while reporting costs are defined via the outlet problem, and are thus in the same units as forecaster utility (i.e. in units of squared forecast errors, not firm values). The model does not supply an exchange rate between these objects: we have no rigorous way to trade off firm value against reporting cost in squared-forecast-error units.³⁵ Abstracting from the information costs of the allocation, as

³³Note that since production depends only on idiosyncratic capital, and investors break even against a fixed risk-free rate, there are no pecuniary links between firms. There is therefore no price that changes with reporting reallocations, so given value functions and the firm distribution, targeting firms with the largest value gains from reporting is sufficient to maximize welfare.

³⁴See Angeletos and Sastry (2025) for a detailed discussion of this ‘paternalistic’ style of welfare analysis in limited-information models, and their alternatives. Note that our case is not quite the same as those they discuss, as for them information costs are borne by the same agents whose utility forms the planner’s objective. For us, costs are borne by media outlets, while utility comes from firms.

³⁵Indeed, since forecast errors concern *squared* market values, there can be no linear exchange rate between κ and market values, which themselves are tightly linked to firm values through equations (13) and (14).

in previous literature, allows us to avoid making an ad-hoc judgment on how to compare the units of these different variables. This same problem technically appears for forecaster utility as well as reporting costs, but since $\bar{U}(\mathcal{M})$ is precisely zero it is immaterial.³⁶

³⁶Total reporting costs may in principle vary across allocations, because in the baseline economy reporting is based on newsworthiness net of κ , so firms with low κ are more likely to be reported. Our constrained-efficient allocation, which ignores reporting costs, therefore comes with a greater average κ than the baseline.

C. A Simplified Model Illustrating Investment Distortions and Media Allocation

This section presents a simple static framework to illustrate the intuitions behind the key mechanisms in this paper. The framework serves two purposes. First, it shows how asymmetric information in the equity market depresses firms' financing and investment. Second, it shows why the distribution of media coverage across firms matters for the aggregate effects of media reporting. Building on this insight, it characterizes the constrained-efficient allocation of scarce reporting capacity.

C.1. Equity Market Equilibrium Under Asymmetric Information

We first study the equity-market equilibrium without media reporting. Under symmetric information, both types choose the efficient investment level. Under asymmetric information, however, low issuance signals high capital quality because high-quality firms bear a greater dilution cost. The resulting separating equilibrium requires the high-quality firm to restrict equity issuance and therefore to underinvest.

C.1.1. Setup

Consider a two-period economy with no discounting. In period 1, a firm has installed capital k and draws its capital quality from two possible values,

$$a \in \{a_L, a_H\}, \quad a_H > a_L.$$

Capital quality is privately observed by the firm and unobserved by outside investors. Let x denote investment and e denote external equity financing. Capital evolves according to

$$k' = ak + x^\theta, \quad \theta \in (0, 1).$$

The firm has no internal liquidity and therefore finances all investment with equity, so $x = e$. In period 2, output is produced one-for-one from capital,

$$y = k',$$

and distributed to shareholders. Thus, the gross terminal value of a type- a firm that raises e is

$$W(k, a, e) = ak + e^\theta.$$

The first term, ak , is the value of installed capital. The second term, e^θ , is the capital created by new investment.

Let $s(k, e)$ denote the number of new shares investors require in exchange for providing e units of funds to an unreported firm with installed capital k . Because existing shares are normalized to one, existing shareholders retain the fraction

$$\frac{1}{1 + s(k, e)}$$

of the firm. Given the share schedule, the payoff to existing shareholders of a type- a firm that issues e is

$$\Pi(e; k, a) = \frac{1}{1 + s(k, e)} W(k, a, e).$$

This setup maps directly into the main model after suppressing productivity and time subscripts, setting depreciation to zero, and normalizing period-2 output to capital.

C.1.2. Symmetric-information benchmark

Under symmetric information, investors observe a , so they price equity fairly. Let $s^{\text{SI}}(k, a, e)$ denote the corresponding share requirement. Investors' break-even condition is

$$\frac{s^{\text{SI}}(k, a, e)}{1 + s^{\text{SI}}(k, a, e)} W(k, a, e) = e.$$

The value retained by existing shareholders is therefore

$$W(k, a, e) - e = ak + e^\theta - e.$$

Because $x = e$, the firm's symmetric-information issuance and investment choices coincide:

$$e^* = x^* = \arg \max_{e \geq 0} \{ak + e^\theta - e\}.$$

The first-order condition is

$$\theta(e^*)^{\theta-1} = 1,$$

so

$$e^* = x^* = \theta^{1/(1-\theta)}.$$

The symmetric-information allocation is independent of a because capital quality affects the value of installed capital but not the marginal return to new investment. This common efficient investment level is not sustainable when capital quality is private information.

C.1.3. Asymmetric-information allocation

Differences in equity-issuance incentives We first compare the two types' incentives to issue equity. A low-quality firm has a stronger incentive to issue because it gives up claims on a less valuable installed capital stock. To establish this single-crossing property, consider two issuance levels $e > e'$ and suppose that higher issuance requires weakly more dilution:

$$s(k, e) \geq s(k, e').$$

The payoff difference between choosing e and e' for type a is

$$\Delta\Pi(a) = \frac{1}{1 + s(k, e)} (ak + e^\theta) - \frac{1}{1 + s(k, e')} (ak + (e')^\theta).$$

Differentiating with respect to a gives

$$\frac{\partial \Delta\Pi(a)}{\partial a} = k \left[\frac{1}{1 + s(k, e)} - \frac{1}{1 + s(k, e')} \right] \leq 0.$$

Thus, as capital quality rises, the relative attractiveness of higher issuance falls. Issuing more equity raises investment but also dilutes claims on installed capital. Because that installed capital is more valuable for the high type, dilution is more costly for the high type, and the low type consequently has a stronger incentive to issue.

This *single-crossing property* guides investors' inference about a firm's type: low issuance signals high quality. It also implies that no positive pooling equilibrium survives the intuitive criterion (Cho and Kreps, 1987).³⁷ The intuition is simple: for any candidate pooling allocation, there is a lower issuance level that

- the high-quality firm strictly prefers when investors interpret the deviation as a high-quality signal, but
- the low-quality firm does not strictly prefer even under this most favorable interpretation.

No-pooling equilibrium We next formalize this argument by showing that every positive pooling allocation admits a lower-issuance deviation that attracts the high type but not the low type.

Proposition 3 (Nonexistence of positive pooling). *There is no equilibrium satisfying the intuitive criterion in which both types issue the same strictly positive amount $e^{\text{pool}} > 0$.*

Proof. Suppose, toward a contradiction, that both types pool at $e^{\text{pool}} > 0$. Let $\bar{a} \in (a_L, a_H)$ denote investors' posterior mean quality at the pooling allocation. The value investors assign to a pooling firm is

$$W^{\text{pool}} = \bar{a}k + (e^{\text{pool}})^\theta.$$

Investor break-even implies that existing shareholders of a type- a firm receive

$$\Pi_a^{\text{pool}} = W(k, a, e^{\text{pool}}) \left(1 - \frac{e^{\text{pool}}}{W^{\text{pool}}} \right).$$

We construct a lower-issuance deviation that the low type does not strictly prefer even under the most favorable belief. Consider $\hat{e} < e^{\text{pool}}$ and suppose investors interpret the deviation as coming from the high type. Under fair high-type pricing, the deviating payoff to type a is

$$\Pi_a^H(\hat{e}) = W(k, a, \hat{e}) \left(1 - \frac{\hat{e}}{W(k, a_H, \hat{e})} \right).$$

³⁷With two types, the intuitive criterion coincides with the D1 criterion of Banks and Sobel (1987), which is the refinement used in the general model.

At $\hat{e} = e^{\text{pool}}$, the low type strictly prefers high-type pricing to pooling because $W(k, a_H, e^{\text{pool}}) > W^{\text{pool}}$, so high-type pricing requires less dilution. At $\hat{e} = 0$, its deviation payoff is $\Pi_{a_L}^H(0) = a_L k$, which cannot exceed its pooling payoff.³⁸ By continuity, there is therefore an $\hat{e} \in [0, e^{\text{pool}})$ that leaves the low type indifferent:

$$\Pi_{a_L}^H(\hat{e}) = \Pi_{a_L}^{\text{pool}}.$$

We next show that the same deviation strictly benefits the high type. The high type's gain relative to the low type's gain is

$$\begin{aligned} & [\Pi_{a_H}^H(\hat{e}) - \Pi_{a_H}^{\text{pool}}] - [\Pi_{a_L}^H(\hat{e}) - \Pi_{a_L}^{\text{pool}}] \\ &= (a_H - a_L)k \left[\frac{e^{\text{pool}}}{W^{\text{pool}}} - \frac{\hat{e}}{W(k, a_H, \hat{e})} \right] > 0. \end{aligned}$$

The inequality follows because $e/W(k, a_H, e)$ is strictly increasing in e , $\hat{e} < e^{\text{pool}}$, and $W(k, a_H, e^{\text{pool}}) > W^{\text{pool}}$. Thus, the deviation strictly benefits the high type while leaving the low type indifferent even under the most favorable possible belief. The intuitive criterion therefore requires investors to interpret the deviation as coming from the high type, under which belief the high type strictly prefers to deviate. This contradicts the candidate pooling equilibrium. $\square$

Separating equilibrium allocation Having ruled out positive pooling, consider the least-cost separating allocation. The low type chooses the symmetric-information issuance level. The high type then chooses the largest lower issuance level that deters the low type from mimicking, thereby minimizing the real cost of separation.

Proposition 4 (Least-cost separating allocation). *In the least-cost separating allocation, the low type issues the symmetric-information amount,*

$$e_L = e^*,$$

whereas the high type issues $e_H < e^$. The value of e_H is the largest issuance level below e^**

³⁸Otherwise, e^{pool} would not be an equilibrium allocation in the first place.

that satisfies the low type's no-mimicking condition

$$W(k, a_L, e^*) - e^* = W(k, a_L, e_H) \left(1 - \frac{e_H}{W(k, a_H, e_H)} \right). \quad (68)$$

Proof. Because issuance reveals type in a separating allocation, investors price both on-path choices fairly. Let s_i denote the new shares issued by type $i \in \{L, H\}$. Investor break-even requires

$$\frac{s_i}{1 + s_i} W(k, a_i, e_i) = e_i, \quad i \in \{L, H\}.$$

In the least-cost separating allocation, the low type faces its symmetric-information problem and chooses $e_L = e^*$. Given single crossing, the high type does not want to imitate this higher-issuance allocation, which entails greater dilution of its more valuable installed capital stock.

If the high type also chose e^* , the low type would mimic it to obtain high-type pricing. Separation consequently requires the high type to issue less. In the least-cost allocation, the low type's no-mimicking condition binds:

$$\begin{aligned} W(k, a_L, e^*) - e^* &= \frac{1}{1 + s_H} W(k, a_L, e_H) \\ &= W(k, a_L, e_H) \left(1 - \frac{e_H}{W(k, a_H, e_H)} \right). \end{aligned} \quad (69)$$

Equation (68) selects the largest $e_H < e^*$ that deters the low type. Single crossing implies that, when the low type is indifferent between the two allocations, the high type strictly prefers its lower-issuance allocation. Hence the high type's incentive constraint also holds. Any smaller e_H would deter imitation but would impose an unnecessarily large signaling cost. $\square$

Firm-level distortion under asymmetric information The separating signal creates a real distortion because equity issuance finances investment one-for-one. The low type remains at the symmetric-information allocation, whereas the high type underinvests:

$$x_L = e_L = e^* = x^*, \quad x_H = e_H < e^* = x^*.$$

As a result, the high type enters period 2 with

$$k'_H = a_H k + e_H^\theta < a_H k + (e^*)^\theta,$$

which is below its capital under symmetric information. Thus, asymmetric information leaves the low type's investment unchanged but depresses the high type's investment. The high type underinvests not because it has a worse investment opportunity, but because restricting equity issuance is the costly signal that prevents imitation by the low type. This is how asymmetric information depresses firm-level investment in the absence of media reporting.

Extension to more than two types The same logic extends recursively to any finite ordered set $a_1 < a_2 < \dots < a_N$. The lowest-quality type chooses the symmetric-information issuance level. Moving upward through the quality distribution, each type n chooses the largest issuance level that prevents the adjacent lower type from imitating it,

$$W(k, a_{n-1}, e_{n-1}) - e_{n-1} = W(k, a_{n-1}, e_n) \left(1 - \frac{e_n}{W(k, a_n, e_n)} \right), \quad (70)$$

subject to not exceeding the symmetric-information level. Under single crossing, these adjacent constraints characterize the least-cost separating allocation. Higher-quality firms consequently issue and invest less because they must distinguish themselves from progressively lower-quality firms.

C.2. Aggregate Relevance of Media Reporting

We now introduce media reporting and use the toy model to establish three results. First, reporting substitutes for costly signaling by revealing a firm's quality directly to investors. Second, its real effects are heterogeneous: reporting raises investment only when asymmetric information distorts a firm's external financing. Third, when reporting capacity is scarce, its aggregate value depends on which firms are covered, not only on the total number of reports.

C.2.1. Media reporting as a substitute for costly signaling

Let $m \in \{0, 1\}$ indicate whether a firm is reported. The media observes the firm's capital quality a , and a report reveals this type to investors before the firm issues equity. We treat

this as a direct information-revelation device: a report certifies the selected firm's type, while the absence of a report carries no type-specific information.³⁹

When $m = 0$, the firm faces the asymmetric-information equilibrium characterized above. The low type issues and invests at the symmetric-information level, whereas the high type restricts issuance to separate itself:

$$e_L(0) = x_L(0) = e^* = x^*, \quad e_H(0) = x_H(0) < e^* = x^*.$$

When $m = 1$, investors observe the firm's type and price its equity with perfect information. Both types can therefore issue and invest at the symmetric-information level:

$$e_L(1) = x_L(1) = e_H(1) = x_H(1) = e^* = x^*.$$

Media reporting does not change the firm's technology or provide it with funds. Instead, it replaces the costly way in which the firm would otherwise communicate with investors. Once the high-quality firm's type is publicly observable, it no longer needs to depress issuance and investment to distinguish itself from the low type.

For any firm j , define the individual treatment effect of reporting on investment in levels as

$$\text{TE}_j^x \equiv x_j(1) - x_j(0).$$

The two types therefore have different treatment effects:

$$\text{TE}_L^x = 0, \quad \text{TE}_H^x = x^* - x_H(0) > 0.$$

Reporting has a real effect only when the firm's unreported allocation is distorted. It leaves the low type's investment unchanged but raises the high type's investment by eliminating the need to signal through under-issuance.

³⁹To focus on the aggregate effects of media reporting, we abstract from the endogenous selection of media reporting in this section and assume being reported as an exogenous outcome.

C.2.2. Heterogeneous treatment effects

The effect of reporting also depends on whether the firm must access the equity market. Private information can distort investment only when desired investment cannot be financed internally. To make this dependence explicit, let $g \in \{C, U\}$ indicate whether a firm is financially constrained or unconstrained, and let $\tau \in \{L, H\}$ denote its quality type. Constrained firms have no internal liquidity and must finance investment by issuing information-sensitive equity. Unconstrained firms begin with internal liquidity ℓ satisfying

$$\ell \geq x^*.$$

They can therefore finance the symmetric-information investment level internally and do not need to issue equity. For either quality type and either reporting status,

$$e_{\tau,U}(m) = 0, \quad x_{\tau,U}(m) = x^*.$$

Private information about capital quality is irrelevant for these firms' investment because they do not need to access the equity market.

For constrained firms, by contrast, equity issuance equals investment expenditure. The separating equilibrium implies

$$x_{L,C}(0) = e^* = x^*, \quad x_{H,C}(0) = e_H(0) < x^*.$$

Reporting eliminates the signaling distortion, so both types choose x^* when reported. The resulting investment effects can be summarized as

Firm state	Investment without reporting	Investment effect of reporting
Constrained high type	$x_{H,C}(0) < x^*$	$\text{TE}_{H,C}^x = x^* - x_{H,C}(0) > 0$
Constrained low type	$x_{L,C}(0) = x^*$	$\text{TE}_{L,C}^x = 0$
Either unconstrained type	$x_{\tau,U}(0) = x^*$	$\text{TE}_{\tau,U}^x = 0$

Thus, high quality alone does not generate a treatment effect. Reporting changes investment only when high quality is combined with dependence on information-sensitive external finance. In the binary model, the constrained high-quality firm is therefore the only respon-

sive state.

C.2.3. Constrained-efficient media allocation

With fixed reporting capacity, aggregate gains depend on which firms receive reports. A constrained planner takes both total media capacity and the equity-market equilibrium as given. It cannot eliminate asymmetric information for every firm or provide firms with additional financing; it can only choose which firms' types the media reveals.

We hold investors' beliefs about unreported firms fixed when reallocating reports. This partial-equilibrium information experiment isolates the direct value of revealing a selected firm's type and abstracts from inference based on the assignment rule itself. Firms that are not reported therefore continue to face the asymmetric-information equilibrium.

To evaluate alternative reporting allocations, define net resource surplus as

$$\mathcal{S}(x; a, k) \equiv ak + x^\theta - x.$$

This object is distinct from the post-issuance firm value W and the value to existing shareholders V used in the main model. The individual surplus gain from reporting firm j is

$$\Delta\mathcal{S}_j = \mathcal{S}(x_j(1); a_j, k_j) - \mathcal{S}(x_j(0); a_j, k_j).$$

For a finite economy with N firms, let $m_j \in \{0, 1\}$ indicate whether firm j receives a report, and collect these reporting choices in $\mathbf{m} = (m_1, \dots, m_N)$. As in the main model, let $r \in (0, 1)$ denote reporting capacity as a fraction of firms, with rN assumed to be an integer. The aggregate surplus gain from an allocation $\mathbf{m}$ is

$$\Delta\mathcal{S}(\mathbf{m}) = \sum_{j=1}^N m_j \Delta\mathcal{S}_j, \quad \sum_{j=1}^N m_j \leq rN.$$

The constrained-efficient allocation therefore solves

$$\mathbf{m}^* \in \arg \max_{\mathbf{m} \in \{0,1\}^N: \sum_{j=1}^N m_j \leq rN} \sum_{j=1}^N m_j \Delta\mathcal{S}_j,$$

and reports the firms with the largest individual surplus gains. In the binary toy model,

$$\Delta\mathcal{S}_{H,C} > 0, \quad \Delta\mathcal{S}_{L,C} = \Delta\mathcal{S}_{H,U} = \Delta\mathcal{S}_{L,U} = 0.$$

Hence, if the media can report only one firm, the planner reports the constrained high-quality firm.

Relationship between Aggregate Welfare and Real Outcomes From a welfare perspective, the appropriate criterion for evaluating a reporting allocation is its effect on aggregate net resource surplus, $\Delta\mathcal{S}(\mathbf{m})$. Investment, capital, and output are real outcomes rather than primitive welfare measures. In this toy model, however, they all rank reporting allocations in the same way. For a responsive constrained firm,

$$\begin{aligned} \Delta\mathcal{S}_j &= [(x^*)^\theta - x^*] - [x_j(0)^\theta - x_j(0)], \\ \text{TE}_j^x &= x^* - x_j(0), \\ \text{TE}_j^{k'} &= (x^*)^\theta - x_j(0)^\theta, \quad \text{TE}_j^y = \text{TE}_j^{k'}. \end{aligned}$$

The relevant unreported firms satisfy $0 \leq x_j(0) \leq x^*$, and $x^\theta - x$ is increasing over this interval. Each of these responses is therefore larger when the firm's initial investment shortfall is larger. Holding reporting capacity fixed, ranking firms by their effects on investment, next-period capital, or output consequently yields the same reporting allocation as ranking them by their surplus gains.

This equivalence need not hold exactly in the general quantitative model, where firms differ in their technologies and states and reporting can have dynamic effects. Nevertheless, investment responses affect future capital and output and are therefore tightly connected to the underlying surplus gains. They are also more directly observable in the data and easier to measure transparently in the quantitative model. We therefore treat net resource surplus as the welfare criterion for constrained efficiency, while using the responses of investment, capital, and output as informative measures of the real effects generated by alternative reporting allocations.

D. Additional Details for the Quantitative Analysis

D.1. Definitions and measurements of target moments

This section describes the construction of target moments reported in Table 4, based on firms in the Compustat sample from 1990 to 2021.

1. Cash flow

- (a) *Operating cash flow*: defined as net cash flow, measured by the operating activities (`oancfy` in the first fiscal quarter and changes in `oancfy` for the second to fourth fiscal quarters)
- (b) *Operating cash flow rate*: defined as operating cash flow normalized by capital stock (measured by lagged total assets `atq`).

2. Investment

- (a) *Investment*: defined as the capital expenditure (`capxy` in the first fiscal quarter and changes in `capxy` for the second to fourth fiscal quarters).
- (b) *Investment rate*: defined as the investment normalized by capital stock
- (c) *Measurement of investment rate dispersion*: Following Sterk et al. (2021), we measure the cross-sectional variation in firms' investment rates by first estimating

$$y_{jt} = \gamma_j + \gamma_{st} + \epsilon_{jt},$$

where y_{jt} denotes investment rate of firm j in quarter t , and $\{\gamma_j, \gamma_{st}\}$ denote firm and sector-by-quarter fixed effects; and then computing the dispersion of investment rate as the standard deviation of ϵ_{jt} .

3. Life-cycle dynamics

- (a) *Age*: the number of years since the CRSP listing.
- (b) *Young and mature firms*: young firms are defined as firms aged 5 years or less; mature firms are defined as firms aged more than 25 years.

- (c) *Growth rate*: the log-difference of the total assets (`atq`).
- (d) *Measurement of age group difference*: To compute the target moment for the life-cycle differences between mature (age > 25) and young (age ≤ 5) firms, we use the coefficient β_5 estimated from

$$y_{jt} = \gamma_j + \gamma_{st} + \beta_1 \cdot \mathbf{1}[5 < Age_{jt} \leq 10] + \beta_2 \cdot \mathbf{1}[10 < Age_{jt} \leq 15] \\ + \beta_3 \cdot \mathbf{1}[15 < Age_{jt} \leq 20] + \beta_4 \cdot \mathbf{1}[20 < Age_{jt} \leq 25] + \beta_5 \cdot \mathbf{1}[Age_{jt} > 25] + \epsilon_{jt},$$

where y_{jt} denotes growth rate or idiosyncratic TFP of firm j in quarter t , Age_{jt} denotes the age of firm j in quarter t , and $\{\gamma_j, \gamma_{st}\}$ denote firm and sector-by-quarter fixed effects.

4. Equity financing

- (a) *Equity issuance*: defined as the sale of common and preferred stock (`sstky` in the first fiscal quarter and changes in the `sstky` for the second to fourth fiscal quarters). Following [McKeon \(2015\)](#), we classify equity issuances that are smaller than 3% of a firm’s market capitalization as zero issuance.
- (b) *Issuance fee ratio*: defined as the fixed issuance cost normalized by the issuance quantity. We use the reported mean in Table 4 of [Lee and Masulis \(2009\)](#) as our target moment.
- (c) *Selling concession ratio*: defined as the log difference between a firm’s stock price before announcing its issuance decision and its post-issuance stock price. We use the reported mean in Table 4 of [Lee and Masulis \(2009\)](#) as our target moment.

5. Media reporting

- (a) *Media-coverage distribution*: measured as the share of all reported firms belonging to each of five market-value groups—the bottom 20%, 20–40%, 40–60%, 60–80%, and top 20%. For each quarter, we calculate the share of reported firms in each group and then average these shares over time.

D.2. Calibration detail for parameters relating to firm dynamics

1. *Cash flow level and dynamics* The aggregate productivity, Z , corresponds to the steady-state level of the average operating cash flow rate. Since the aggregate productivity determines firms' average level of internal financing, we calibrate it to match the average operating cash flow rate in the data. The idiosyncratic productivity shock, z , is the source of the cash flow risk firms face, which shapes the firms' ex-post heterogeneity and the precautionary motives in their investment decisions. We calibrate the idiosyncratic productivity's persistence and volatility to match the empirical estimates from [İmrohoroglu and Tüzel \(2014\)](#).
2. *Investment technology and capital accumulation* We calibrate the depreciation rate, δ , to match the average investment rate at which firms replenish their depreciated capital and grow. The return-to-scale of investment technology, θ , governs the sensitivity of firms' investments to their capital profitability. We set $\theta = 0.81$ to target the cross-sectional standard deviation of firms' investment rates.
3. *Life-cycle dynamics* The ex-post heterogeneity among firms is shaped by the dynamics of the firms' idiosyncratic productivity and the distribution of the entrants. Two parameters of the entrant distribution, $\{\mu_{\log z}^{\text{entrant}}, \mu_{\log k}^{\text{entrant}}\}$, govern the variation across firms in different age groups.⁴⁰ Therefore, we calibrate the entrant distribution parameters to match the differences in the growth rate and the measured idiosyncratic TFP between young firms ($\text{age} \leq 5$) and mature firms ($\text{age} > 25$).

D.3. Numerical solution

D.3.1. Overview of the solution method

We solve the stationary equilibrium using a nested fixed-point algorithm. The nesting reflects the feedback between firms' financing decisions and media outlets' reporting decisions.

In the inner layer, we take the reporting probability function $R(k, z)$ as given. This function determines the probability that investors observe a firm's capital quality and there-

⁴⁰We parameterize the entrant distribution $\mathcal{F}^{\text{entrant}}(z, k)$ as a mixture of two independent normal distributions of the firms' log productivities and log sizes: $\log z \sim \mathcal{N}(\mu_{\log z}^{\text{entrant}}, 0.01)$ and $\log k \sim \mathcal{N}(\mu_{\log k}^{\text{entrant}}, 0.01)$. The standard deviation is set to be 0.01, which is a sufficiently small value to smooth the distribution without affecting the results.

fore whether the firm receives full-information equity pricing. Conditional on $R(k, z)$, we jointly solve the firms' value and policy functions, the equity-market equilibrium, and the stationary distribution of firms.

In the outer layer, we update the reporting probability. The market values obtained from the inner layer determine the uncertainty about each firm's value and hence its newsworthiness. Newsworthiness is then mapped into an updated reporting probability. Because this updated function generally differs from the initial conjecture, we iterate between the inner firm equilibrium and the outer media problem until the reporting probability converges.

Economically, the fixed point operates through two related channels. First, media coverage relaxes the information friction in the equity market and changes firms' equity issuance, investment, next-period capital, and market values. The resulting policy functions determine the stationary distribution of firms. Second, the dispersion in market values determines newsworthiness and therefore the distribution of media coverage. The equilibrium reporting function is the one that is consistent with the firm behavior and market values it induces.

D.3.2. Detailed algorithm

Equilibrium objects For a given parameterization, the numerical algorithm solves the following objects:

- *Firm value functions:*
 - the post-issuance value function, $W(k, z, a, e)$; and
 - the value accruing to existing shareholders, $V(k, z, a, m)$, where $m \in \{0, 1\}$.
- *Firm policy functions:*
 - equity issuance, $\mathbf{e}(k, z, a, m)$;
 - investment, $\mathbf{x}(k, z, a, m)$;
 - dividend payments, $\mathbf{div}(k, z, a, m)$; and
 - next-period capital, $\mathbf{k}'(k, z, a, m)$.
- *Equity-market objects:*

- investors’ posterior beliefs about the capital quality of unreported firms, $\mathcal{B}(a \mid k, z, e)$;
 - the equity issuance price function, $P(k, z, a, m, e)$; and
 - the stock market value, $MV(k, z, a, m)$.
- *Firm distribution:* the stationary distribution over the persistent firm states, $\mathcal{F}(k, z)$. Capital quality does not enter this stationary distribution because a is independently drawn from $\mathcal{G}(a)$ in every period.
 - *Media-reporting objects:*
 - the newsworthiness function, $N(k, z)$; and
 - the reporting probability function, $R(k, z)$.

Both functions depend only on the publicly observable states (k, z) in the baseline model.

Discretization and function approximation The numerical approximation has four components:

1. *Productivity grid.* We approximate the AR(1) process for log idiosyncratic productivity using the Rouwenhorst method. This produces five productivity states and the transition matrix $\Gamma^z(z' \mid z)$. We exponentiate the productivity states and normalize them so that their stationary mean equals one.
2. *Capital-quality grid.* We approximate the capital-quality distribution $\mathcal{G}(a)$ using the ordered finite grid

$$\{a_1 < a_2 < \cdots < a_{N_a}\}, \quad N_a = 20. \quad (71)$$

The grid points are evenly spaced in capital-quality levels over a truncated approximation interval for $\mathcal{G}(a)$, with lower and upper bounds corresponding to three standard deviations below and above the mean of log capital quality, and the probability assigned to each point equals the mass of its surrounding interval. This construction preserves the ordering used in the separating equity-market recursion.

3. *Value-function approximation.* We approximate the firm value functions using tensor-product splines. Productivity and capital quality are evaluated on their discrete grids. The continuous dimensions include capital and the resources available for investment and dividend payments after the equity issuance decision. We concentrate grid points near the lower bounds of these state spaces, where financing and investment policies change more rapidly, and increase the distance between adjacent points geometrically toward the upper bounds. We use 29 spline breakpoints for capital, which generate 30 nodes under the quadratic spline basis.
4. *Stationary-distribution grid.* We use a separate, finer grid to compute the stationary distribution. This grid is approximately uniform in log capital and includes the discrete productivity states. The log-capital dimension contains 50 grid points, so together with the five productivity states, the distribution is computed over 250 (k, z) grid points. When next-period capital falls between two grid points, we distribute its probability mass between the adjacent points using linear interpolation.

Nested stationary-equilibrium algorithm Let ℓ index the outer iteration. We solve the stationary equilibrium as follows.

1. *Initialize the equilibrium.* We first solve the model under symmetric information. The resulting value functions, policy functions, and stationary distribution provide initial guesses for the asymmetric-information model. We also construct an initial conjecture for the reporting probability function,

$$R^{(0)}(k, z). \tag{72}$$

2. *Inner layer: solve the stationary firm equilibrium.* At outer iteration ℓ , take $R^{(\ell)}(k, z)$ as given and solve the following inner fixed point.

- (a) *Post-issuance firm problem.* Given a conjecture for the continuation value, solve $W^{(\ell)}(k, z, a, e)$. This problem determines investment, dividends, and next-period capital conditional on the firm's equity issuance and capital quality.

(b) *Full-information issuance benchmark.* For every capital-quality type a_ι , compute

$$\mathbf{e}^{*,(\ell)}(k, z, a_\iota) \in \arg \max_{e \geq 0} \{W^{(\ell)}(k, z, a_\iota, e) - e\}. \quad (73)$$

When $m = 1$, investors observe capital quality and price equity using full information. Hence,

$$\mathbf{e}^{(\ell)}(k, z, a_\iota, 1) = \mathbf{e}^{*,(\ell)}(k, z, a_\iota), \quad (74)$$

$$V^{(\ell)}(k, z, a_\iota, 1) = \max_{e \geq 0} \{W^{(\ell)}(k, z, a_\iota, e) - e\}. \quad (75)$$

(c) *Separating equilibrium for unreported firms.* When $m = 0$, investors infer capital quality from equity issuance. For each pair of publicly observable states (k, z) , proceed recursively along the ordered grid $a_1 < \dots < a_{N_a}$.

The lowest-quality type faces no lower-quality imitator and therefore chooses its full-information issuance,

$$\mathbf{e}^{(\ell)}(k, z, a_1, 0) = \mathbf{e}^{*,(\ell)}(k, z, a_1). \quad (76)$$

Now suppose the equilibrium issuance of type a_ι is positive, and write it as

$$e_\iota^{(\ell)} \equiv \mathbf{e}^{(\ell)}(k, z, a_\iota, 0). \quad (77)$$

The upper bound $\bar{e}_{\iota+1}^{(\ell)}$ on the issuance of type $a_{\iota+1}$ is determined by the adjacent lower type's binding no-mimicking condition:

$$\begin{aligned} & W^{(\ell)}(k, z, a_\iota, e_\iota^{(\ell)}) - e_\iota^{(\ell)} \\ &= W^{(\ell)}(k, z, a_\iota, \bar{e}_{\iota+1}^{(\ell)}) \left[1 - \frac{\bar{e}_{\iota+1}^{(\ell)}}{W^{(\ell)}(k, z, a_{\iota+1}, \bar{e}_{\iota+1}^{(\ell)})} \right]. \end{aligned} \quad (78)$$

The left side is type a_ι 's payoff from its own allocation. The right side is its payoff from imitating type $a_{\iota+1}$ and receiving type- $a_{\iota+1}$ pricing. The candidate issuance

of the next type is therefore

$$\tilde{e}_{i+1}^{(\ell)} = \min \left\{ \mathbf{e}^{*,(\ell)}(k, z, a_{i+1}), \bar{e}_{i+1}^{(\ell)} \right\}. \quad (79)$$

Type a_{i+1} chooses this candidate if it yields a higher payoff than zero issuance; otherwise, it chooses zero. Continue recursively until all types have been assigned an issuance choice. Once one type chooses zero issuance, all higher-quality types also choose zero.

Each positive on-equilibrium-path issuance identifies the issuing type. Investors' beliefs are therefore degenerate at that type, and investor break-even implies

$$P^{(\ell)}(k, z, e_i^{(\ell)}) = W^{(\ell)}(k, z, a_i, e_i^{(\ell)}) - e_i^{(\ell)}. \quad (80)$$

If several types choose zero issuance, beliefs at $e = 0$ condition on the set of zero-issuance types. These beliefs and prices determine $V^{(\ell)}(k, z, a, 0)$ for every unreported type.

- (d) *Complete the firm policies and market values.* The post-issuance problem supplies $\mathbf{x}^{(\ell)}$, $\mathbf{div}^{(\ell)}$, and $\mathbf{k}'^{(\ell)}$ at the equilibrium issuance choice. Combining these policies with the reported and unreported equity-market outcomes determines $V^{(\ell)}(k, z, a, m)$ and $MV^{(\ell)}(k, z, a, m)$.
- (e) *Update the continuation value.* Combine the reported and unreported outcomes using $R^{(\ell)}(k, z)$. Integrate over next-period productivity, capital quality, firm exit, and next-period media reporting. Repeat the preceding inner steps until the value and policy functions converge for the current reporting probability.
- (f) *Compute the stationary distribution.* Using the converged policies, construct the transition matrix for (k, z) . This transition matrix combines the capital choices of reported and unreported firms, the reporting probability $R^{(\ell)}(k, z)$, the capital-quality distribution $\mathcal{G}(a)$, the productivity transition matrix $\Gamma^z(z' | z)$, and firm exit and entry. Solve the resulting law of motion to obtain $\mathcal{F}^{(\ell)}(k, z)$.

3. *Outer layer: update newsworthiness and reporting probabilities.* Given the market

values from the inner layer, update newsworthiness according to

$$N^{(\ell+1)}(k, z) = \text{Var}_{a \sim \mathcal{G}} \left[\text{MV}^{(\ell)}(k, z, a, 1) \mid k, z \right]. \quad (81)$$

The parameterized reporting rule then gives

$$R^{(\ell+1)}(k, z) = \lambda_p + (1 - \lambda_p) \left[1 - \exp \left\{ - \left(\frac{N^{(\ell+1)}(k, z)}{\lambda_\alpha} \right)^{\lambda_\xi} \right\} \right]. \quad (82)$$

The stationary distribution supplies the cross-sectional weights for the implied aggregate reporting rate,

$$\bar{r}^{(\ell+1)} = \int R^{(\ell+1)}(k, z) \mathcal{F}^{(\ell)}(k, z) dk dz. \quad (83)$$

Under the parameterized rule used in the quantitative model, the reporting-rule parameters discipline this aggregate reporting rate in the calibration; the outer iteration does not solve an additional reporting threshold.

4. *Check convergence.* Compare the updated reporting probability with the previous conjecture:

$$\sup_{k, z} |R^{(\ell+1)}(k, z) - R^{(\ell)}(k, z)|. \quad (84)$$

We also check changes in the value functions and stationary distribution across outer iterations. If all changes are below their respective tolerances, the algorithm stops. Otherwise, set

$$R^{(\ell)}(k, z) \leftarrow R^{(\ell+1)}(k, z), \quad (85)$$

increase ℓ by one, and return to the inner layer.

The converged objects constitute the stationary equilibrium.

D.4. Additional quantitative results

D.4.1. Rationale for targeting the distribution of reported firms

Our data cover only three newspapers. The reporting probability measured in our sample therefore understates the probability that a firm would be reported in the broader newspaper universe, which contains far more than three newspapers. We therefore target the distribution of reported firms across market-value groups. Specifically, for each quarter, we calculate each group's share of all reported firms and then average these quarterly shares over time.

To see why this choice is robust to the number of newspapers, let $\bar{p}_{j,t}$ denote the probability that firm j is reported by one newspaper in quarter t . If our sample contains n newspapers, let $p_{j,n,t}$ denote the probability that firm j is reported by at least one newspaper in our sample. Assuming that reporting events are independent across newspapers,

$$p_{j,n,t} = 1 - (1 - \bar{p}_{j,t})^n.$$

When the per-newspaper reporting probabilities are small, this expression satisfies

$$p_{j,n,t} = 1 - (1 - \bar{p}_{j,t})^n \approx n\bar{p}_{j,t}.$$

If G denotes a firm group, its expected share of reported firms in quarter t is therefore approximately

$$s_{G,n,t} \equiv \frac{\sum_{j \in G} p_{j,n,t}}{\sum_j p_{j,n,t}} \approx \frac{n \sum_{j \in G} \bar{p}_{j,t}}{n \sum_j \bar{p}_{j,t}} = \frac{\sum_{j \in G} \bar{p}_{j,t}}{\sum_j \bar{p}_{j,t}}.$$

The factor n cancels between the numerator and denominator, so the quarterly group share is approximately independent of the number of newspapers. We therefore use the distribution of reported firms across market-value groups, constructed by averaging these quarterly shares over time, as the target moments in our calibration.

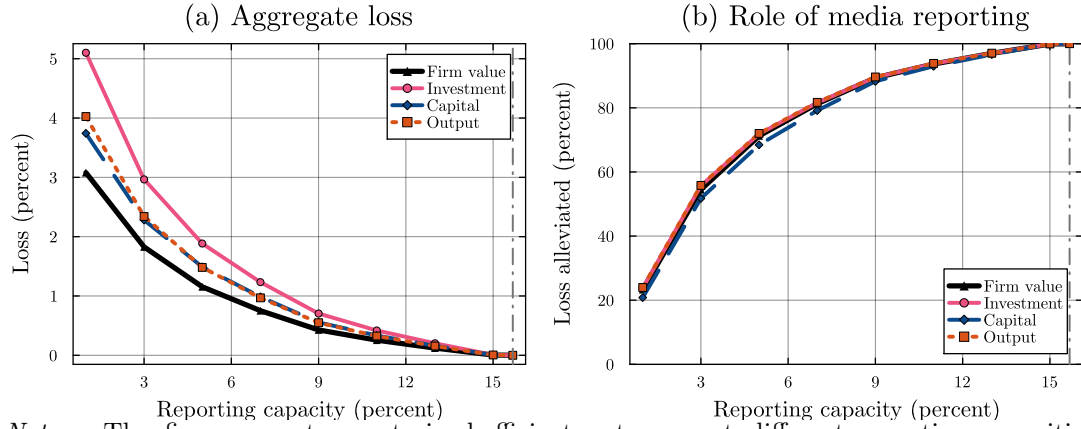
D.4.2. The aggregate role of media under alternative reporting capacities

We conduct a series of counterfactual analyses of the aggregate role of media under constrained-efficient allocations with varying reporting capacity. At each capacity level, reports are allocated to the firms with the largest gains in firm value from coverage. This exercise traces how reporting capacity affects the aggregate role of media when reports are allocated efficiently.

Figure D.1 reports the results. Panel (a) shows the remaining aggregate loss relative to the symmetric-information benchmark, while panel (b) shows the fraction of the corresponding no-media loss eliminated by reporting. Even at a reporting capacity of 1%, the constrained-efficient allocation already eliminates roughly one-fifth of the aggregate loss. As capacity increases, the role of media becomes progressively larger, and the remaining aggregate loss falls essentially to zero once total capacity approaches the baseline capacity level marked by the vertical line.

The strong effect at low capacity results from two mechanisms. First, selection: the constrained-efficient allocation directs reports to firms with the largest gains from coverage. Second, firm dynamics: these firms tend to rely most heavily on external finance and to be smaller and younger. By relaxing their financing constraints, reporting allows these firms to grow faster, thereby reshaping firm life-cycle dynamics and the resulting firm distribution. This highlights that the aggregate effect of media operates not only through the direct firm-level effects of reporting, but also through the resulting changes in firm dynamics and the firm distribution.

Figure D.1: Constrained-efficient outcomes across reporting capacities



Notes: The figure reports constrained-efficient outcomes at different reporting capacities. Panel (a) shows aggregate loss, defined as the percentage shortfall relative to the symmetric-information benchmark. Panel (b) shows, for each outcome, the fraction of the corresponding no-media loss eliminated by reporting. The horizontal axis measures total reporting capacity as a percentage of firms. The lines show firm value, investment, capital, and output; the vertical line marks baseline reporting capacity in the calibrated economy.

E. A Model with Investor-Led Media Demand

In this section, we provide an alternative microfoundation for the media’s reporting policy. We incorporate a media sector in a variant of the static [Grossman and Stiglitz \(1980\)](#) model. Unlike our main quantitative model, noise traders prevent the perfect aggregation of information in asset prices. This causes investors to value information coming from the media. To solve this model, we abstract from the firm block of our quantitative model. Instead, a firm’s equity generates a payoff that is independent of media decisions but initially unknown to investors.

E.1. Environment

Assets There is a risk-free asset with fixed return r and a price of 1 (the numeraire). There are N firms. The firms’ equities consist of risky assets with payoffs given by the $N \times 1$ vector $f \sim N(\bar{f}, \Sigma_f)$, where Σ_f is diagonal (firm payoffs are independent). The prices of these risky assets are collected in the $N \times 1$ vector p . f is exogenous, but p will be determined in equilibrium by investors’ behavior.

Media There is a representative media outlet, which observes the realization of f before the market opens. The outlet produces a publication in which they report the realized payoffs from a subset of firms’ equities. As in [Section 3](#), the outlet has a space constraint

$$\sum_{j=1}^N m_j \leq N_r \quad (86)$$

where m_j is an indicator equal to 1 if the outlet reports on firm j , and $N_r < N$. The outlet sells this publication to investors at a price $c > 0$.

Investors There is a unit mass of investors indexed by i , with exponential utility over final wealth, W_i , net of the costs of any information acquired, cL_i .

$$U_i = -\exp(-\rho(W_i - cL_i)), \quad (87)$$

where $\rho > 0$ governs the risk aversion and $L_i \in \{0, 1\}$ is an indicator for whether the investor purchased the media publication.

Each investor has an endowment of W_0 units of the risk-free asset. Let q_i be the $N \times 1$ vector of the quantities of each risky asset purchased by investor i . To buy this portfolio, they must sell $q_i'p$ units of the risk-free endowment. Their end-of-period wealth is therefore

$$W_i = (W_0 - q_i'p)r + q_i'f. \quad (88)$$

Investors can observe which firms are reported before they choose whether to purchase the media publication, but they can only see the information in the publication if they purchase it. If an investor purchases the publication, they can only process a limited amount of information from its contents. We model this fixed information capacity with the constraint

$$|\Sigma_i^{-1}| \leq e^{2K} |\Sigma_f^{-1}|, \quad (89)$$

where Σ_i is the variance-covariance matrix of investor i 's beliefs after processing the information found in the publication, but before observing p . The constant $K > 0$ determines the investor's information capacity.⁴¹

Market clearing The supply of each risky asset is constant. The demand for risky assets comes from investors and noise traders, who add a random component x to the asset demand. Market clearing therefore requires

$$\int_0^1 q_i di + x = \bar{x}, \quad x \sim N(0, \sigma_x^2 I), \quad (90)$$

where $\bar{x}$ and x are $N \times 1$ vectors of the asset supplies and the noise trader shocks, respectively. The noise trader shocks have a common variance $\sigma_x^2 \geq 0$.

Timing The model consists of five stages:

1. f is realized. The media outlet observes this and chooses which firms to report.

⁴¹With Gaussian priors and posteriors (verified below), this constraint implies that the mutual information between the priors and the posteriors is $\leq K$, as is standard in rational inattention models (Maćkowiak et al., 2023).

2. Investors decide whether they wish to purchase the publication and (conditional on purchasing) how to allocate their information capacity.
3. Investors observe the realization of their chosen signals.
4. Asset markets open. Investors observe the asset prices and choose among the portfolios. Simultaneously, prices are determined as a function of investor demand.
5. Payoffs are realized.

E.2. Equilibrium with given information sets

We begin by solving for the asset demand and equilibrium for any given information set. In stage 4 of the model timing, equilibrium is a set of asset demands, q_i , and prices, p , such that

1. q_i maximizes investor i 's expected utility, conditional on the information they processed and any information contained in p .
2. p is such that asset markets clear.

Portfolio choice When the asset market opens, investors observe p . We summarize any other information they may have obtained from the media in $\mathcal{I}_i$. Their expected utility at this point, after substituting out for W_i using the budget constraint (88) and simplifying, is

$$\mathbb{E}_i[U_i|p, \mathcal{I}_i] = -\exp(-\rho r W_0) \exp(\rho c L_i) \mathbb{E}_i[\exp(-\rho q'_i(f - pr))|p, \mathcal{I}_i]. \quad (91)$$

The first two exponential terms are known positive constants, so they do not affect the portfolio choice problem. Since f is normally distributed, $\exp(-\rho q'_i(f - pr))$ has a log-normal distribution. Assuming all signals from prices and purchased information preserve this distribution (we will verify later), the expectation in equation (91) can be written as

$$\mathbb{E}_i[U_i|p, \mathcal{I}_i] \propto -\mathbb{E}_i[\exp(-\rho q'_i(f - pr))|p, \mathcal{I}_i] = -\exp\left(-\rho q'_i(\mathbb{E}_i[f|p, \mathcal{I}_i] - pr) + \frac{\rho^2}{2} q'_i \mathbb{V}_i[f|p, \mathcal{I}_i] q_i\right) \quad (92)$$

where $\mathbb{V}_i[f|p, \mathcal{I}_i]$ is the $(N \times N)$ posterior variance of investor i 's beliefs about f .

Maximizing this with respect to q_i gives the asset demand equation

$$q_i = \frac{1}{\rho} (\mathbb{V}_i[f|p, \mathcal{I}_i])^{-1} (\mathbb{E}_i[f|p, \mathcal{I}_i] - pr) \quad (93)$$

Prior information All investors know the distribution of f . If investors have paid for information, then they also observe a vector of noisy signals, before the markets open, of the form

$$s_i = f + \varepsilon_i, \quad \varepsilon_i \sim N(0, \Sigma_{\varepsilon i}), \quad (94)$$

where the noise ε_i is idiosyncratic to investor i and independent of f .

For simplicity we restrict our attention to cases where $\Sigma_{\varepsilon i}$ is diagonal (i.e., the signal noise is independent across assets). Incorporating these signals using Bayes' rule, investor i 's beliefs about f before the market opens are normally distributed, with

$$\mathbb{V}_i[f|\mathcal{I}_i] \equiv \Sigma_i = (\Sigma_f^{-1} + \Sigma_{\varepsilon i}^{-1})^{-1} \quad (95)$$

$$\mathbb{E}_i[f|\mathcal{I}_i] \equiv \mu_i = \Sigma_i (\Sigma_f^{-1} \bar{f} + \Sigma_{\varepsilon i}^{-1} s_i). \quad (96)$$

If investor i does not purchase the (media) information, they do not observe the signals, so $\Sigma_{\varepsilon i}^{-1}$ is a matrix of 0s. If an asset j is not reported by the media, then the j, j 'th element of $\Sigma_{\varepsilon i}^{-1}$ is 0 for all investors, as no one is able to learn about asset j .

Information in prices We solve for the prices, p , by guessing and verifying that they are a linear function of the payoffs and the shocks, where

$$p = A + Bf + Cx \quad (97)$$

for some diagonal $N \times N$ matrices A, B, C .

We split the assets into two groups, depending on whether or not they are reported in the media. Without loss of generality, we index the assets that are reported in the media by $n \in \{1, \dots, N_r\}$ and those that are unreported by $n \in \{N_r + 1, \dots, N\}$.

(a) Unreported firms: Since no investors have information on the realized f_n for unreported

firms, the prices are uninformative and the final $N - N_r$ rows and columns of B contain only 0s. As a result, the beliefs about f_n depend on the underlying payoff distribution only. The demand for the equity of an unreported firm is therefore identical across investors and is given by

$$q_{ni} = \frac{\bar{f}_n - r p_n}{\rho \sigma_{f_n}^2}, \quad (98)$$

where $\sigma_{f_n}^2$ is the n th diagonal element of Σ_f . Substituting this into the market clearing in equation (90) for firm n 's equity and rearranging yields

$$p_n = \frac{\bar{f}_n - \rho \sigma_{f_n}^2 \bar{x}_n}{r} - \frac{\rho \Sigma_{f_n}}{r} x_n \quad (99)$$

This is of the form in equation (97), with the n th diagonal element of B equal to 0.

- (b) Reported firms: The asset prices of the reported firms contain some information. Let z_r denote a $N_r \times 1$ vector consisting of the first N_r elements of any vector z , that is, f_r denotes the payoffs of the reported assets; similarly, A_r , B_r , C_r , Σ_{rf} , and Σ_{ri} denote $N_r \times N_r$ matrices, consisting of the first N_r rows and columns of A , B , C , Σ_f , and Σ_i , respectively.

From the price law of motion, investors can construct an unbiased signal about f_r as

$$B_r^{-1}(p_r - A_r) = f_r + B_r^{-1}C_r x \sim N(f_r, \Sigma_{rp}), \quad (100)$$

$$\Sigma_{rp} \equiv \sigma_x^2 B_r^{-1} C_r (B_r^{-1} C_r)'. \quad (101)$$

After incorporating this signal, investors' posteriors are normally distributed, with

$$\mathbb{V}_i[f|p, \mathcal{I}_i] \equiv \hat{\Sigma}_{ri} = (\Sigma_{ri}^{-1} + \Sigma_{rp}^{-1})^{-1}, \quad (102)$$

$$\mathbb{E}_i[f|p, \mathcal{I}_i] \equiv \hat{\mu}_{ri} = \hat{\Sigma}_{ri}(\Sigma_{ri}^{-1}\mu_{ri} + \Sigma_{rp}^{-1}B_r^{-1}(p_r - A_r)). \quad (103)$$

Substituting $\hat{\mu}_{ri}$ and $\hat{\Sigma}_{ri}$ into equation (93) we obtain the asset demand

$$q_{ri} = \frac{1}{\rho} \Sigma_{ri}^{-1} \mu_{ri} + \frac{1}{\rho} (\Sigma_{rp}^{-1} (B_r^{-1} - r I_r) - r \Sigma_{ri}^{-1}) p_r - \frac{1}{\rho} \Sigma_{rp}^{-1} B_r^{-1} A_r. \quad (104)$$

Substituting out for μ_{ri}, Σ_{ri} using equations (95) and (96) and aggregating across investors, the market clearing becomes

$$\frac{1}{\rho}(\Sigma_{rf}^{-1}\bar{f}_r - \Sigma_{rp}^{-1}B_r^{-1}A_r) + \frac{1}{\rho}\bar{\Sigma}_{r\epsilon}^{-1}f_r + \frac{1}{\rho}(\Sigma_{rp}^{-1}(B_r^{-1} - rI_r) - r\Sigma_{rf}^{-1} - r\bar{\Sigma}_{r\epsilon}^{-1})p + x_r = \bar{x}_r, \quad (105)$$

where $\bar{\Sigma}_{r\epsilon}^{-1} = \int_0^1 \Sigma_{r\epsilon i}^{-1} di$ is the average precision of the investors' signals. This rearranges to the form in equation (97), confirming our guess. Matching the coefficients yields solutions for A, B, C .

E.3. Information choice

We now go back a step and solve for the information choices, taking the media reporting as given.

Indirect expected utility Equation (91) defines the expected utility conditional on observing $p, \mathcal{I}$. Substituting out for the expectation, using (92), and for q_i using (93), this becomes

$$\mathbb{E}_i[U_i|p, \mathcal{I}_i] = -\exp(-\rho r W_0) \exp(\rho c L_i) \left[\exp \left(-\frac{1}{2} (\mathbb{E}_i[f|p, \mathcal{I}_i] - pr)' \mathbb{V}_i[f|p, \mathcal{I}_i]^{-1} (\mathbb{E}_i[f|p, \mathcal{I}_i] - pr) \right) \right]. \quad (106)$$

When the investor makes their information choice, they have not yet observed $p, \mathcal{I}$. We therefore need to take the expectation of equation (106) over these objects, or equivalently over the posterior expectation $\mathbb{E}_i[f|p, \mathcal{I}_i]$.⁴² This is an expectation of an exponential of a squared Gaussian distribution, which is given by (see Veldkamp, 2023, ch. 7.3)

$$\begin{aligned} \mathbb{E}_i[U_i] = & -\exp(-\rho r W_0) \exp(\rho c L_i) \left(\frac{|\mathbb{V}_i[f|p, \mathcal{I}_i]|}{|\Sigma_f|} \right)^{\frac{1}{2}} \\ & \cdot \left[\exp \left(-\frac{1}{2} \mathbb{E}_i[\mathbb{E}_i[f|p, \mathcal{I}_i] - pr]' \Sigma_f^{-1} \mathbb{E}_i[\mathbb{E}_i[f|p, \mathcal{I}_i] - pr] \right) \right]. \end{aligned} \quad (107)$$

The final bracketed term of this expression consists of investors' expectations of posterior

⁴²The posterior variance $\mathbb{V}_i[f|p, \mathcal{I}_i]$ is unaffected by the realization of the signals or the prices. When they make that decision, investors know the $\mathbb{V}_i[f|p, \mathcal{I}_i]$ they will face with and without purchasing the information.

beliefs and prices. Investors know that information will make their beliefs more precise, but ex-ante they do not expect it to make their beliefs systematically more or less optimistic. Whether or not they purchase this information, this final term is therefore constant. As a result, only the terms in L_i and $(|\mathbb{V}_i[f|p, \mathcal{I}_i]|/|\Sigma_f|)^{-\frac{1}{2}}$ are affected by the information choice.

The expected utility for informed investors who purchase media and for uninformed investors who do not is therefore (respectively) proportional to

$$\mathbb{E}_i[U_i] \propto -e^{\rho c} \left(\frac{|\mathbb{V}_i[f|p, \mathcal{I}_i]|}{|\Sigma_f|} \right)^{\frac{1}{2}}, \quad \mathbb{E}_U[U_U] \propto - \left(\frac{|\mathbb{V}[f|p]|}{|\Sigma_f|} \right)^{\frac{1}{2}}, \quad (108)$$

where $\mathbb{V}_i[f|p, \mathcal{I}_i]$ may differ across informed investors, i , depending on how they choose to allocate their information capacity.

Information capacity allocation An investor who purchases the media publication chooses the properties of their noisy signals (94) to maximize their expected utility (107) subject to their capacity constraint (89). Since the priors are Gaussian, equation (94) is the optimal signal structure, and investors only have to choose the noise variance matrix $\Sigma_{\varepsilon i}$.

The important step here is to note, as shown in e.g., Veldkamp (2023), that the objective function is convex, implying there are gains to specialization. The optimal information capacity allocation is for the investor to devote all of their capacity to learning about a single firm's equity. In other words, the investor's signal is such that all elements of $\Sigma_{\varepsilon i}^{-1}$ are zero except for one. If an investor learns about firm n^* , the capacity constraint implies

$$\sigma_{\varepsilon i n^*}^{-2} = (e^{2K} - 1) \sigma_{f n^*}^{-2}, \quad (109)$$

where $\sigma_{\varepsilon i n^*}^2, \sigma_{f n^*}^2$ are the n^* th diagonal elements of $\Sigma_{\varepsilon i}$ and Σ_f , respectively.

Since Σ_f and $\mathbb{V}_i[f|p, \mathcal{I}_i]$ are diagonal, $\mathbb{E}_i[U_i]$ from equation (108) can be written as

$$\mathbb{E}_i[U_i] \propto -e^{\rho c} \prod_{n=1}^N \left(\frac{\mathbb{V}_i[f_n|p_n, \mathcal{I}_i]}{\sigma_{f n}^2} \right)^{\frac{1}{2}} = -e^{\rho c} \prod_{n=1}^{N_r} \left(\frac{\sigma_{f n}^{-2} + \sigma_{\varepsilon i n}^{-2} + \sigma_{p n}^{-2}}{\sigma_{f n}^{-2}} \right)^{-\frac{1}{2}} \quad (110)$$

$$= -e^{\rho c} \left(\frac{|\mathbb{V}[f|p]|}{|\Sigma_f|} \right)^{\frac{1}{2}} \left(\frac{\sigma_{f n^*}^{-2} e^{2K} + \sigma_{p n^*}^{-2}}{\sigma_{f n^*}^{-2} + \sigma_{p n^*}^{-2}} \right)^{-\frac{1}{2}}. \quad (111)$$

The first of these equalities uses the observation that for all unreported firms, $\mathbb{V}_i[f_n|p_n, \mathcal{I}_i] =$

σ_{fn}^2 . The second uses the fact that investor i uses all of their information capacity to learn about a single firm, denoted n^* , with the information precision given in equation (109).

Investors therefore learn about the firm with the highest “learning index,” $\mathcal{L}_n$, defined as

$$\mathcal{L}_n \equiv \frac{\sigma_{fn}^{-2} e^{2K} + \sigma_{pn}^{-2}}{\sigma_{fn}^{-2} + \sigma_{pn}^{-2}}. \quad (112)$$

This is strictly increasing in σ_{fn}^{-2} and strictly decreasing in the precision of the information contained in the prices, σ_{pn}^{-2} . We show below that if more investors learn about asset n , σ_{pn}^{-2} rises. Investors therefore prefer to learn about the assets that other investors are not learning about.

Mixed strategy equilibrium We follow [Van Nieuwerburgh and Veldkamp \(2010\)](#) and look for an equilibrium in mixed strategies. Since investors wish to learn about the assets that other investors are not learning about, ex-ante identical investors specialize by randomizing the use of their information capacity.

Suppose that conditional on buying the publication, investors devote their information capacity to learning about asset n with probability π_n . For such a strategy to be optimal, investors must be indifferent between learning about all firms in the publication, given all other investors are playing the same mixed strategy. From equation (111), this implies⁴³

$$\mathcal{L}_n = \mathcal{L}_{n'} \quad \text{for all } (n, n') \text{ such that } \pi_n, \pi_{n'} > 0. \quad (113)$$

Learning indices in equilibrium Let λ_n be the fraction of investors who process information about firm n , equal to π_n multiplied by the fraction of investors purchasing the media publication. The average precision of the investors’ signals about firm n is then given by

$$\bar{\sigma}_{n\epsilon}^{-2} = \lambda_n (e^{2K} - 1) \sigma_{fn}^{-2}. \quad (114)$$

⁴³This is exactly as in [Van Nieuwerburgh and Veldkamp \(2010\)](#), except for the extra constraint that investors can only learn about the assets that are reported in the media and only if they purchase the media publication.

Substituting this into row n of equation (105) and rearranging the leads to

$$p_n = (\sigma_{pn}^{-2}(b_n^{-1} - r) - r\sigma_{fn}^{-2}(1 + \lambda_n(e^{2K} - 1)))^{-1} \left[(\rho\bar{x}_n + \sigma_{pn}^{-2}b_n^{-1}a_n - \sigma_{fn}^{-2}\bar{f}_n) - \lambda_n\sigma_{fn}^{-2}(e^{2K} - 1)f_n - \rho x_n \right]. \quad (115)$$

Matching the coefficients between equations (97) and (115), we obtain

$$b_n = -\frac{\lambda_n\sigma_{fn}^{-2}(e^{2K} - 1)}{\sigma_{pn}^{-2}(b_n^{-1} - r) - r\sigma_{fn}^{-2}(1 + \lambda_n(e^{2K} - 1))} \quad (116)$$

$$c_n = b_n \cdot \frac{\rho}{\lambda_n\sigma_{fn}^{-2}(e^{2K} - 1)}. \quad (117)$$

Equation (101), combined with the fact that all of the matrices here are diagonal, implies that the variance of the noise in the price of asset n is

$$\sigma_{pn}^2 = \sigma_x^2(b_n^{-1}c_n)^2 = \frac{\rho^2\sigma_x^2\sigma_{fn}^4}{\lambda_n^2(e^{2K} - 1)^2}, \quad (118)$$

which confirms our earlier statement that σ_{pn}^{-2} is increasing in λ_n . Substituting this into equation (112) and simplifying, we can express the learning index as

$$\mathcal{L}_n = 1 + \frac{e^{2K} - 1}{1 + \lambda_n^2\sigma_{fn}^{-2}\sigma_x^{-2}\rho^{-2}(e^{2K} - 1)^2}. \quad (119)$$

Many of the elements of this formula for the learning index are common across assets. Condition (113) is therefore satisfied if and only if

$$\frac{\lambda_n^2}{\sigma_{fn}^2} = \frac{\lambda_{n'}^2}{\sigma_{fn'}^2} \quad \text{for all } (n, n') \text{ such that } \lambda_n, \lambda_{n'} > 0. \quad (120)$$

This is the key indifference condition for the mixed strategy equilibrium. For two assets with the same prior variance, the fraction of the informed investors λ_n must be equal. Otherwise, the assets with greater prior uncertainty will have a greater proportion of informed investors.

A final implication of these results is that the investors learn about all of the firms that are included in the media publication, with positive probability. To see this, suppose that no investor learns about firm n_0 , so $\lambda_{n_0} = 0$. In equation (119), that firm's learning index

would be $\mathcal{L}_{n_0} = \exp(2K)$, which is strictly greater than the learning index for any firm with a positive λ_n . As a result, if $\lambda_{n_0} = 0$, then an investor could always increase their expected utility by deviating from the mixed strategy of the other investors and from learning about n_0 with probability 1. It is therefore not possible for a mixed strategy equilibrium to entirely exclude some reported firms.

Media purchase Using equations (108) and (111), the expected utility gain from purchasing the media publication is given by

$$\mathbb{E}_i[U_i] - \mathbb{E}_U[U_U] = \left(\frac{|\mathbb{V}[f|p]|}{|\Sigma_f|} \right)^{\frac{1}{2}} (1 - e^{\rho c} \mathcal{L}_{n^*}^{-\frac{1}{2}}), \quad (121)$$

where $\mathcal{L}_{n^*}$ is the learning index of any of the assets over which investors mix.

Investors purchase the information if the expected utility gain is positive. The proportion of the investors who purchase the publication is such that investors are indifferent between purchasing or not purchasing the information, which occurs at $\mathcal{L}_{n^*} = e^{2\rho c}$. Therefore, a given value of c pins down a unique learning index.

Media-reporting decision The media outlet chooses which firms to report on to maximize their profits. Let Q be the proportion of investors who purchase the outlet's publication, so the latter's profits are revenues, cQ , minus costs, which we assume are independent of the outlet's reporting decisions. We take c as given, so the outlet chooses reporting to maximize its readership, Q .

To find the optimal reporting strategy, it is helpful to note that condition (120) implies that λ_n can be expressed as

$$\lambda_n = \lambda_0 \sigma_{fn}, \quad (122)$$

where λ_0 is identical for all firms, n . Substituting this into equation (119), we find that λ_0 is uniquely determined by the parameters that are common to all n and by the learning index $\mathcal{L}_n$, which in turn is fixed by c . We can therefore treat λ_0 as fixed.

Recall that λ_n is the proportion of investors who process information about firm n , which is given by the proportion who buy the media publication, multiplied by the probability an

informed investor devotes their information capacity to that firm: $\lambda_n \equiv Q\pi_n$. Summing over all reported firms and using the fact that $\sum_{n=1}^{N_r} \pi_n = 1$, we therefore have

$$Q \sum_{n=1}^{N_r} \pi_n = \sum_{n=1}^{N_r} \lambda_n \implies Q = \lambda_0 \sum_{n=1}^{N_r} \sigma_{fn}. \quad (123)$$

Since λ_0 is fixed by c , the outlet maximizes Q by reporting on the N_r firms with the most volatile payoffs, that is, those firms with the highest σ_{fn} .

Relationship to the quantitative model To solve this model, we abstracted from firms' decisions. The variance of the payoffs from holding equity of firm n is thus fixed at σ_{fn}^2 . In the quantitative model, media reporting affects firms' decisions and so affects that variance.

The appropriate analogue to the reporting policy derived here is that media outlets report on the firms with large payoff variances *conditional* on being reported. To see why, consider an outlet choosing between reporting on firms j and j' . If the outlet reports on firm j , investors observe that and evaluate the benefits of purchasing the outlet's publication, based on the resulting variance of asset j 's payoff. If the outlet does not report on j but instead reports on j' , then the value of the publication, to investors, is determined by the variance of asset j' , given that j' was reported. The appropriate comparison is therefore between the variances of the payoffs conditional on the firm being reported. This is exactly the reporting policy derived in Section 3.2 in the quantitative model.

F. Quantitative Analysis with Noisy Media Reports

The baseline model assumes that a media report reveals a firm's capital quality exactly. This appendix relaxes that assumption. We first describe how noisy reporting changes firms' signaling problem and the media's newsworthiness calculation, and then quantify whether reporting accuracy changes our conclusions about the aggregate relevance of media reporting allocation. All other elements of the model are unchanged.

F.1. Setup

We allow for two dimensions of noise: one concerning the outlets' own observations, and another concerning the transmission from outlets to forecasters.

Noise dimension 1: outlets. Instead of observing capital quality a precisely, as in the main text, we now assume that media outlets observe only a noisy signal about a . This signal takes the form of a *segment indicator*.

Specifically, let G denote the cumulative distribution function of capital quality a . For a given integer $N \geq 1$, partition the support of a into N intervals with equal prior probability,

$$\mathcal{S}_\ell^N = \left[G^{-1} \left(\frac{\ell-1}{N} \right), G^{-1} \left(\frac{\ell}{N} \right) \right], \quad \ell = 1, \dots, N. \quad (124)$$

Outlets observe the segment to which each firm belongs, but not their precise a within the segment. All outlets therefore observe the same signals as one another. As $N \rightarrow \infty$, segments become arbitrarily small, and we return to our exact-observation model from the main results.

Noise dimension 2: forecasters. Along with imprecise outlet observations, we also allow for noise in how forecasters read outlet reports. To do this, we assume that if an outlet reports a firm, the segment its forecaster reads, $\hat{s}$, is correct with probability $p \in (1/N, 1]$. Conditional on receiving an *incorrect* segment signal, signal probabilities are uniform over the other $N - 1$ segments. Given this, we define the ‘confusion matrix’ $\Pi_{\ell'|\ell}$ as

$$\Pi_{\ell'|\ell} = \Pr(\hat{s} = \ell' \mid a \in \mathcal{S}_\ell^N) = \begin{cases} p, & \ell' = \ell, \\ \frac{1-p}{N-1}, & \ell' \neq \ell. \end{cases} \quad (125)$$

We recover the perfect-transmission model whenever $p = 1$. Importantly, this transmission noise is forecaster-specific, and errors are independent across forecasters.

Investors. When a firm is reported, investors observe the interval $\mathcal{S}_\ell^N$ containing its capital quality, rather than its exact quality.⁴⁴ Thus, N indexes the precision of a report.⁴⁵ When

⁴⁴Formally, if a firm is reported in equilibrium, investors observe an infinite number of noisy $\hat{s}$ signals, from which the modal signal realization reveals the true segment s^* . By the law of large numbers the cross-outlet frequency of signal ℓ' equals $\Pi_{\ell'|s^*}$. Since we assume $p > 1/N$, the diagonal entry is the unique largest, so $\text{mode}_{\text{outlets}} \hat{s} = s^*$ almost surely..

⁴⁵A conventional alternative would model a noisy report as a continuous signal and index its accuracy by the variance of the signal noise. That formulation is less useful for our mechanism. Because the equilibrium is separating, reducing the variance of a signal whose noise has full support does not change which types remain possible after any report. As the toy-model argument in Section F.2 makes clear, what relaxes the information friction is removing potential mimickers. Our interval formulation captures this force directly:

$N = 1$, the report provides no information about capital quality; as N becomes large and the intervals become arbitrarily fine, the model converges to the exact-report benchmark in the main text.

Because a noisy report reveals only an interval, investors continue to use equity issuance to infer a reported firm's type within that interval. Let $MV_t^N(k, z, a, 1)$ denote the reported firm's equilibrium market value under reporting precision N , and define the expected market value associated with report interval ℓ as

$$\overline{MV}_{t\ell}^N(k, z) = \mathbb{E} [MV_t^N(k, z, a, 1) \mid a \in \mathcal{S}_\ell^N, k, z]. \quad (126)$$

The relevant uncertainty resolved by the report is now the dispersion in expected market values across report intervals.

Outlets. Appendix F.5 proves that the news reporting equilibrium in this noisy-reporting model has the same form as Proposition 1. The only difference is that the newsworthiness function in equation (25) becomes

$$\begin{aligned} N_t^N(k, z) &= \mathbb{V}_\ell [\overline{MV}_{t\ell}^N(k, z) \mid k, z] \\ &= \frac{\lambda^2}{N} \sum_{\ell=1}^N \left(\overline{MV}_{t\ell}^N(k, z) - \frac{1}{N} \sum_{s=1}^N \overline{MV}_{ts}^N(k, z) \right)^2. \end{aligned} \quad (127)$$

where $\lambda \equiv \frac{pN-1}{N-1} \in (0, 1]$ captures the degree of transmission noise to forecasters.

The law of total variance clarifies the distinction from exact reporting:

$$\mathbb{V}_a [MV_t^N(k, z, a, 1) \mid k, z] = N_t^N(k, z) + \frac{1}{N} \sum_{\ell=1}^N \mathbb{V}_a [MV_t^N(k, z, a, 1) \mid a \in \mathcal{S}_\ell^N, k, z]. \quad (128)$$

Only the first term is resolved by the report; the second is residual uncertainty within the reported interval. Holding the underlying distribution of market values fixed, a coarser report therefore makes a firm less newsworthy because it resolves less uncertainty about the firm's market value.

The media's reporting rule retains the form used in the quantitative model, but it is

a more accurate report narrows the set of possible types and can therefore eliminate a larger share of the loss generated by the information friction.

evaluated using $N_t^N(k, z)$. In particular, the reporting probability is

$$R_t^N(k, z) = \lambda_p + (1 - \lambda_p) \left\{ 1 - \exp \left[- \left(\frac{N_t^N(k, z)}{\lambda_\alpha} \right)^{\lambda_\xi} \right] \right\}. \quad (129)$$

Importantly, notice that transmission noise to forecasters shifts the newsworthiness of all firms by a factor of λ^2 , and thus only appears in the reporting probability function relative to the scale parameter λ_α . Since λ_α is chosen to match the moments of the empirical reporting distribution, any change in forecaster noise will be immediately offset by a change in λ_α . Our second dimension of noise, in the transmission from outlets to forecasters, is therefore quantitatively irrelevant.

The first dimension of noise, in outlet observations, does however alter the equilibrium. Changes in *reporting* precision affect not only how much information a report conveys, but also which firms the media chooses to report on.

F.2. Intuition from the toy model

The ordered-type extension of the toy model in Appendix C clarifies the firm-level effect of noisy reports. Without a report, all types form a single signaling chain. The lowest-quality type issues at the symmetric-information level, and the no-mimicking condition in equation (70) recursively determines the issuance of each higher-quality type. Signaling distortions can therefore accumulate along the full support of the distribution: the issuance of a high-quality firm may ultimately be constrained by a long chain of lower-quality types.

Now suppose a report places the firm in interval $\mathcal{S}_\ell^N$. Let $a_{\ell,1} < \dots < a_{\ell,n_\ell}$ denote the types in this interval, ordered by capital quality. The report rules out all types outside the interval as potential mimickers, so the least-cost separating allocation is solved anew within the interval:

$$\begin{aligned} e_{\ell,1}^N &= e^*, \\ e_{\ell,j}^N &= \bar{e}_{\ell,j}(e_{\ell,j-1}^N), \quad j = 2, \dots, n_\ell, \end{aligned} \quad (130)$$

where $\bar{e}_{\ell,j}(e_{\ell,j-1}^N)$ is the largest issuance by type $a_{\ell,j}$ that prevents the adjacent lower type $a_{\ell,j-1}$ from mimicking it. A noisy report does not necessarily restore symmetric-information

issuance for every reported firm, because types within the interval must still separate. It does, however, break the original signaling chain into shorter chains. The lowest-quality type in each report interval is reset to its symmetric-information choice, and signaling distortions accumulate only among the types within that interval. More precise reports create more reset points and reduce underissuance and underinvestment among reported firms.

F.3. Quantitative analysis

Our empirical impulse responses capture the dynamic association between media coverage and firm outcomes, but do not isolate the causal effect of reporting. We therefore do not use these responses as identifying moments for the accuracy of the information conveyed by a report. Instead, we conduct a series of quantitative sensitivity exercises for $N \in \{2, 5, 10, 15\}$ to assess how reporting accuracy shapes the firm-level and aggregate implications of media coverage.

Implementing these exercises requires recalibrating and re-solving the model for each value of N ; it is not equivalent to changing only the information received by investors. Equations (127) and (129) show that N also changes firms' newsworthiness and hence the equilibrium allocation of coverage. We therefore recalibrate the model separately for each N using the same procedure as in the baseline calibration. Table F.1 reports the resulting parameter values and model fit across the full set of target moments.

F.4. Quantitative results

For each value of N , we use the corresponding recalibrated economy to evaluate the aggregate role of media reporting in three steps. First, we compare the baseline equilibrium with the corresponding no-media economy and calculate the fraction of each aggregate loss eliminated by reporting. Second, holding reporting capacity fixed, we reallocate reports according to the constrained-efficient allocation (CEA) described in Section 4.3 and calculate the same fraction under this allocation. Third, we take the ratio of the CEA fraction to the baseline fraction to measure the amplification achieved through reallocation.

Table F.2 reports the results. Under the baseline allocation, media reporting eliminates only 1.38% of the output loss from asymmetric information when $N = 2$ and 4.05% when $N = 15$. The accompanying decline in absolute losses as N increases is consistent with

Table F.1: Model calibration under alternative reporting accuracies

	Data	Baseline	Noisy reports			
		$N = \infty$	$N = 2$	$N = 5$	$N = 10$	$N = 15$
Panel A: Parameters						
<i>Cash flow dynamics</i>						
ρ_z : idiosyncratic productivity, persistence	–	0.95	0.95	0.95	0.95	0.95
σ_z : innovation standard deviation	–	0.11	0.11	0.11	0.11	0.11
<i>Investment technology</i>						
δ : depreciation rate	–	3.30%	3.30%	3.30%	3.30%	3.30%
θ : return to scale	–	0.81	0.81	0.81	0.81	0.81
<i>Life-cycle dynamics</i>						
$\mu_{\log z}^{\text{entrant}}$: entrants' average log productivity	–	-0.175	-0.175	-0.175	-0.175	-0.175
$\mu_{\log k}^{\text{entrant}}$: entrants' average log size	–	-1.761	-1.761	-1.761	-1.761	-1.761
<i>Information and financial friction</i>						
σ_a : dispersion of capital-quality shock	–	0.18	0.18	0.18	0.18	0.18
ϕ^e : fixed cost of issuing equity	–	0.015%	0.015%	0.015%	0.015%	0.015%
<i>Selective media reporting</i>						
$(\lambda_\alpha, \lambda_p)$: location parameters (scaled by 1000)	–	(4.7, 2.0)	(2.9, 3.0)	(4.0, 3.0)	(5.1, 4.0)	(5.7, 4.0)
λ_ξ : curvature	–	1.80	1.80	1.80	1.90	1.90
Panel B: Target moments						
<i>Cash flow dynamics (annual, %)</i>						
Idiosyncratic TFP, persistence	0.70	0.69	0.68	0.68	0.69	0.70
–, standard deviation	0.38	0.37	0.37	0.37	0.38	0.38
<i>Investment (annual, %)</i>						
Investment rate, mean	6.00	6.12	6.10	6.04	6.17	6.08
–, standard deviation	5.53	5.40	5.33	5.33	5.39	5.43
<i>Life-cycle dynamics (%)</i>						
Growth rate, young minus mature firms	10.55	10.49	10.47	10.48	10.49	10.49
Idiosyncratic TFP, –	-7.52	-7.50	-7.50	-7.50	-7.50	-7.50
<i>Financing (%)</i>						
Fraction of firms issuing equity, annual mean	17.87	17.97	18.01	18.05	17.78	17.83
Issuance-fee ratio, mean	2.17	2.21	2.23	2.22	2.21	2.21
Selling-concession ratio when issuing, mean	2.97	3.02	3.04	3.03	3.03	3.01
<i>Media-report distribution (share of all reports, %)</i>						
Top 20% of market value	87.22	86.18	86.29	85.87	86.70	86.69
60–80% of market value	8.45	10.74	10.55	10.59	9.88	9.93
40–60% of market value	2.54	2.18	2.08	2.42	2.11	2.09
20–40% of market value	1.30	0.56	0.61	0.67	0.70	0.69
Bottom 20% of market value	0.50	0.35	0.48	0.46	0.60	0.60

Notes: Panel A reports the full set of internally calibrated parameters. The exact-report column gives the baseline calibration used in the main text. In each noisy-report calibration, the parameters unrelated to media reporting retain their baseline values, while the parameters governing the reporting-probability function are recalibrated for the indicated value of N . Panel B reports the data targets and their model counterparts under each calibration. All columns use the same calibration procedure and targets.

the toy-model implication that more accurate reports generate larger firm-level effects and therefore eliminate more aggregate loss. Nevertheless, the aggregate role of media remains modest: across investment, the capital stock, output, and firm value, the baseline allocation eliminates no more than 4.09% of the corresponding no-media loss for any value of N .

Holding reporting capacity fixed, the CEA substantially increases the role of media. Reporting now eliminates 35.33% of the output loss when $N = 2$ and 91.44% when $N = 15$,

with similarly large roles across the other aggregate outcomes.

The central result comes from comparing the role of media across the two allocations. Across every outcome and value of N , the CEA amplifies the aggregate role of media by more than twentyfold relative to the baseline allocation. This uniformly large amplification shows that the paper’s main conclusion does not depend on reports revealing firms’ capital quality exactly. Instead, the limited aggregate relevance of media reporting arises primarily from how scarce coverage is allocated: who receives coverage is quantitatively more important than whether a report reveals a firm’s capital quality exactly or only locates it within a coarse interval.

Table F.2: Aggregate relevance of noisy media reports

N	Outcome	Aggregate loss (%)		Fraction of no-media loss eliminated (%)		Amplification from reallocation ($\times$)
		Baseline	CEA	Baseline	CEA	
2	Investment	6.60	4.35	1.38	35.07	25.35
	Capital stock	4.67	3.06	1.22	35.34	28.96
	Output	5.22	3.42	1.38	35.33	25.65
	Firm value	3.96	2.59	1.31	35.62	27.18
5	Investment	6.48	1.93	3.21	71.24	22.18
	Capital stock	4.59	1.35	2.83	71.54	25.27
	Output	5.13	1.51	3.19	71.52	22.40
	Firm value	3.89	1.14	3.06	71.74	23.45
10	Investment	6.43	0.71	3.98	89.39	22.46
	Capital stock	4.56	0.50	3.51	89.47	25.50
	Output	5.09	0.56	3.94	89.45	22.69
	Firm value	3.86	0.42	3.78	89.57	23.69
15	Investment	6.42	0.58	4.09	91.39	22.34
	Capital stock	4.56	0.40	3.61	91.45	25.37
	Output	5.08	0.45	4.05	91.44	22.57
	Firm value	3.86	0.34	3.88	91.50	23.59

Notes: “Aggregate loss” is the percentage shortfall relative to the symmetric-information economy. The baseline columns evaluate the equilibrium allocation of reports in each recalibrated economy. The constrained-efficient allocation (CEA) columns evaluate a reallocation that maximizes the aggregate effect of reporting while holding fixed the reporting capacity in the corresponding baseline economy. “Fraction of no-media loss eliminated” is the share of the corresponding no-media loss removed by reporting under each allocation. “Amplification from reallocation” is the ratio of this fraction under the CEA to that under the baseline allocation.

F.5. Proof of the noisy reporting probability equation

We show that there is a unique reporting policy that can be sustained as a symmetric equilibrium. To find this, we begin by considering an arbitrary candidate reporting policy. We then show that there is a unique candidate reporting policy from which no outlet would find it optimal to deviate.

The candidate reporting policy is characterized by a vector of reporting choices $\mathbf{m}_t =$

$\{m_{jt}\}_{j=0}^1$, which satisfies the space constraint (16). Without loss of generality, assume that $\mathbf{m}_t$ involves all outlets reporting on firm j and all outlets not reporting on firm j' .

Notation For notational simplicity below, we define the following objects

$$\bar{\mu}(k, z) = N^{-1} \sum_{\ell=1}^N \overline{MV}_{t\ell}^N(k, z), \quad \sigma_\ell^2 = \mathbb{V} [MV_t^N(k, z, a, 1) \mid a \in \mathcal{S}_\ell^N, k, z] \quad (131)$$

Optimal forecasts For a firm j that is reported on by outlet i , forecaster i forecasts market value using

$$\text{Pred}_t(k_{jt}, z_{jt}, 1, \hat{s}_i(a_{jt}) = \ell') = \lambda \overline{MV}_{t\ell'}^N(k_{jt}, z_{jt}) + (1 - \lambda) \bar{\mu}(k_{jt}, z_{jt}). \quad (132)$$

Substituting this forecast into equation (22) reveals that when $m_{ijt}^o = 1$, forecaster i makes the forecast error:

$$\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, 1, \hat{s}_i(a_{jt}) = \ell') = [\lambda \overline{MV}_{t\ell'}^N(k_{jt}, z_{jt}) + (1 - \lambda) \bar{\mu}(k_{jt}, z_{jt}) - MV_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2. \quad (133)$$

For a firm that is not reported by outlet i , the optimal forecast by forecaster i is given by⁴⁶

$$\text{Pred}_t(k_{jt}, z_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}} \mid m_{ijt}^o = 0) = \mathbb{E} [MV_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}) \mid k_{jt}, z_{jt}, m_{jt}]. \quad (134)$$

implying forecast error

$$\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \text{no rep}) = [\mathbb{E} [MV_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}) \mid k_{jt}, z_{jt}, m_{jt}] - MV_t(k_{jt}, z_{jt}, a_{jt}, m_{jt})]^2. \quad (135)$$

Both forecasts (with and without reporting) are uncertain, because the forecaster is uncertain about a_{jt} . In general, this will lead forecaster i to make non-zero forecast errors.⁴⁷ For the

⁴⁶This will only equal $\bar{\mu}(k_{jt}, z_{jt})$ if $m_{jt} = 1$.

⁴⁷Even if investors did not have any superior media reports than the forecaster, realized a_{jt} could still affect the realized market value indirectly, through firm j 's equity issuance (see Section 3.1.1). Since forecasters make predictions before equity markets open, they cannot observe equity issuance. They do not therefore know what the investors' posteriors $\mathcal{B}(a_{jt})$ will be and so cannot be certain about the realization of the market value.

forecaster who does not receive a report, these errors come entirely from a lack of information about a_{jt} . For forecasters who do receive a report, their posterior over a_{jt} is narrower, though still not a singleton unless each segment is a point. They do, however, have an extra source of uncertainty, stemming from the knowledge that their signal may have implied the wrong segment (if $p < 1$).

Symmetry At the reporting stage, each outlet knows k, z, κ for each firm. This information set is common to all outlets. Since they face the same problem, and we do not allow for mixed strategies, all outlets must make the same reporting choices in equilibrium. The pure strategy equilibrium is therefore symmetric.

Forecaster utility at equilibrium As established above, the reporting policy is symmetric, i.e. all outlets make the same reporting decisions. This means that in expectation, all forecasters make the same forecast errors. As a result,

$$\mathbb{E} \text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) = \mathbb{E} \overline{\text{FE}}_{-it}(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{-it}^{\text{news}}), \quad (136)$$

and thus $\mathbb{E} U_{it} = 0$.

Outlet deviations A minimal deviation from $\mathbf{m}_t$ consists of an outlet i ceasing to report on firm j and, instead, reporting on firm j' . $\mathbf{m}_t$ can only be sustained in equilibrium if no outlet finds it optimal to deviate in this way. Since in the absence of any deviation we have obtained that $\mathbb{E} U_{it} = 0$, a sufficient condition for $\mathbf{m}_t$ to be an equilibrium is that

$$\mathbb{E} \hat{U}_{it}(j, j') + \kappa_{jt} - \kappa_{j't} \leq 0, \quad (137)$$

where $\hat{U}_{it}(j, j')$ is the utility of forecaster i if outlet i deviates. If this condition holds for all pairs of the reported and unreported firms, j, j' , the news outlets never deviate and $\mathbf{m}_t$ is an equilibrium.

We now proceed to find an expression for $\mathbb{E} \hat{U}_{it}(j, j')$. First, notice that the deviation would have no effect on firms other than j and j' . From the definition of the forecaster's

utility in equation (21), we therefore have

$$\begin{aligned} \mathbb{E} \hat{U}_{it}(j, j') = & -\mathbb{E} [\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) - \overline{\text{FE}}_{-it}(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{-it}^{\text{news}})] \\ & -\mathbb{E} [\text{FE}_t(k_{j't}, z_{j't}, a_{j't}, m_{j't}, \mathcal{I}_{it}^{\text{news}}) - \overline{\text{FE}}_{-it}(k_{j't}, z_{j't}, a_{j't}, m_{j't}, \mathcal{I}_{-it}^{\text{news}})] . \end{aligned} \quad (138)$$

To evaluate this, we will first evaluate the expected utility change conditional on the asset quality segments of firms j and j' . With this in hand, we can then take the expectation over those segments.

The first two terms of equation (33) give the utility change from no longer reporting on firm j . Other forecasters are still reporting on j and so it remains the case that $m_{jt} = 1$ and the realized market value of firm j is unchanged. The forecast errors of the other forecasters, therefore, remain unchanged at equation (29). However, forecaster i 's prediction does change, as their information set no longer contains a_{jt} . The optimal prediction for firm j 's market value, in the case where $m_{jt} = 1$ but forecaster i receives no signal, is simply $\bar{\mu}(k_{jt}, z_{jt})$. Therefore:

$$\begin{aligned} & -\mathbb{E} [\text{FE}_t(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{it}^{\text{news}}) - \overline{\text{FE}}_{-it}(k_{jt}, z_{jt}, a_{jt}, m_{jt}, \mathcal{I}_{-it}^{\text{news}}) | s_{jt}^*, s_{j't}^*] \\ & = \sum_{\ell'} \Pi_{\ell' | s_{jt}^*} \mathbb{E}_{a_{jt} \in S_{s_{jt}^*}} [\tilde{\mu}_{\ell'}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 \\ & \quad - \mathbb{E}_{a_{jt} \in S_{s_{jt}^*}} [\bar{\mu}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2. \end{aligned} \quad (139)$$

where $\tilde{\mu}_{\ell'}(k_{jt}, z_{jt}) \equiv \lambda \overline{\text{MV}}_{t\ell'}^N(k_{jt}, z_{jt}) + (1 - \lambda) \bar{\mu}(k_{jt}, z_{jt})$ is the posterior mean of an informed forecaster who observes $\hat{s} = \ell'$, and s_{jt}^* is the true segment to which a_{jt} belongs.

The second two terms of equation (33) give the utility change due to reporting on firm j' . Recall that in the noiseless model, investors observe a firm's asset quality if at least one outlet reports it (equation (20)). In this extension to noisy signals, that is no longer guaranteed, because investors extract the true segment only from the mode over many outlet reports. We therefore need to specify here what outlets believe will happen to investor beliefs in this off-equilibrium deviation. For simplicity, and maximum comparability with the noiseless model, we assume that the investors learn the true segment of firm j' without noise, even if only one outlet reports it. This would be consistent, for example, with a microfoundation in which the noise in transmission from outlet to forecaster comes about due to forecaster

inattention, not anything written by the outlet: in this case, the investor pays full attention, and recovers $s_{j't}^*$ with certainty, implying $m_{j't} = 1$. As a result, forecaster i forms their prediction as in equation (29).

However, unlike for firm j , forecasters other than i now see their forecast errors change, since $m_{j't}$ has changed, and thus so has the realized market value.

$$\overline{\text{FE}}_{-it}(k_{j't}, z_{j't}, a_{j't}, 1, \mathcal{I}_{-it}^{\text{news}}) = \int_{i' \neq i} [\text{Pred}_t(k_{j't}, z_{j't}, 1, \mathcal{I}_{i',t}^{\text{news}} | m_{i'j't}^o = 0) - \text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1)]^2 di' \quad (140)$$

All outlets, i' , are identical, so when using the same steps as those used to derive equation (37), the expectation of this average forecast error becomes

$$\mathbb{E} [\overline{\text{FE}}_{-it}(k_{j't}, z_{j't}, a_{j't}, 1, \mathcal{I}_{-it}^{\text{news}}) | s_{j't}^*] = \mathbb{E}_{a_{j't} \in S_{s_{j't}^*}} [\bar{\mu}(k_{j't}, z_{j't}) - \text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1)]^2. \quad (141)$$

Substituting these results into equation (33), the expected utility of deviating in this way (conditional on $s_{jt}^*, s_{j't}^*$) is

$$\begin{aligned} \mathbb{E} (\hat{U}_{it}(j, j') | s_{jt}^*, s_{j't}^*) = & \left(\sum_{\ell'} \Pi_{\ell' | s_{jt}^*} \mathbb{E}_{a_{jt} \in S_{s_{jt}^*}} [\tilde{\mu}_{\ell'}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 \right. \\ & \left. - \mathbb{E}_{a_{jt} \in S_{s_{jt}^*}} [\bar{\mu}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 \right) \\ & - \left(\sum_{\ell'} \Pi_{\ell' | s_{j't}^*} \mathbb{E}_{a_{j't} \in S_{s_{j't}^*}} [\tilde{\mu}_{\ell'}(k_{j't}, z_{j't}) - \text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1)]^2 \right. \\ & \left. - \mathbb{E}_{a_{j't} \in S_{s_{j't}^*}} [\bar{\mu}(k_{j't}, z_{j't}) - \text{MV}_t(k_{j't}, z_{j't}, a_{j't}, 1)]^2 \right). \quad (142) \end{aligned}$$

Note that the sum over ℓ' appears in each pair of terms for different reasons: in the first pair, it is present due to the averaging over many forecasters (all forecasters other than i). In the second pair, it arises from taking expectations over all possible signal draws forecaster i may observe, given the true segment observed by the outlet.

We now take the first pair of terms, and simplify it. Since the second pair of terms has the same structure, the simplifications will apply there too. Start with term 2, which by

expanding the square and rearranging we can write as:

$$\begin{aligned}
& -\mathbb{E}_{a_{jt} \in S_{s_{jt}}^*} [\bar{\mu}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 \\
& = (\mathbb{E}[\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 - \mathbb{E}[\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 - (\bar{\mu}(k_{jt}, z_{jt}) - \mathbb{E}[\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2) \\
& = -(\sigma_{s_{jt}}^2 + (\overline{MV}_{ts_{jt}}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2)
\end{aligned} \tag{143}$$

where the second line omits the $a_{jt} \in S_{s_{jt}}^*$ subscripts on the expectations for notational simplicity. This is the familiar identity that mean-squared error is equal to bias-squared plus variance.

Applying the same steps to the expectations in the first term of equation (40), we obtain:

$$\begin{aligned}
& \sum_{\ell'} \Pi_{\ell' | s_{jt}^*} \mathbb{E}_{a_{jt} \in S_{s_{jt}}^*} [\tilde{\mu}_{\ell'}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 \\
& = \sum_{\ell'} \Pi_{\ell' | s_{jt}^*} (\sigma_{s_{jt}}^2 + (\tilde{\mu}_{\hat{s}}(k_{jt}, z_{jt}) - \overline{MV}_{ts_{jt}}^N(k_{jt}, z_{jt}))^2) \\
& = \sigma_{s_{jt}}^2 + \sum_{\ell'} \Pi_{\ell' | s_{jt}^*} (\tilde{\mu}_{\hat{s}}(k_{jt}, z_{jt}) - \overline{MV}_{ts_{jt}}^N(k_{jt}, z_{jt}))^2
\end{aligned} \tag{144}$$

Combining these expressions yields:

$$\begin{aligned}
& \sum_{\ell'} \Pi_{\ell' | s_{jt}^*} \mathbb{E}_{a_{jt} \in S_{s_{jt}}^*} [\tilde{\mu}_{\ell'}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 - \mathbb{E}_{a_{jt} \in S_{s_{jt}}^*} [\bar{\mu}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 \\
& = \sum_{\hat{s}} \Pi_{\hat{s} | s_{jt}^*} (\tilde{\mu}_{\hat{s}}(k_{jt}, z_{jt}) - \overline{MV}_{ts_{jt}}^N(k_{jt}, z_{jt}))^2 - (\overline{MV}_{ts_{jt}}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2.
\end{aligned} \tag{145}$$

Going further, use the definition of $\tilde{\mu}$ to write

$$\begin{aligned}
\tilde{\mu}_{\hat{s}}(k_{jt}, z_{jt}) - \overline{MV}_{ts_{jt}}^N(k_{jt}, z_{jt}) & = \lambda \overline{MV}_{t\hat{s}}^N(k_{jt}, z_{jt}) + (1 - \lambda) \bar{\mu}(k_{jt}, z_{jt}) - \overline{MV}_{ts_{jt}}^N(k_{jt}, z_{jt}) \\
& = \lambda (\overline{MV}_{t\hat{s}}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt})) - (\overline{MV}_{ts_{jt}}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))
\end{aligned} \tag{146}$$

Substituting this into equation (145), we obtain

$$\begin{aligned}
& \sum_{\ell'} \Pi_{\ell'|s_{jt}^*} \mathbb{E}_{a_{jt} \in S_{s_{jt}^*}} \left[\tilde{\mu}_{\ell'}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) \right]^2 \\
& \quad - \mathbb{E}_{a_{jt} \in S_{s_{jt}^*}} \left[\bar{\mu}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) \right]^2 \\
& = \sum_{\hat{s}} \Pi_{\hat{s}|s_{jt}^*} (\lambda (\overline{MV}_{t\hat{s}}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt})) - (\overline{MV}_{ts_{jt}^*}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt})))^2 \\
& \quad - (\overline{MV}_{ts_{jt}^*}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2
\end{aligned} \tag{147}$$

To make further progress, we now note that:

$$\begin{aligned}
& \sum_{\hat{s}} \Pi_{\hat{s}|s_{jt}^*} (\overline{MV}_{t\hat{s}}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt})) \\
& = p(\overline{MV}_{ts_{jt}^*}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt})) \\
& \quad + \frac{1-p}{N-1} \left(\sum_{\hat{s}} (\overline{MV}_{t\hat{s}}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt})) - (\overline{MV}_{ts_{jt}^*}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt})) \right) \\
& = \lambda(\overline{MV}_{ts_{jt}^*}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))
\end{aligned} \tag{148}$$

where the second equality uses the observation that $\sum_{\hat{s}} (\overline{MV}_{t\hat{s}}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt})) = 0$.

Following the same steps, we also find:

$$\begin{aligned}
& \sum_{\hat{s}} \Pi_{\hat{s}|s_{jt}^*} (\overline{MV}_{t\hat{s}}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2 \\
& = \lambda(\overline{MV}_{ts_{jt}^*}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2 + \frac{1-\lambda}{n} \sum_{\hat{s}} (\overline{MV}_{t\hat{s}}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2
\end{aligned} \tag{149}$$

where this time the summation term does not evaluate to 0.

Expanding the square in equation (147), and applying these two identities, implies

$$\begin{aligned}
& \sum_{\ell'} \Pi_{\ell'|s_{jt}^*} \mathbb{E}_{a_{jt} \in S_{s_{jt}^*}} \left[\tilde{\mu}_{\ell'}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) \right]^2 - \mathbb{E}_{a_{jt} \in S_{s_{jt}^*}} \left[\bar{\mu}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) \right]^2 \\
& = -\lambda^2 \left[(2-\lambda)(\overline{MV}_{ts_{jt}^*}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2 - (1-\lambda) \frac{1}{n} \sum_{\ell} (\overline{MV}_{t\ell}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2 \right]
\end{aligned} \tag{150}$$

We now take the expectation of this over the possible segments s_{jt}^* . Since a_{jt} is i.i.d and

independent of (k_{jt}, z_{jt}) , and the segments are equal-mass, s_{jt}^* is uniform on $\{1, \dots, n\}$ and independent of (k_{jt}, z_{jt}) . The expectation is then a simple unweighted average over segments, implying

$$\begin{aligned}
& \mathbb{E}_{s_{jt}^*} \left[\sum_{\ell'} \Pi_{\ell'} | s_{jt}^* \mathbb{E}_{a_{jt} \in S_{s_{jt}^*}} [\tilde{\mu}_{\ell'}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 \right. \\
& \quad \left. - \mathbb{E}_{a_{jt} \in S_{s_{jt}^*}} [\bar{\mu}(k_{jt}, z_{jt}) - \text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1)]^2 \right] \\
&= -\lambda^2 \left[(2 - \lambda) \frac{1}{n} \sum_{\ell} (\overline{MV}_{t\ell}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2 - (1 - \lambda) \frac{1}{n} \sum_{\ell} (\overline{MV}_{t\ell}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2 \right] \\
&= -\lambda^2 \frac{1}{n} \sum_{\ell} (\overline{MV}_{t\ell}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2 \tag{151}
\end{aligned}$$

In fact, we can simplify even further, using the definition of $\tilde{\mu}$ again to write:

$$\lambda^2 \frac{1}{n} \sum_{\ell} (\overline{MV}_{t\ell}^N(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2 = \frac{1}{n} \sum_{\ell} (\tilde{\mu}_{\ell}(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2 = \mathbb{V}_{\ell}(\tilde{\mu}_{\ell}(k_{jt}, z_{jt})) \tag{152}$$

Spelling this out more explicitly, we can expand out our $\tilde{\mu}$ notation to write:

$$\mathbb{V}_{\ell}(\tilde{\mu}_{\ell}(k_{jt}, z_{jt})) = \mathbb{V}_{\ell} \left(\mathbb{E}[\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) | k_{jt}, z_{jt}, \ell] | k_{jt}, z_{jt} \right) \tag{153}$$

where the outer variance is taken over the report ℓ the forecaster reads (equivalently, over $(a, \text{transmission noise})$).

Condition (32) is therefore satisfied, and the candidate, $\mathbf{m}_t$, can be sustained as a symmetric equilibrium, if and only if

$$N_t(k_{j't}, z_{j't}) - \kappa_{j't} \leq N_t(k_{jt}, z_{jt}) - \kappa_{jt} \tag{154}$$

for all pairs of reported and unreported firms j, j' , where

$$N_t(k_{jt}, z_{jt}) \equiv \frac{1}{n} \sum_{\ell} (\tilde{\mu}_{\ell}(k_{jt}, z_{jt}) - \bar{\mu}(k_{jt}, z_{jt}))^2 \equiv \mathbb{V}_{\ell} \left(\mathbb{E}[\text{MV}_t(k_{jt}, z_{jt}, a_{jt}, 1) | k_{jt}, z_{jt}, \ell] | k_{jt}, z_{jt} \right). \tag{155}$$

The reporting policy for which this condition holds is the one described in Proposition [B.1](#), with the threshold N_t^* determined by:

$$\int_0^1 \mathbb{1}(\mathbb{N}_t(k_{jt}, z_{jt}) - \kappa_{jt} \geq N_t^*) dj = r. \quad (156)$$